

Electromagnetic fields as in vitro monotherapy on cancer cells: a systematic analysis of the existing evidence

Abstract

This report presents an analysis of the literature on the use of electromagnetic fields as monotherapy on cancer cell cultures in vitro, together with a discussion of their selectivity. The analysis is conducted on a dataset built from 49 primary studies.

The following sections describe the structure of the dataset and the distribution of the different field types used; subsequently, a detailed analysis is carried out for each of the macro-categories of field identified (ELF-EMF, LF-EMF, SMF, RF-EMF, PMF, Electric Fields).

1 Dataset structure

The dataset `FinalTable.xlsx` consists of 439 rows and 132 columns. An expanded version, that is, `FinalTable_expanded.xlsx`, was produced by adding two columns reporting the title and DOI of the reference paper for each entry, and the one-hot encodings of the `Genomic_List`, `Proteomic_List`, and `Cell_Cycle_Phase` columns, reaching a total of 273 columns. Both files are available at <https://github.com/silviadenti24/Master-Thesis-Project.git>. Throughout this report, papers are cited by their numerical identifier in square brackets; the corresponding title and DOI can be retrieved from `FinalTable_expanded.xlsx`.

1.1 Row organization

Each row in the dataset represents a single experimental condition. The same study can contribute multiple rows whenever it tests different combinations of cell line, field type, field parameters (frequency, intensity, waveform, and other variables that depend on the specific type of field considered) or exposure times. This choice is important because aggregating at the study level would erase the differences that the dataset is instead meant to highlight: the same study may report opposite effects on different cell lines or under different exposure parameters, so each of these combinations must be considered as a unique observation.

1.2 Column organization

The columns are organized into four thematic blocks:

- **Study and experimental setup.** The ID of the study from which the information was extracted, the tumor and/or healthy cell lines used, the type of field applied and its parameters.
- **Cell fate outcomes.** The seven outcomes (viability, proliferation, apoptosis, necrosis, ferroptosis, senescence, motility) that describe the fate of the cells following exposure to the field and represent the main indicators of the observed effect.
- **Mechanistic outcomes.** ROS, antioxidants, membrane potential, and other mechanisms involved in the cellular response to the field.
- **Molecular outcomes.** For each gene or protein reported in the individual experimental conditions, the dataset indicates whether it was measured in the specific condition and, if so, in which direction (significantly increased, significantly decreased, or unchanged).

2 Focus on non-thermal mechanisms

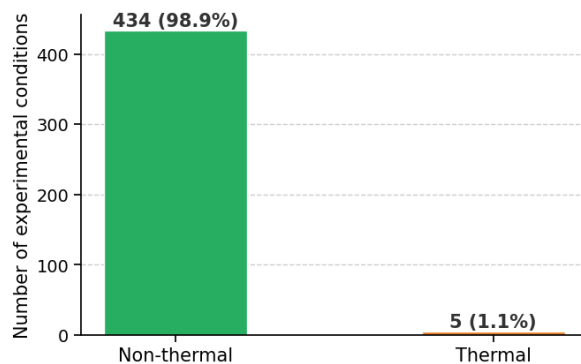


Figure 1: Distribution of experimental conditions by mechanism type. The majority of conditions in the dataset (434, 98.9%) are associated with non-thermal mechanisms, while only 5 conditions (1.1%) involve a thermal contribution and have been excluded from the subsequent analyses.

The whole analysis is conducted under experimental conditions in which the observed effect is not attributable to a thermal mechanism, that is, to the simple heating of cells induced by the absorption of electromagnetic energy. The purpose of the research is, in fact, to understand whether biophysical interactions exist between the field and the biological system that operate beyond the thermal effect.

For this reason, the rows in which the authors report a thermal contribution have been excluded from the dataset used for the analysis. As can be seen in Figure 1, their number is very low and marginal and therefore does not distort the overall picture of the analysis.

3 Types of electromagnetic field

3.1 Distribution of experimental conditions

Within the dataset, various experimental conditions test different types of electromagnetic field. Figure 2 shows, for each category of electromagnetic field present in the dataset, the corresponding number of experimental conditions.

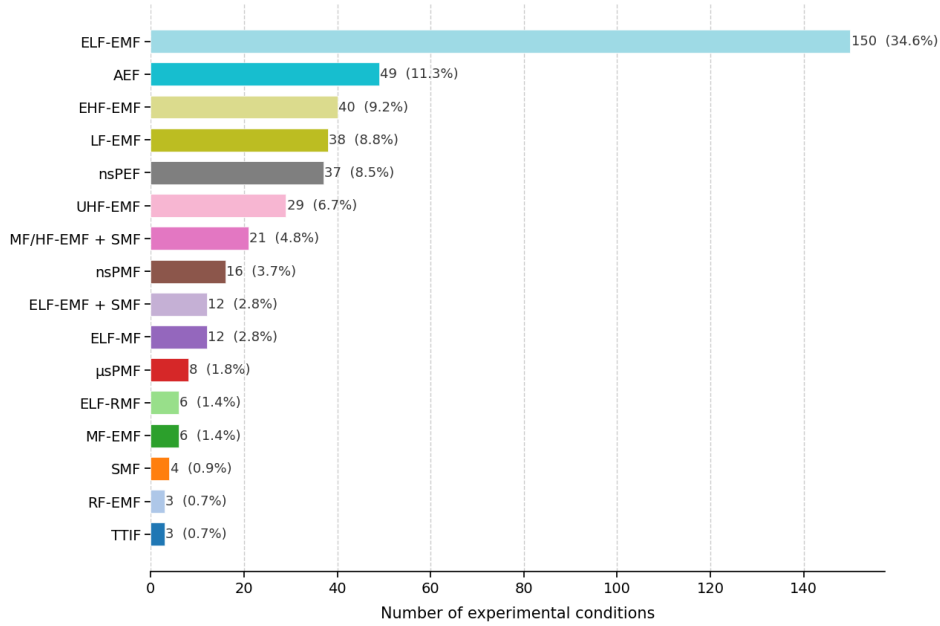


Figure 2: Number of experimental conditions for each type of electromagnetic field present in the dataset. The acronyms are: Extremely Low Frequency Electromagnetic Fields (ELF-EMF), Alternating Electric Fields (AEF), Extremely High Frequency Electromagnetic Fields (EHF-EMF), Low Frequency Electromagnetic Fields (LF-EMF); nanosecond Pulsed Electric Fields (nsPEF); Ultra High Frequency Electromagnetic Fields (UHF-EMF); combined Medium/High Frequency Electromagnetic Fields with Static Magnetic Fields (MF/HF-EMF + SMF); nanosecond Pulsed Magnetic Fields (nsPMF); combined Extremely Low Frequency Electromagnetic Fields with Static Magnetic Fields (ELF-EMF + SMF); Extremely Low Frequency Magnetic Fields (ELF-MF); microsecond Pulsed Magnetic Fields (μ sPMF); Extremely Low Frequency Rotating Magnetic Fields (ELF-RMF); Medium Frequency Electromagnetic Fields (MF-EMF); Static Magnetic Fields (SMF); Radio Frequency Electromagnetic Fields (RF-EMF); and Tumor Treating Induced Fields (TTIF).

As can be seen, the most widely used field type is Extremely Low Frequency Electromagnetic Fields with 150 experimental conditions, followed by Alternating Electric Fields (49 experimental conditions) and Extremely High Frequency Electromagnetic Fields (40 experimental conditions). The characteristics of all these fields and their parameters will be described in their dedicated sections.

3.2 Distribution of independent studies

The number of experimental conditions per field category is a first indicator, but it has to be read bearing in mind that a particularly prolific study can generate many of them within the same field type. For this reason, the counting by conditions is complemented by a counting at the study level: for each category, the number of distinct studies that contributed to it is reported.

Figure 3 shows, for each type of electromagnetic field, the number of studies that include at least one experimental condition tested with that specific field type. It should be noted that if a study tests two or more types of field belonging to different categories, it is counted in each of the corresponding categories; for this reason, in Fig. 3, the sum of the counts across categories exceeds 49, which is the total number of studies analyzed. The studies that investigated more than one type of field are

4, namely [8] (ELF-EMF, ELF-MF), [18] (ELF-EMF + SMF, SMF), [22] (nsPMF, μ sPMF), and [44] (LF-EMF, SMF).

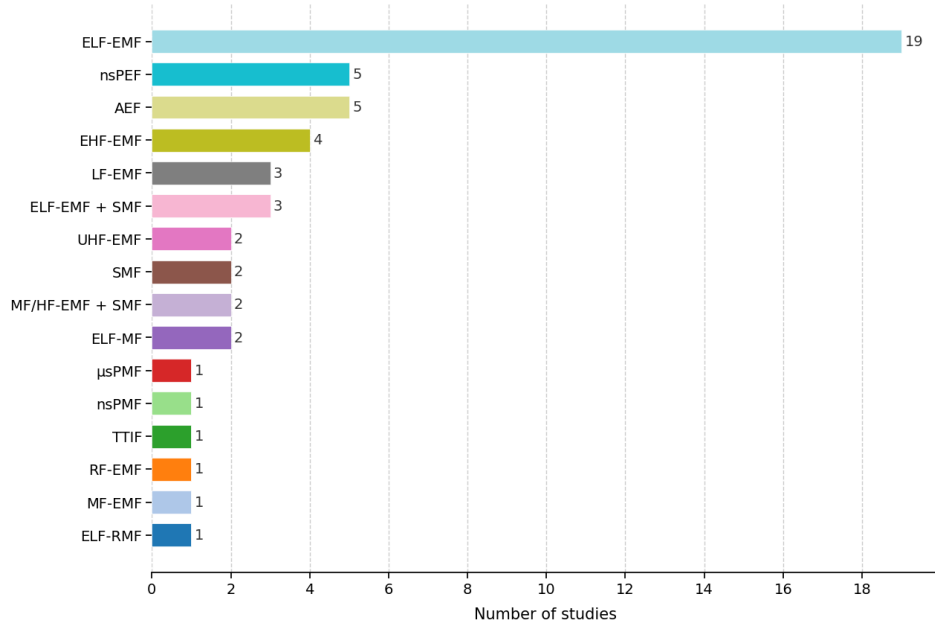


Figure 3: Number of distinct studies that include at least one experimental condition tested with each type of electromagnetic field. Studies that investigated more than one field type belonging to different categories are counted once in each corresponding category, so the sum across categories exceeds the total number of studies in the dataset (49).

4 Grouping into macro-categories

The field categories identified are numerous and in part heterogeneous with respect to minor technical differences (for example, variants of the same ELF family are indicated with different acronyms depending on the conventions used by the authors). In order to make the downstream analyzes more readable and to obtain subgroups with a minimum useful size, the categories have been grouped into 6 macro-categories, as shown in Table 1. Each macro-category has its own dedicated section in the remainder of the report.

Macro-category	Field categories included
ELF-EMF	ELF-EMF, ELF-EMF + SMF, ELF-RMF, ELF-MF, AMF / ELF-MF
SMF	SMF
LF-EMF	LF-EMF
RF-EMF	MF/HF-EMF + SMF, MF-EMF, UHF-EMF, EHF-EMF, RF-EMF
PMF	nsPMF, μ sPMF
Electric Fields	AEF, TTIF, nsPEF

Table 1: Grouping of field categories into the six macro-categories used in the downstream analyses.

Figures 4 and 5 show, respectively, the number of experimental conditions that fall into each of the six macro-categories listed above and the number of independent studies contributing to each one. The Extremely Low Frequency Electromagnetic Fields category is the most represented both in terms of experimental conditions (180) and in terms of contributing studies (24), confirming that its dominance is not driven by a single prolific study but reflects a broader interest across the literature.

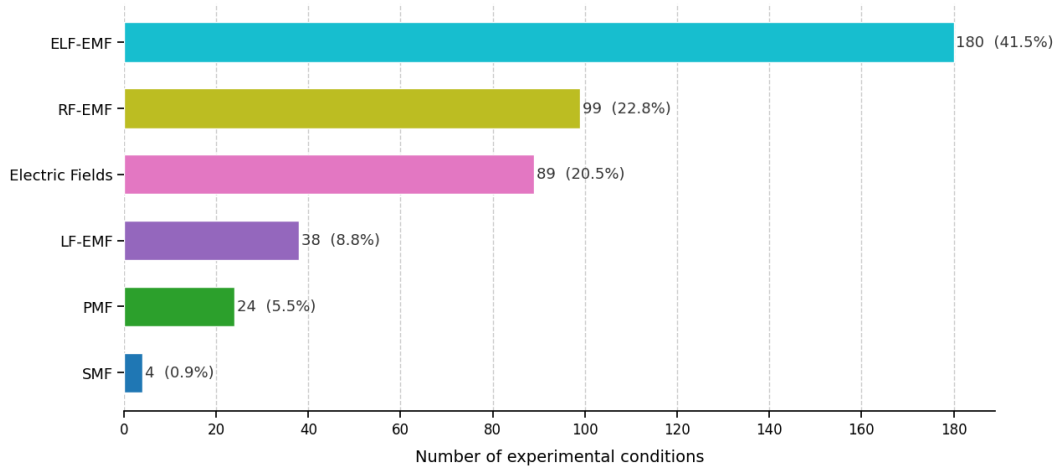


Figure 4: Number of experimental conditions falling into each of the six macro-categories defined in Table 1.

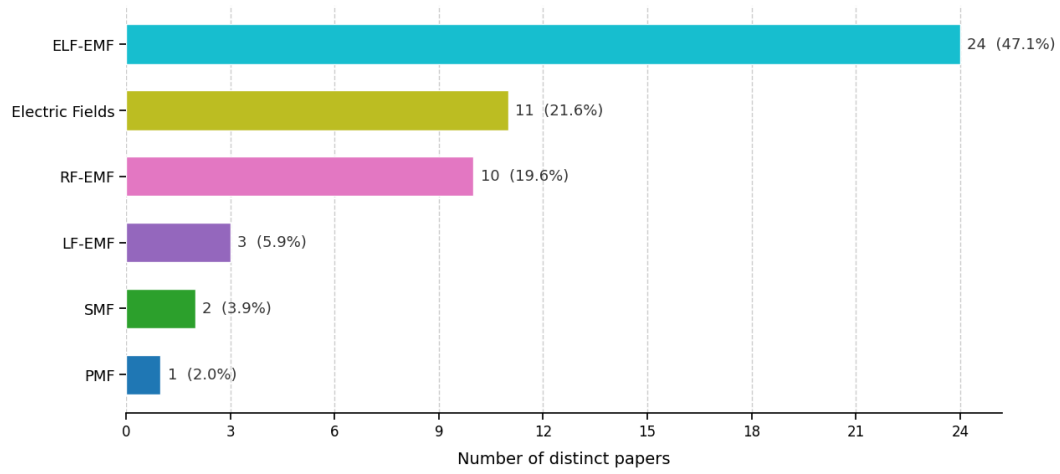


Figure 5: Number of distinct studies contributing to each EMF macro-category. Studies that investigated field types belonging to more than one macro-category are counted once in each corresponding subgroup, so the sum across categories exceeds the total of 49 studies.

5 Gap identification

Before moving on to the analysis for each macro-category, it is important to show that within each of them there is a marked imbalance in how many experimental conditions address which biological mechanisms. This gap identification is carried out in order to highlight one of the main limitations of this analysis: the lack of a consistent set of investigated mechanisms across the different macro-categories. This limits the possibility of comparing the macro-categories with one another and, in many cases, restricts the ability to build a coherent picture for a given macro-category.

The two heat maps at the experimental condition level (Figures 6 and 7) show the amount of data available for each macro-category-mechanism combination of the EMF, while those at the study level (Figures 8 and 9) allow one to assess how many independent research groups have addressed each combination, correcting for the distortion introduced by very prolific studies. It is important to consider these four heat maps together, because otherwise these indicators, taken separately, can lead to misleading conclusions.

The most evident result is the broader coverage of viability and proliferation, which are the two most widely tested outcomes across almost all macro-categories. However, looking at these two outcomes, a significant difference emerges between them: viability is covered more uniformly across macro-categories, whereas proliferation already shows gaps (for example in RF-EMF and PMF), suggesting that not even the best-known outcomes are consistently applied.

Apoptosis represents the third most covered outcome, but in a non-uniform way: it is well represented in ELF-EMF (44 conditions, 24% of the subgroup) and in RF-EMF (39 conditions, 39% of the subgroup), while it is almost absent in LF-EMF, SMF and PMF. This heterogeneity has consequences: in several macro-categories it is not possible to verify whether a decrease in viability is attributable to a programmed cell death mechanism or to non-specific causes, and this limits the possibility of formulating solid mechanistic hypotheses.

The situation is worse for mechanistic and molecular outcomes, such as calcium signaling, membrane potential, metabolic reprogramming, cell cycle analysis, genomic analysis, and proteomic analysis. Figure 6 shows that the coverage of these outcomes is concentrated almost exclusively in ELF-EMF and RF-EMF, while the other macro-categories display largely empty rows. For LF-EMF, SMF, PMF and Electric Fields the characterization is fragmentary or absent. This represents the most serious gap emerging from the dataset.

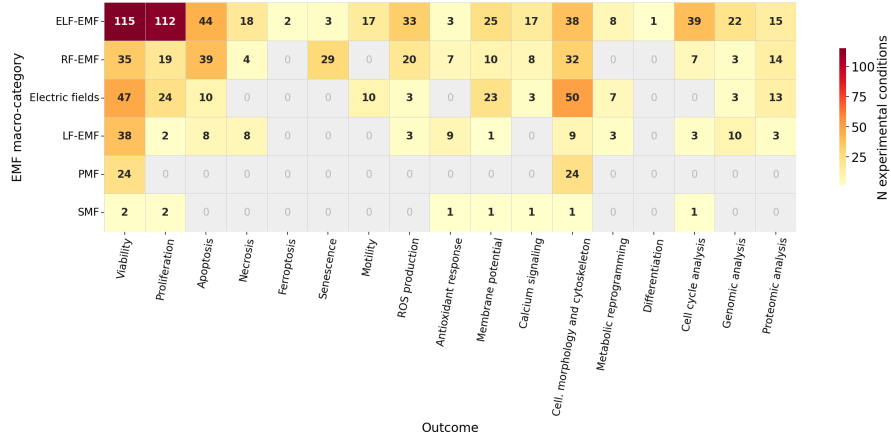


Figure 6: Absolute number of experimental conditions in which each biological mechanism was investigated, broken down by EMF macro-category. Empty cells (zero conditions) are shown in grey to highlight the gaps in the literature.

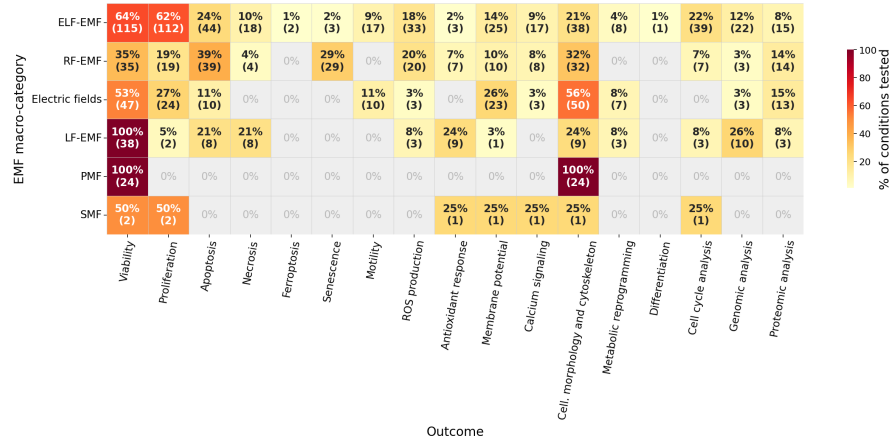


Figure 7: Percentage of experimental conditions in which each biological mechanism was investigated, relative to the total number of conditions within each EMF macro-category. Absolute counts are shown in parentheses.

Outcomes such as ferroptosis, differentiation, senescence, and necrosis are only rarely covered. Ferroptosis, in particular, appears only in the ELF-EMF subgroup, with only 2 experimental conditions from a single study (Figures 6 and 8).

The comparison between the heat map normalized by conditions (Figure 7) and the one normalized by studies (Figure 9) reveals that some percentages, taken on their own, would be misleading. For example, PMF shows values of 100% for viability and cell morphology and cytoskeleton (Figure 6); however, in absolute terms, these correspond to only 24 experimental conditions, all traceable to a single study (Figures 7 and 9). The same applies to SMF, where the 25–50% percentages derive from only 2 conditions overall.

Moving from the condition level to the study level makes it possible to assess how solid the observations that can be drawn actually are. Comparing Figures 6 and 8, it can be seen that, even where the number of experimental conditions is high, the number of independent studies is often much lower; this leads to a downsizing of the possible conclusions.

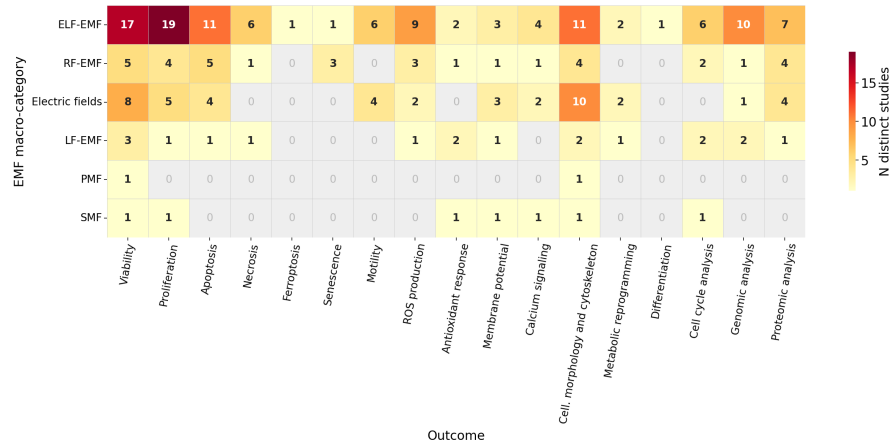


Figure 8: Number of distinct studies in which each biological mechanism was investigated at least once, broken down by EMF macro-category. Compared to the condition-level counts (Fig. 6), this view reflects the breadth of independent scientific interest in each mechanism-field combination, regardless of how many experimental conditions each study contributed.

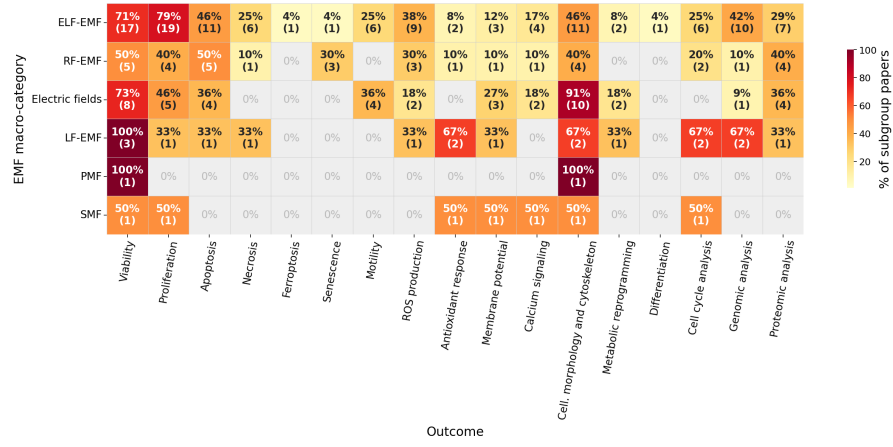


Figure 9: Percentage of distinct studies within each EMF macro-category that investigated each biological mechanism at least once. Absolute paper counts are shown in parentheses. The denominator for each row is the total number of studies contributing to that specific macro-category, not the full set of 49 studies in the dataset.

6 Extremely Low Frequency Electromagnetic Fields

According to the general categorization of electromagnetic frequencies by the IEEE (Institute of Electrical and Electronics Engineers), ELF is Extremely Low electromagnetic Frequencies that cover the interval of 3 Hz – 30 kHz [10]. Other studies often refer to ELF as the narrower frequency range of 0 Hz – 300 Hz, following a convention commonly adopted in the biomedical literature [13], [28].

According to previous studies on childhood leukemia risk, due to residential exposure to ELF-EMF, in 2002 IARC classified ELF-EMF in Group 2B (possibly carcinogenic to humans). Nevertheless, many studies have reported the positive effects and therapeutic uses of ELF-EMF, e.g. wound repair, bone repair, pain management, and Alzheimer’s disease (AD) [13].

In recent years, there has been growing interest in the possible effects of ELF-EMF on tumor cells, and many studies have been conducted to evaluate their efficacy and explore alternative approaches to cancer treatment.

6.1 Dataset overview and cell line distribution

The ELF-EMF macro-category constitutes the largest dataset of the entire analysis, with 178 experimental conditions from 24 independent studies. Of these 178, 131 (73.6%) concern tumor cell lines and 47 (26.4%) healthy cell lines.

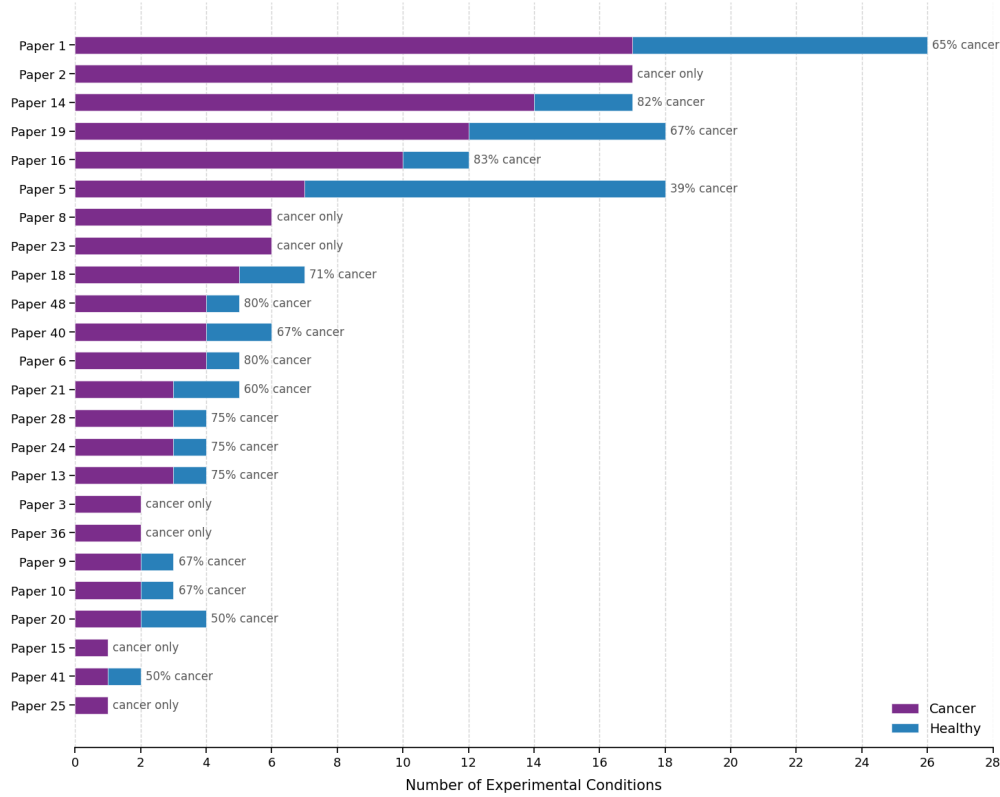


Figure 10: Cancer vs healthy cell line distribution across the ELF-EMF studies.

Figure 10 shows the ratio between tumor lines and healthy lines tested for each study. Note that 17 of 24 include both types, allowing a direct evaluation of the selectivity of ELF-EMFs to target tumor cells while preserving healthy ones, a fundamental requirement for a potential therapeutic application in oncology.

Tables 2 and 3 report respectively the tumor lines and healthy lines exposed to ELF-EMFs. A total of 38 different cancer cell lines and 19 healthy cell lines were tested. The majority experimental conditions focus on breast cancer (MDA-MB-231, MCF-7) and tumors of the central nervous system (N2A, CT2A), joined by the corresponding healthy lines (MCF-10A, C8D1A). As can be seen in the

two figures, numerous other tumor categories have been studied, although with a smaller number of experimental conditions, making the landscape heterogeneous and offering a broader view, not limited to only two neoplastic types.

Cell line	Type	Category	Origin	Exp. cond.
MDA-MB-231	Triple-negative breast cancer (TNBC)	Breast	Human	20
MCF-7	Breast cancer	Breast	Human	13
N2A	Neuroblastoma	Brain & CNS	Mouse	9
CT2A	Glioblastoma	Brain & CNS	Mouse	8
PANC-1	Pancreatic cancer	Pancreatic	Human	8
Raji	Human lymphoma cells (Burkitt lymphoma)	Hematological	Human	6
HT-1080	Fibrosarcoma	Sarcoma	Human	5
PC12	Pheochromocytoma	Adrenal	Rat	5
SKBR3	Breast cancer	Breast	Human	5
A549	Lung adenocarcinoma	Lung	Human	4
1627	Glioblastoma	Brain & CNS	Human	4
HeLa	Cervical adenocarcinoma	Cervical	Human	4
B16-BL6	Melanoma	Melanoma	Mouse	3
DIPG	Diffuse Intrinsic Pontine Glioma	Brain & CNS	Human	3
U373 MG	Glioblastoma astrocytoma	Brain & CNS	Human	3
DU-145	Prostate cancer	Prostate	Human	3
B16F10	Melanoma	Melanoma	Mouse	3
GBM115	Glioblastoma	Brain & CNS	Human	3
HepG2	Human liver cancer cells	Liver	Human	2
MC-38-Luc	Colon adenocarcinoma	Colorectal	Mouse	2
MC4-L2	Breast duct carcinoma (ER+)	Breast	Mouse	1
SiHa-LUC	Cervical cancer	Cervical	Human	1
CaSki-LUC	Cervical cancer	Cervical	Human	1
HeLa-LUC	Cervical cancer	Cervical	Human	1
HT-29	Colorectal carcinoma (primary tumor)	Colorectal	Human	1
Caki1	Metastatic clear cell renal (ccRCC)	Kidney	Human	1
769-P	Clear cell renal cell carcinoma (ccRCC)	Kidney	Human	1
786-O	Clear cell renal cell carcinoma (ccRCC)	Kidney	Human	1
U251-P53	Glioblastoma (P53 variant)	Brain & CNS	Human	1
LLC	Lung carcinoma	Lung	Mouse	1
MDA-MB-468	Triple-negative breast cancer (TNBC)	Breast	Human	1
BT549	Triple-negative breast cancer (TNBC)	Breast	Human	1
U87	Glioblastoma	Brain & CNS	Human	1
HT-1197	Bladder cancer	Bladder	Human	1
K562	Chronic myelogenous leukemia	Hematological	Human	1
U251	Glioblastoma	Brain & CNS	Human	1
G401	Nephroblastoma (Wilms tumor)	Kidney	Human	1
HeLa-LUC (co-culture with CD8+ T cells)	Cervical cancer (co-culture)	Cervical	Human	1

Table 2: Cancer cell lines used in the ELF-EMF macro-category: type, category, origin, and number of experimental conditions.

Cell line	Type	Category	Origin	Exp. cond.
C8D1A	Astrocytes	Brain & CNS	Mouse	9
MCF-10A	Breast epithelial cells	Breast	Human	8
A9	Mouse fibroblasts	Connective Tissue	Mouse	3
3T3	Fibroblast	Connective Tissue	Mouse	3
HEK293T	Embryonic kidney cells	Kidney	Human	2
CV-1	Kidney fibroblasts	Kidney	African green monkey	2
COS-1	SV40-transformed kidney cells	Kidney	African green monkey	2
PAM	Alveolar macrophages (primary cells)	Lung	Porcine	2
HEK293	Embryonic kidney epithelial	Kidney	Human	2
HaCaT	Keratinocyte (skin)	Skin	Human	2
293T	Normal human renal epithelial cells	Kidney	Human	2
HDF	Fibroblast	Connective Tissue	Human	2
NIH/3T3	Normal fibroblast	Connective Tissue	Mouse	2
RPE1	Retinal pigment epithelial	Eye	Human	1
hLEC	Lung lymphatic microvascular endothelial cells	Lung	Human	1
HUVEC	Umbilical vein endothelial cells	Vascular	Human	1
FF95	Normal dermal fibroblast	Skin	Human	1
Normal human fibroblast	Normal fibroblast	Connective Tissue	Human	1
CD8+ T cells	Cytotoxic T lymphocytes	Blood/Immune	Human	1

Table 3: Healthy cell lines used in the ELF-EMF macro-category: type, category, origin, and number of experimental conditions.

6.2 Exposure parameters

The parameters characterizing an ELF-EMF are three: frequency, intensity, and waveform. The frequency indicates the number of oscillations of the field per unit of time, expressed in Hertz (Hz). In the case of ELF-EMFs, it falls within the extremely low frequency region (see the discussion above). In this analysis, intensity refers to the intensity of the magnetic field, expressed in microtesla (μT). In particular, in Figure 12 reference is made to the maximum intensity reached by the field during exposure. Although these are electromagnetic fields, therefore also possessing an electrical component, the focus here is on the magnetic component, as it is more intense and capable of penetrating deeper into biological tissues. In some experimental conditions, in addition to the alternating magnetic field (AC component), a static magnetic field (DC component) is also applied to study the effects on cell lines when the two components are superimposed. The waveform describes the temporal evolution of the field. The shapes most commonly adopted in the considered studies are sinusoidal, square, and pulsed, each characterized by a different temporal profile.

C.ID	Paper	Field Type	Wave Type	Freq. (Hz)	Max AC (μT)	DC (μT)
C1	13	ELF-EMF	Sine	0.01	100000	
C2	13	ELF-EMF	Sine	1	100000	
C3	6, 25	ELF-EMF		1	100000	
C4	23	Gradient RMF	Periodic	2.5	410000	

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C.ID	Paper	Field Type	Wave Type	Freq. (Hz)	Max AC (μ T)	DC (μ T)
C5	21	ELF-EMF	Sine	3	1	
C6	2	ELF-EMF	Sine	3	70	45
C7	2	ELF-EMF	Sine	4	70	45
C8	40	LF-RMF	Sine	4.2	100000	
C9	40	LF-RMF	Sine	4.2	400000	
C10	23	Gradient RMF	Periodic	5	410000	
C11	23	Gradient RMF	Periodic	5	40000	
C12	23	Gradient RMF	Periodic	5	200000	
C13	2	ELF-EMF	Sine	6	70	45
C14	36	ELF-EMF + SMF	Sine	6	100	45
C15	20	SREMF	Square	7.83	200	
C16	20	SREMF	Square	7.83	600	
C17	2	ELF-EMF	Sine	8	70	45
C18	10	ELF-PEMF	Full square (pulsed)	8	11000	
C19	2	ELF-EMF	Sine	10	70	45
C20	2	ELF-EMF	Sine	14	70	45
C21	18	ELF-EMF + SMF	Sine	14.5	10	45
C22	18	ELF-EMF + SMF	Sine	15	10	45
C23	18	ELF-EMF + SMF	Sine	15.5	10	45
C24	18	ELF-EMF + SMF	Sine	16.5	18	45
C25	18	ELF-EMF + SMF	Sine	16.5	18.5	45
C26	48	PEMF	Sine	20	3000	
C27	16	ELF-EMF	Sine-like (not explicitly specified)	20	5000	
C28	1	ELF-EMF	Square pulse	20	100	
C29	36	ELF-EMF + SMF	Sine	24	100	45
C30	14	ELF-EMF	Square	30	100	
C31	15	ELF-PEMF		30	1500–16000	
C32	5	MEMP	Pulsed (multifrequency)	<30	>2e+06	
C33	1	ELF-EMF	Square pulse	30	100	
C34	18	ELF-EMF + SMF	Sine	33	10	45
C35	1	ELF-EMF	Square pulse	40	100	
C36	1	ELF-EMF	Square pulse	45	100	
C37	14	ELF-EMF	Square	45	100	
C38	19	ELF-MF		50	1000	
C39	14	ELF-EMF	Square	50	100	
C40	24	ELF-EMF		50; 385	20000	
C41	9	ELF-EMF + SMF	Sine	50	1000–2500	3000–4000
C42	3	ELF P-EMF	Sine	50	35000	
C43	3	ELF P-EMF	Sine	50	22600	
C44	41	ELF-EMF	Sine	50	1000	
C45	1	ELF-EMF	Square pulse	50	100	
C46	19	ELF-MF		50	100	
C47	28	ELF-EMF	Sine	50	4500	
C48	14	ELF-EMF	Square	55	100	
C49	1	ELF-EMF	Square pulse	55	100	
C50	48	PEMF	Sine	60	1200	
C51	1	ELF-EMF	Square pulse	60	100	
C52	14	ELF-EMF	Square	80	100	

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C.ID	Paper	Field Type	Wave Type	Freq. (Hz)	Max AC (μ T)	DC (μ T)
C53	8	Spinning Osc. Mag. Field	Sine	137	5000	
C54	8	Osc. Mag. Field - Helmholtz coil	Sine	137	5000	
C55	8	Spinning Osc. Mag. Field	Sine	272	5000	
C56	21	ELF-EMF	Complex patterned (see notes)			1

Table 4: ELF-EMF exposure conditions lookup table.

Table 4 summarizes all the experimental conditions (from C1 to C56) that correspond to the different combinations of these parameters. It is important to note that not all experimental conditions were tested in all cell lines introduced in Section 6.1. Since the data come from independent studies, each condition was applied only to the subset of cell lines used in the corresponding study. In some cases, different studies used the same cell lines, allowing cross-comparisons. With the exception of condition C3, used in two distinct studies, each experimental condition can be traced back to a single study; conversely, the same study may have explored multiple experimental conditions, as can be seen in the second column of the Table 4.

The exposure times of the cell lines to the field are not included in the table and will be discussed later.

Finally, it should be noted that, although dealing only with the ELF-EMF macro-category, the table includes the Field Type column: within this macro-category, the various authors have named the fields used with different names, and, for reasons of transparency with respect to the original sources, it was chosen to report the denomination exactly as provided by the authors themselves.

6.2.1 Frequency distribution and rationales

Figure 11 provides an overview of the cell lines tested at each frequency. The figure does not show the exact cell lines tested at each frequency; it illustrates how frequently each frequency has been used in the macro-category. What immediately stands out is that 50 Hz has been tested in a larger number of cell lines (both cancer and healthy) than any other frequency. This is because 50 Hz corresponds to the standard domestic power frequency [3], [9] and is the frequency of ELF magnetic fields generated by electric devices and equipment in clinical practice, industrial environments, and common domestic situations in Europe [19], [41]. The choice of this frequency is therefore driven by environmental relevance rather than by a specific biomedical rationale.

Another frequency frequently tested is 20 Hz. The rationale reported by the authors is that the choice was motivated by the good results obtained in previous studies [16], or by previous studies that demonstrated their effectiveness in modulating cellular functions [48].

In [1] and [14], the objective was to explore the biological window between 20 and 100 Hz.

In [13], multiple frequencies (0.01 Hz, 0.1 Hz, 1 Hz, 10 Hz) were tested to identify the frequency producing the most pronounced biological effect for each cell line in the study.

In [23], the frequencies 2.5 Hz and 5 Hz were tested, stopping at the latter due to limitations of the device used to generate the field.

In [21], the exposure signal was designed to replicate a physiological pattern: the EMF was modeled after the physiological activation pattern of intercellular calcium waves in murine cells located in the longitudinal layer of the cecum and colon. The pattern was designed to have the same amplitude, frequency, and temporal characteristics as a naturally occurring calcium wave. The frequency of 3 Hz was then used as a comparison.

Particularly notable are the rationales behind the use of 7.83 Hz (the Schumann Resonance frequency), which closely approximates the resonance frequency of calcium ions (Ca^{2+}) within some cancer cells. Low-intensity EMFs at this frequency have been shown to significantly inhibit cancer cell proliferation, probably through modulation of Ca^{2+} influx [20]. A similar approach is found in [10], where the frequency was approximated to 8 Hz using the rationale for the magnetic frequency of the Earth.

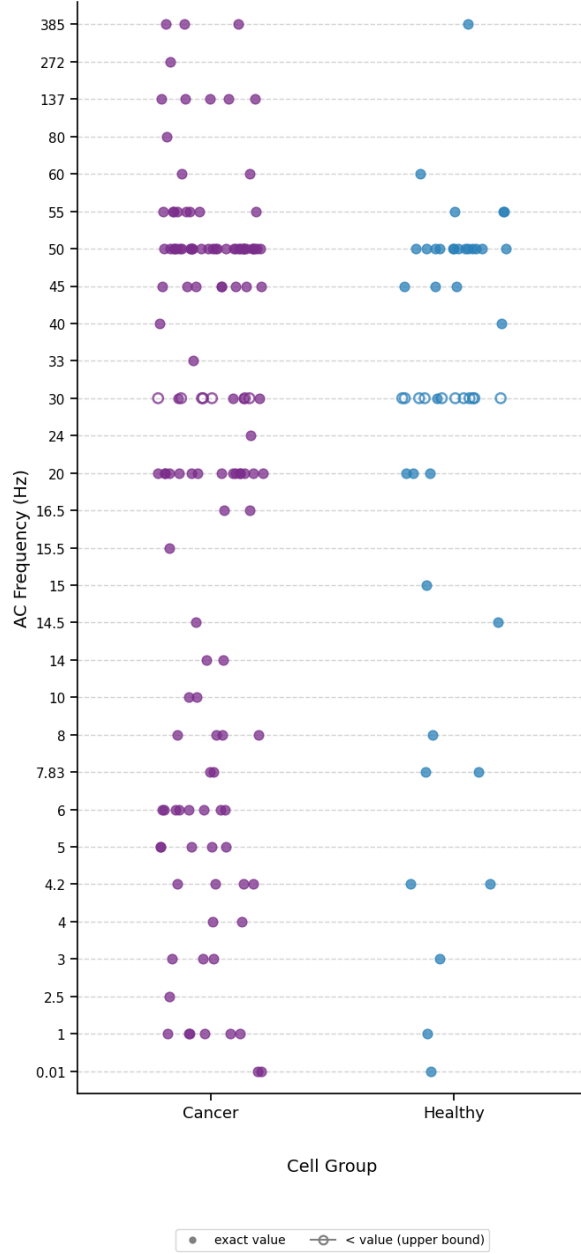


Figure 11: Distribution of tested frequencies by cancer and healthy cells.

In [18], a different approach was adopted: frequencies were chosen based on proton-proton J-coupling constants (13–18 Hz range) observed in NMR spectroscopy for biological molecules, particularly trans-olefinic protons with a J value of 16.5 Hz, and to investigate resonance phenomena in feedback systems with time delays in biological processes.

In [5], the equipment uses a non-ionizing multifrequency electromagnetic field with a frequency lower than 30 Hz, designed to interact with the electronegative charge of cell cultures in a cell-type-dependent manner.

Finally, a particularly interesting approach is found in [2] and [36], where a thermodynamic model was used to calculate a cell line specific frequency (frequency is the inverse of the characteristic time τ defined by the thermal exchange equation $\nu \sim \frac{1}{\tau}$). This represents one of the most compelling strategies adopted by researchers to identify the optimal frequency at which to expose cells to the field.

6.2.2 Intensity distribution and rationales

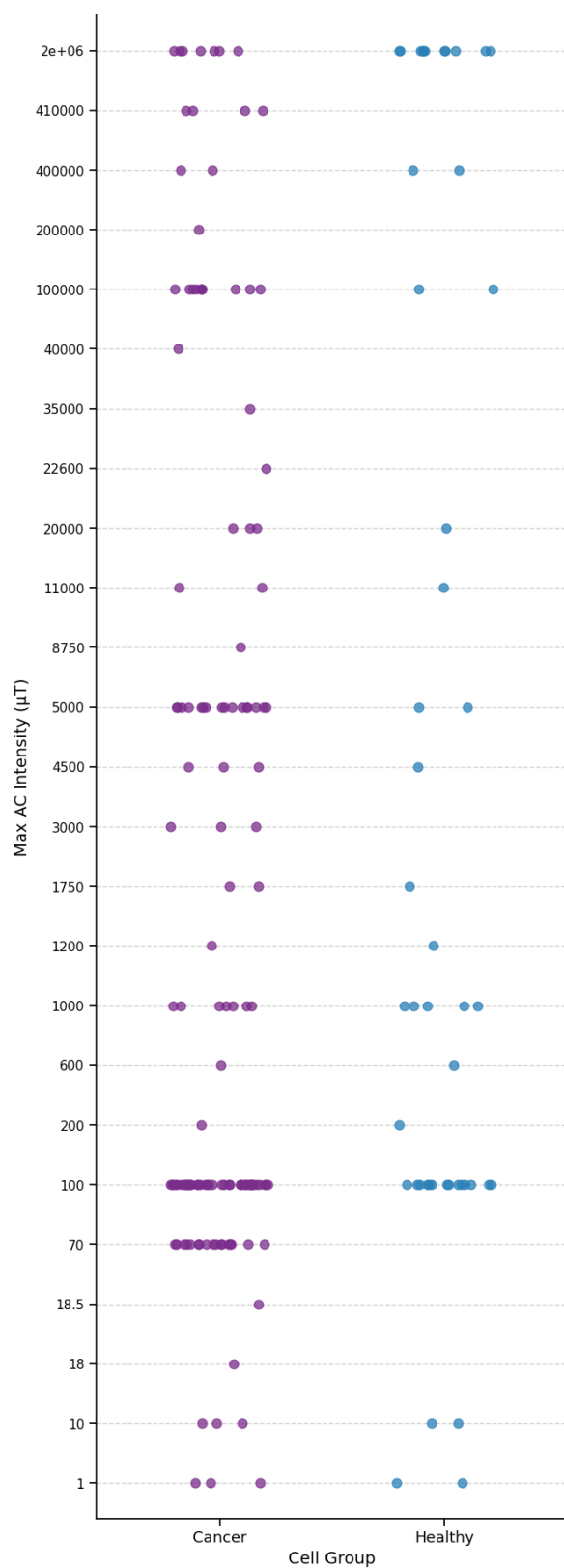


Figure 12: Distribution of tested max AC intensities by cancer and healthy cells.

The maximum AC field intensities used across the dataset span microtesla (μT) up to the Tesla (T) scale. The rationales behind the choice of these intensities vary considerably.

In [1], [14], and [19], an intensity of 100 μT was chosen because it corresponds to the exposure limit established by EU Recommendation 1999/519/EC in Spain for a frequency of 50 Hz, chosen as standard reference intensity for bioelectromagnetic in vitro studies.

Below this threshold, an intensity of 1 μT was also used, which coincides with the intensities of numerous biomolecular pathways [21].

Intensities of 10 μT and 18.5 μT were tested, combined with a DC field intensity of 45 μT , so the AC and DC field intensities were deliberately chosen to lie significantly below the ionizing frequency thresholds and thermal noise levels, ensuring that the MF strength was insufficient to directly break chemical bonds, so that the observed effects likely result from changes in reaction rates mediated by the alignment of proton nuclear spins [18].

An AC intensity of 100 μT was also tested in combination with a DC field of 45 μT , where the DC component of 45 μT represents the average value of the Earth’s magnetic field, used to simulate the geomagnetic background [36].

Despite the limits imposed by the EU Recommendation, the majority of the intensities tested across the dataset exceed the established threshold. In [3] and [24], the intensities were dictated by the technical limitations of the devices used.

In [8], multiple peak-to-peak amplitude values (~ 0.42 , ~ 1.16 , ~ 5.46 , ~ 58.3 mT) were tested by varying the distance between cell culture plates and the oncoscillator (7, 5, 3 and 1.4 cm respectively) to determine the effective magnetic field strength range for ROS induction based on the radical pair mechanism.

An intensity of 0.011 T was employed in the context of stochastic resonance, which was considered critical for cellular effects without increase in energy [10].

Intensities of 0.1 T and 0.4 T were used to investigate the effects on F-actin and breast cancer metastasis, with these two intensities selected to compare dose-dependent effects [40].

In [5], a field intensity greater than 2 T was applied, designed to generate electronic perturbations that can alter the electronic distribution of biomolecules in a cell-type-dependent manner.

The remaining studies selected their intensities on the basis of results obtained in previous work or following preliminary comparisons among different intensities, ultimately adopting the one that proved most effective for their specific purposes.

6.3 Cell fate outcomes

Cell fate outcome	Cancer exp. conditions	Healthy exp. conditions	Total
Viability	79	36	115
Proliferation	83	29	112
Apoptosis	37	7	44
Necrosis	14	4	18
Ferroptosis	2	0	2
Senescence	2	1	3
Motility	15	2	17

Table 5: Number of ELF-EMF macro-category experimental conditions assessing each cell fate outcome, broken down by cancer and healthy cell lines.

Table 5 reports, for each cell fate outcome, the number of experimental conditions under which that specific outcome was assessed, divided by cancer and healthy cell lines. The last column represents the sum of the two.

Viability and proliferation are the outcomes that are the most frequently tested in both cancer and healthy cells. Apoptosis ranks as the third most studied outcome within the ELF-EMF subgroup. The remaining outcomes (necrosis, ferroptosis, senescence, and motility) have been investigated under considerably fewer experimental conditions, which may limit the robustness of subsequent analyzes.

6.3.1 Viability

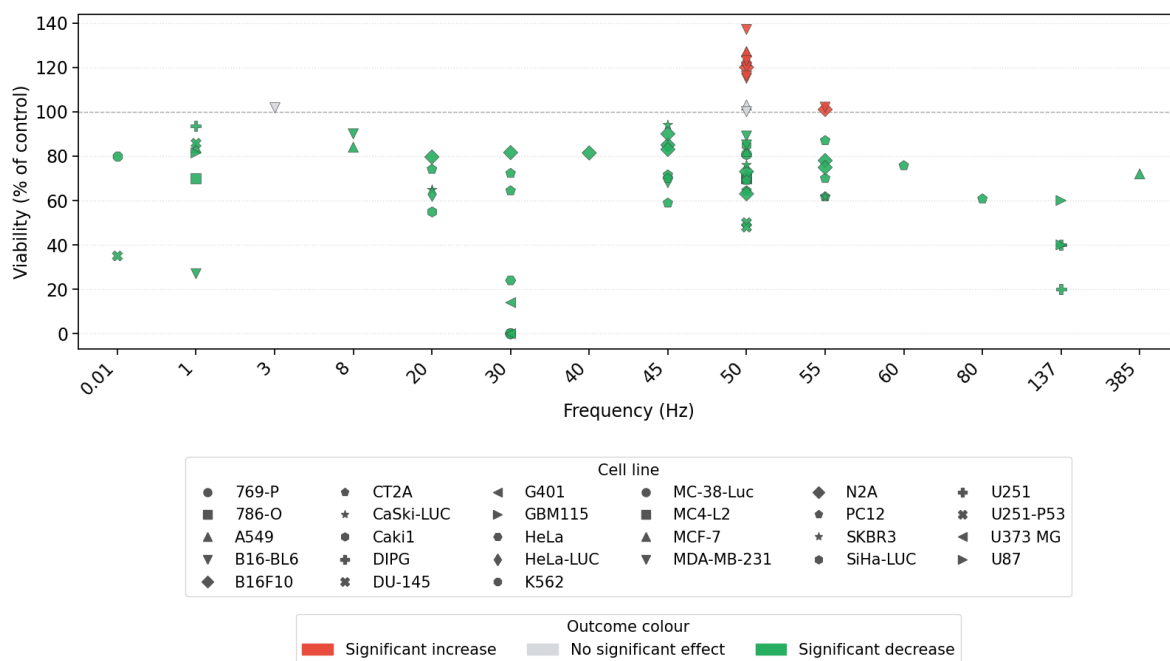


Figure 13: Residual viability (% of control) of cancer cell lines as a function of frequency across ELF-EMF exposure conditions. Each marker represents a distinct cell line; color indicates the direction and significance of the effect.

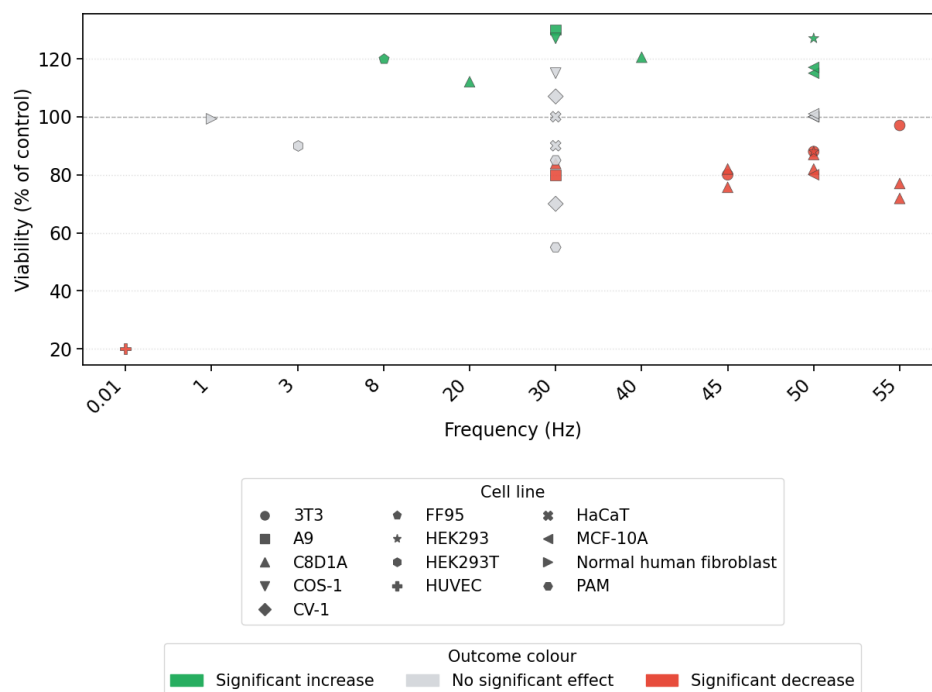


Figure 14: Residual viability (% of control) of healthy cell lines as a function of frequency across ELF-EMF exposure conditions. Each marker represents a distinct cell line; color indicates the direction and significance of the effect.

Viability data comprise 115 experimental conditions, 79 cancer cell lines and 36 healthy cell lines.

Effective conditions. From Figure 13, the most effective condition in reducing cancer cell viability is C32 (Multifrequency Electromagnetic Pulses, <30 Hz, >2 T): in the cancer cell lines U373 MG, HeLa and MC-38-Luc, viability reaches 0% compared to the control groups after only 5 minutes of exposure, and already at 2.5 minutes the values are extremely low (0–24% compared to the control). This result must be put into context, as the field intensity employed (>2 T) is orders of magnitude higher than that of the other experimental conditions and exceeds any regulatory exposure limit. Nevertheless, from Figure 14, the most interesting aspect is that healthy cells HaCaT, CV-1, COS-1, A9 and PAM tested under the same conditions do not show a significant decrease in viability and even reach viability percentages that are significantly higher than those of the control group (127% for COS-1 after 5 minutes of exposure and 130% for A9 after 2.5 minutes of exposure). However, for A9, the situation reverses at 5 minutes of exposure, where a significant decrease is observed, reaching 80% compared to the control group. The authors of this study do not report the precise details of their device, stating only that modulator equipment was designed and developed by the North American corporation Paso Alto Biotechnology (Inc.), under industrial secret. They also argue that the electronic perturbation induced by this MEMP in the culture medium may alter the electronic distribution of the biomolecules in a cell type dependent manner.

Condition C53 (Spinning Oscillating Magnetic Fields, 137 Hz, 5 mT) shows significant reductions in glioblastoma cell lines: DIPG reaches 20% viability compared to the control group with intermittent stimulation: T_{on} 250 ms, T_{off} 750 ms, for 4 h total. This corresponds to 19200 pulse trains. GBM115 reaches 40%. The field used in the study (sOMF) is generated by rapidly rotating neodymium permanent magnets. The field has two components: sinusoidal magnetic field oscillations and cyclically changing angles of the field axis.

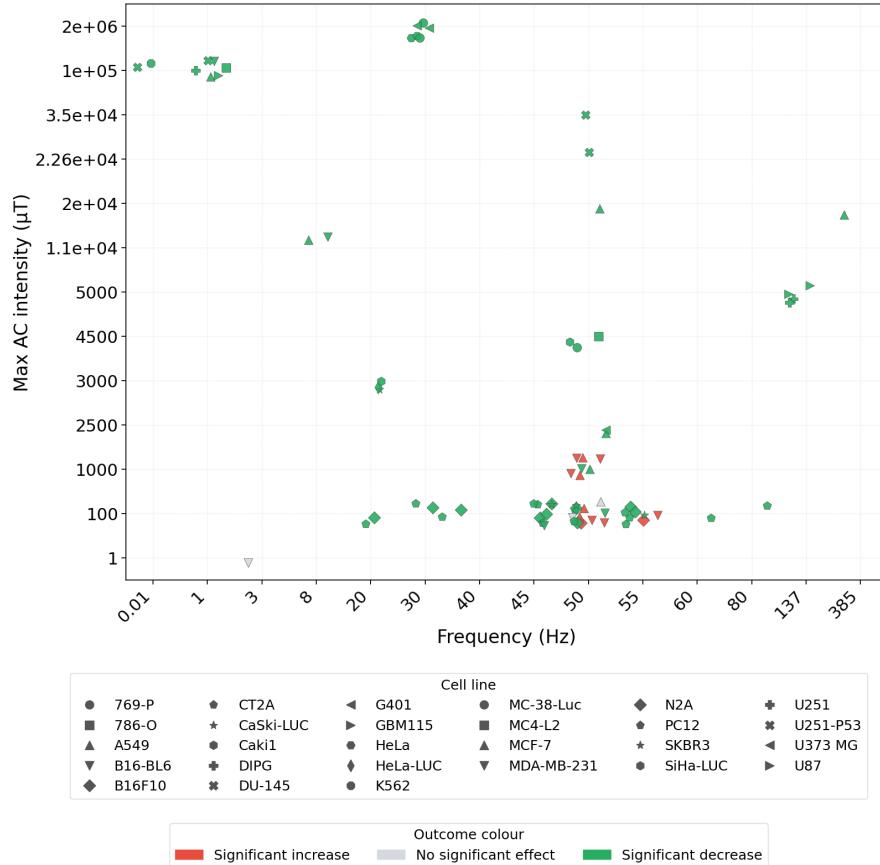


Figure 15: Viability direction with respect to the control group of cancer cell lines as a function of the frequency and the maximum AC intensity (μT) across ELF-EMF exposure conditions. Color indicates the direction and significance of the effect.

Conditions at very low frequency and high intensity also prove effective in reducing cancer cell viability: at 0.01 Hz and 100 mT (C1), after 120 hours of field exposure, DU-145 drops to 35% compared to the control group; K562 cells, cultured in suspension, also show a significant reduction, although smaller than the other cell line (80% compared to the control).

At 1 Hz and 100 mT (C3), the MDA-MB-231 cell line reaches 27% viability compared to the control after 120 hours (5 days) of field exposure.

From Figure 15, it can be seen that at an intensity of 100 μ T several frequencies are tested. The best results are observed on CT2A cells at 55 Hz (C49, viability 61.7% compared to the control at 72 hours of exposure) and at 45 Hz (C36, 58.9% at 72 hours), with a dose-time effect: by extending the exposure from 24 hours to 72 hours, the viability decreases progressively (CT2A at 45 Hz: 71.5% at 24 h, 58.9% at 72 h; at 55 Hz: 70.0% at 24 h, 61.7% at 72 h). The same trend is observed for N2A at 50 Hz (73% at 24 h, 63% at 48 h). PC12 cells at 80 Hz (C52) also reach a significantly low viability of 60.74% after 72 hours of exposure.

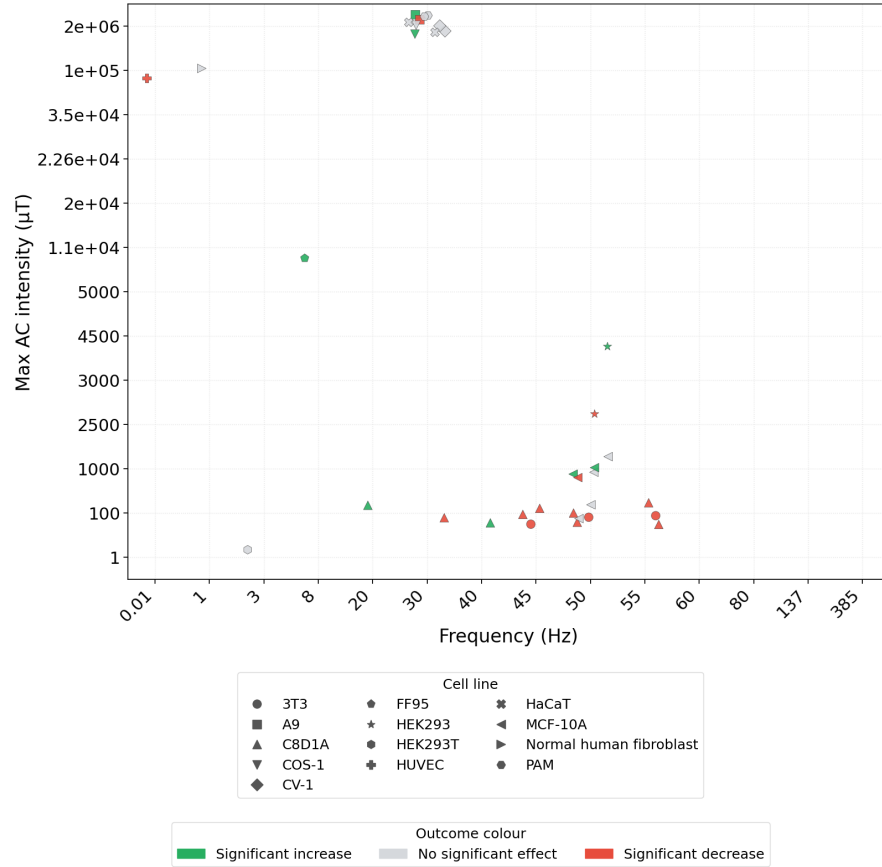


Figure 16: Viability direction with respect to the control group of healthy cell lines as a function of the frequency and the maximum AC intensity (μ T) across ELF-EMF exposure conditions. Color indicates the direction and significance of the effect.

Counterproductive conditions. From Figures 13 and 14, it can also be observed that the frequency of 50 Hz can prove problematic in several cases. For MDA-MB-231 and MCF-7 cell lines, using intensities of 0.1 mT (C46) and 1 mT (C38), viability increases compared to the control group, reaching 137% (MDA-MB-231, 1 mT, 4 hours of field exposure) and 127% (MCF-7, 0.1 mT and 1 mT, 8 hours of exposure). The latter condition shows a reversal at 24 hours of exposure, when viability at 1 mT drops to 82% compared to the control group. This suggests that short exposures at 50 Hz at these intensities may have a pro-proliferative effect on breast carcinoma cells, representing a potential risk.

B16F10 (C39) at 50 Hz and 100 μ T also shows an increase in viability (120% compared to the control at 24 hours).

Selectivity. The conditions showing the best selectivity, targeting cancer cells while leaving healthy cells unaffected, are condition C32 (<30 Hz and >2 T), as already discussed; C28 (20 Hz and 100 μ T), where cancer cells CT2A drop to 74% while healthy C8D1A cells rise to 112%; C18 (8 Hz and 0.011 T), where the two cancer cell lines MDA-MB-231 and MCF-7 reach 90% and 84% viability compared to the control, respectively, while healthy FF95 cells reach 120%.

In contrast, at 45 Hz and 100 μ T (C36/C37), the selectivity is minimal: healthy C8D1A cells reach 82% compared to the control group after 24 hours of exposure and 75.69% after 72 hours, while 3T3 cells reach 80% at 72 hours, values comparable to those observed in cancer cell lines at the same frequency. Condition C1 (0.01 Hz, 100 mT) also shows negative selectivity: healthy HUVEC cells drop to 20%, a value lower than those of cancer cell lines tested under the same condition.

6.3.2 Proliferation

Proliferation data comprise 112 experimental conditions, 83 on cancer cell lines and 29 on healthy cell lines.

Effective conditions. The most striking antiproliferative effects are observed under condition C32 (<30 Hz, >2 T): colony formation capacity in U373 MG, HeLa and MC-38-Luc drops to approximately 20%, 15%, and 5% respectively after 2.5 minutes of exposure, and collapses to zero colonies after 5 minutes. Healthy cells HaCaT, CV-1, COS-1, and A9 show no significant reduction in colony formation under the same condition. Condition C18 (8 Hz and 0.011 T) produces a dramatic and selective reduction in cancer cell proliferation: MDA-MB-231 drops to 16% of the control and MCF-7 to 33%, while healthy FF95 cells show an increase to 123%. This represents one of the most pronounced selectivity profiles in the entire dataset for this outcome. At condition C3 (1 Hz and 100 mT), significant antiproliferative effects are observed in glioblastoma lines: U87 reaches 83% and U251 68% of control after 5 days of exposure. U251-P53 cells, carrying a P53 mutation, show a non-significant decrease (70%), consistent with the P53-dependence of the antiproliferative response. Normal human fibroblasts exposed to the same condition also show a significant decrease (86%), limiting the selectivity of this condition for the proliferation endpoint. At low frequency with thermodynamically calculated parameters (C6 at 3 Hz, 70 μ T + 45 μ T DC; C7 at 4 Hz), PANC-1 and 1627 glioblastoma cells show significant reductions of 13–22% in proliferation. MCF-7 cells are significantly inhibited at condition C14 (6 Hz, 70 μ T + 45 μ T DC, reaching 76%). Notably, in the same study HepG2 cells at condition C27 (20 Hz, 5 mT) show significant inhibition in adherent culture (82%), while Raji suspended lymphoma cells are not affected when dispersed; however, when Raji cells are maintained in aggregated form in a restricted geometry (25-cm² flask), inhibition rises to 36% by day 9. The authors attribute this to cell-cell contact facilitated by the field, proposing that physical proximity is required to maintain the inhibitory stimulus. Conditions C15 and C16 (7.83 Hz, 200 and 600 μ T) produce strong colony formation inhibition in HeLa cells (49% and 38% respectively), while NIH/3T3 healthy cells remain completely unaffected at both intensities. At 50 Hz, the picture is heterogeneous. In breast cancer cell lines, conditions C46 (0.1 mT) and C38 (1 mT) show opposing patterns depending on exposure time: at 4 hours MDA-MB-231 and MCF-7 proliferation is above control (110–115%); at 24 hours the effect either persists or reverses depending on the intensity, with MCF-7 at 1 mT dropping to 77–82%. This temporal reversal is consistent with the viability data already discussed. Under condition C44 (50 Hz and 1 mT), MDA-MB-231 instead shows a clear pro-proliferative effect (+35%), while healthy MCF-10A cells are significantly inhibited (75%). This constitutes a further signal that the 50 Hz / 1 mT parameter window is problematic for triple-negative breast cancer and should be avoided therapeutically. At condition C47 (50 Hz, 4.5 mT), proliferation was not directly measured; the cell-count data in the viability section are used as a proxy. At condition C50 (60 Hz, 1.2 mT), PEMF exposure of CD8+ T cells leads to a 40% reduction in proliferation in co-cultured HeLa-LUC cancer cells, consistent with the enhanced cytotoxic activity of the activated T cells.

Counterproductive conditions. Beyond the 50 Hz / 1 mT case already discussed, condition C18 with reversed polarity in HT-1080 fibrosarcoma cells (study in the SMF subgroup) serves as a reminder that at lower frequencies and sub-threshold intensities, stimulatory effects can emerge even in cell lines where inhibition is the goal.

6.3.3 Apoptosis

Apoptosis data comprise 44 experimental conditions, 37 on cancer cell lines and 7 on healthy cell lines.

The most prominent apoptotic response in the subgroup is observed in MC4-L2 ductal breast carcinoma cells under condition C3 (1 Hz and 100 mT, 10 hours of exposure): the apoptotic population increases approximately 9-fold compared to the control group. Western blot analysis confirms the activation of both the intrinsic pathway (cleaved caspase-9, 13-fold increase) and the extrinsic pathway (cleaved caspase-8 and cleaved caspase-3, each approximately 11.6-fold), establishing a mixed apoptotic and necroptotic pattern. Normal human fibroblasts exposed to the same condition show no significant increase in apoptosis (142%, not significant), confirming the selectivity of C3 for this endpoint. In the three glioblastoma cell lines tested under C3, apoptotic responses differ markedly depending on P53 status. U87 cells (wild-type P53) reach 256% of control and MCF-7 reaches 509%; U251 (mutant P53) shows a weaker but still significant increase (198%); and U251-P53 cells, in which wild-type P53 was reintroduced by transfection, reach 598%, the highest apoptotic response among the glioblastoma lines. Under condition C41 (50 Hz, 1.0–2.5 mT + SMF), G401 and A549 cancer cells show significant increases in apoptosis (200% and 275% respectively) on the third day of exposure, consistent with the ferroptosis data that accompany this condition. Under conditions C42 and C43 (50 Hz at 35 mT and 22.6 mT), DU-145 prostate cancer cells show strong apoptotic induction (481% and 270% respectively), with the higher intensity producing the stronger effect. Under conditions C53 and C54 (spinning oscillating magnetic field at 137 Hz), a modest but significant increase in caspase-3 activation is observed in GBM115 and DIPG glioblastoma cells, consistent with the viability reductions reported under these conditions. Under conditions C15 and C16 (7.83 Hz at 200 and 600 μ T), HeLa cells exhibit progressive morphological features of apoptosis, and at 50 Hz and 1 mT (C44), MDA-MB-231 shows no apoptotic response, consistent with the pro-proliferative and pro-migratory phenotype already described for this condition.

The apoptotic response was not significant in cancer cell lines K562 and MDA-MB-231 under condition C1 (0.01 Hz, 100 mT) at 120 hours, despite substantial ROS induction, suggesting that the mechanism of viability reduction under this condition is not primarily apoptotic. For HUVEC healthy cells under C1, apoptosis was significantly elevated at 24 and 48 hours but not at 120 hours; taken together with the negative selectivity in viability and ROS already described, this confirms that C1 is not a selective condition.

Under condition C50 (60 Hz, 1.2 mT), cancer cell apoptosis is induced indirectly: when CD8+ T cells are exposed and then co-cultured with HeLa-LUC cells, apoptosis in the cancer line increases to 195% of control, confirmed by c-CASP3/CASP3 upregulation and Bcl-2 downregulation. IL-37 overexpression in T cells abolishes the effect, identifying IL-37 suppression as the upstream mechanism.

6.3.4 Necrosis

Necrosis was analyzed under 18 experimental conditions, 14 in cancer cell lines and 4 in healthy cell lines.

Cell lines U87, U251, U251-P53, MCF-7 and normal human fibroblasts were exposed to condition C3 (1 Hz and 100 mT, 10 hours of field exposure). A significant increase in necrosis is observed compared to the corresponding control groups in three cancer cell lines: U87 (336%), U251-P53 (635%) and MCF-7 (303%). The cancer cell line U251 shows a non-significant increase (248%). The selectivity of this condition is good, as normal fibroblasts do not only not show an increase in necrosis, but exhibit a slight decrease (77.5% compared to the control group).

The same condition C3 is used in the ductal breast carcinoma cell line MC4-L2. In this case, necroptosis (a regulated form of necrosis) is observed, confirmed by the increased phosphorylation of RIPK1, RIPK3 and MLKL. These are three key proteins of the necroptotic signalling cascade: RIPK1 (Receptor-Interacting Protein Kinase 1) and RIPK3 (Receptor-Interacting Protein Kinase 3) are serine/threonine kinases that form the necrosome complex upon activation, which in turn phosphorylates MLKL (Mixed Lineage Kinase Domain-Like protein), the terminal effector that translocates to the plasma membrane and causes its permeabilization, leading to cell death. ROS accumulation and calcium overload are required to induce ELF-EMF-mediated necroptosis.

An increase in necrosis is reported in cancer cell lines A549 (126%), LLC (115%) and MDA-MB-231 (135%) under condition C40 (50/385 Hz, 20 mT, 10 minutes of exposure), while healthy hLEC cells remain unchanged (100%). The authors attribute this effect to the interaction of the pulsed magnetic

field with sulphated glycosaminoglycans (heparan sulfate) present on the cell membrane. Heparan sulfate chains carry a high density of negative charges due to their sulfate and carboxyl groups; cancer cells are known to overexpress these anionic surface molecules compared to healthy cells. The authors hypothesize that the pulsed magnetic field exerts a mechanical or electrical perturbation on these charged structures, preferentially compromising membrane integrity in cells with a higher density of sulphated glycans, which would explain the selective cytotoxic effect observed on the cancer cell lines.

Necrosis at 50 Hz and 4.5 mT (C47, 2.5 hours of exposure) is not significant in any of the cell lines tested (786-O, 769-P, Caki1 and healthy HEK293).

Conditions C42 (50 Hz and 35 mT, 30 minutes of exposure) and C43 (50 Hz and 22.6 mT, 30 minutes of exposure) do not show significant changes in necrosis compared to the control group on DU-145 cells.

A noteworthy finding emerges: at 50 Hz and 1 mT (C44, 4 hours of exposure), healthy MCF-10A cells show a significant increase in necrosis (200% compared to control), while cancer MDA-MB-231 cells remain unchanged (106%, not significant). This is the only condition in the entire subgroup where necrosis preferentially affects healthy cells, representing a case of negative selectivity.

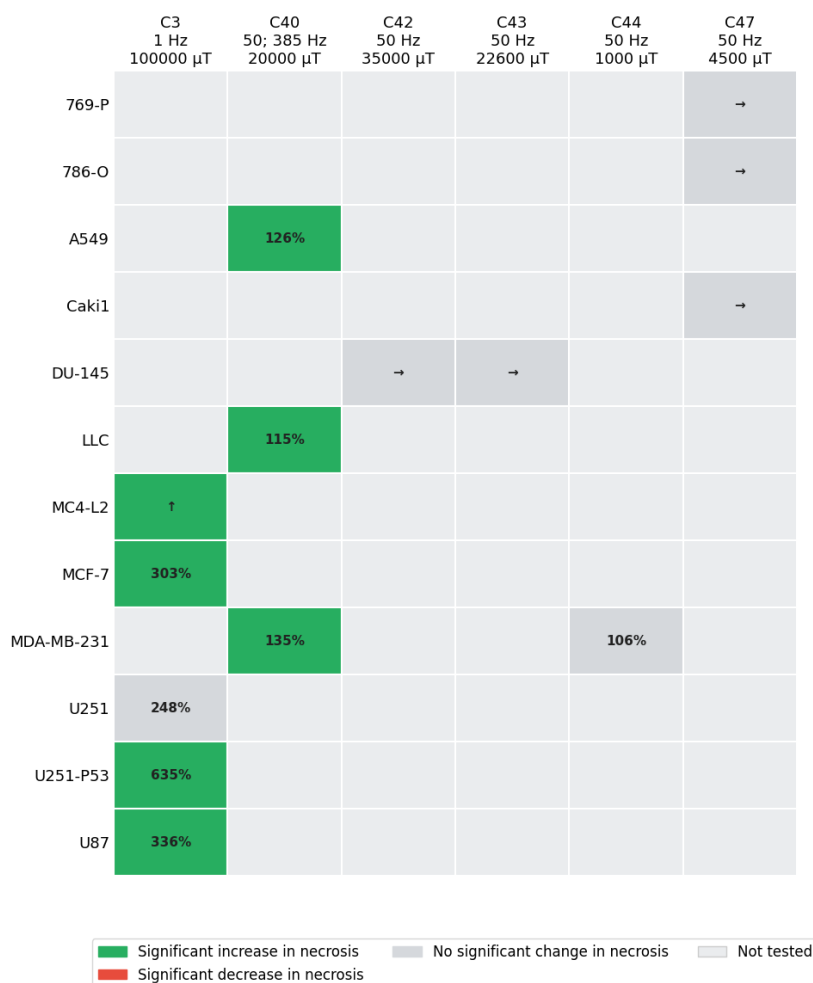


Figure 17: Necrosis heatmap for cancer cell lines. Green indicates a significant increase in necrosis, red indicates a significant decrease, dark gray indicates no significant change, and light gray denotes untested combinations. Percentages relative to the control group are reported inside each cell, when available.

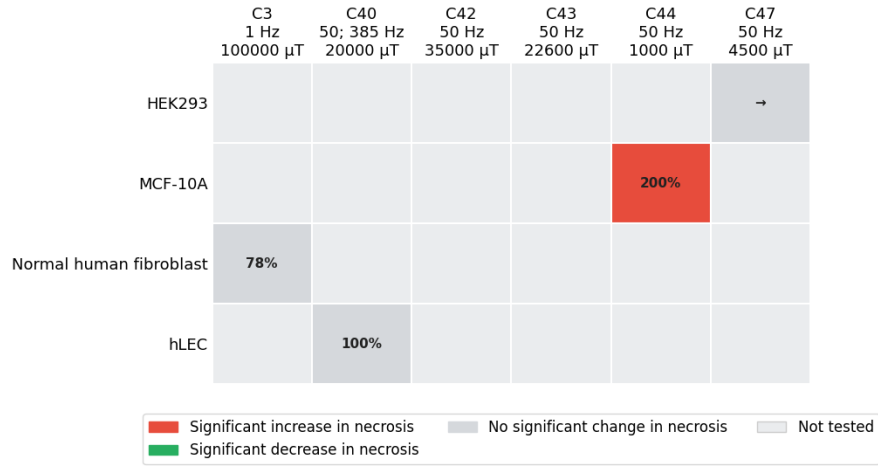


Figure 18: Necrosis heatmap for healthy cell lines. Red indicates a significant increase in necrosis, green indicates a significant decrease, dark gray indicates no significant change, and light gray denotes untested combinations. Percentages relative to the control group are reported inside each cell, when available.

6.3.5 Ferroptosis

Ferroptosis was measured exclusively on the cancer cell lines G401 and A549, both exposed to condition C41. The field used in this study is a combined modulated magnetic field: a 50 Hz ELF-EMF modulated by a static magnetic field (SMF). Intensities vary periodically through 8 rounds (T1–T8) with AC (ELF-EMF) values of 1.5, 2.5, 1.5, 2.5, 1, 1.5, 1, 1.5 mT and DC (SMF) values of 3, 4, 3, 4, 3, 4, 3, 4 mT. The combined time-averaged intensity is 5.1 mT. Each round lasts 3.5 min, making each session 30 min, with a total daily exposure of 2h (4 sessions).

The study reports that Ferrostatin-1 (Fer-1) partially reversed the growth inhibition induced by the magnetic field on the third day of exposure (2 hours per day for three consecutive days). Fer-1 is a specific inhibitor that exclusively blocks ferroptosis without interfering with other cell death mechanisms: by adding it to the treatment, the growth inhibition of the two cell lines was reduced, indicating that when ferroptosis is blocked, cells survive to a greater extent. This confirms that ferroptosis contributes to the antitumor mechanism. This is the first report of MF-induced ferroptosis.

The increase in ferroptosis is accompanied by a significant reduction in viability and proliferation in both cell lines and by a significant increase in apoptosis compared to the control group.

6.3.6 Senescence

Senescence was measured in only 3 experimental conditions (2 cancer and 1 healthy), all under condition C18 (8 Hz and 0.011 T, full square wave, 30 minutes of field exposure twice a day for 5 days). Cancer cells MDA-MB-231 show a 952% increase in senescence compared to the control group and MCF-7 a 400% increase, whereas healthy FF95 cells remain unchanged (100%). These results indicate that condition C18 is capable of inducing senescence in breast carcinoma cells with a highly precise selectivity over healthy cells. This finding is consistent with what was reported for viability (90% and 84% for cancer cell lines and 120% for the healthy cell line) and proliferation (16% and 33% for cancer cell lines and 123% for the healthy cell line).

6.3.7 Motility

Motility was analyzed under 17 experimental conditions, 15 in cancer cell lines and 2 in healthy cell lines, from 6 independent studies. A brief commentary for each study is reported below.

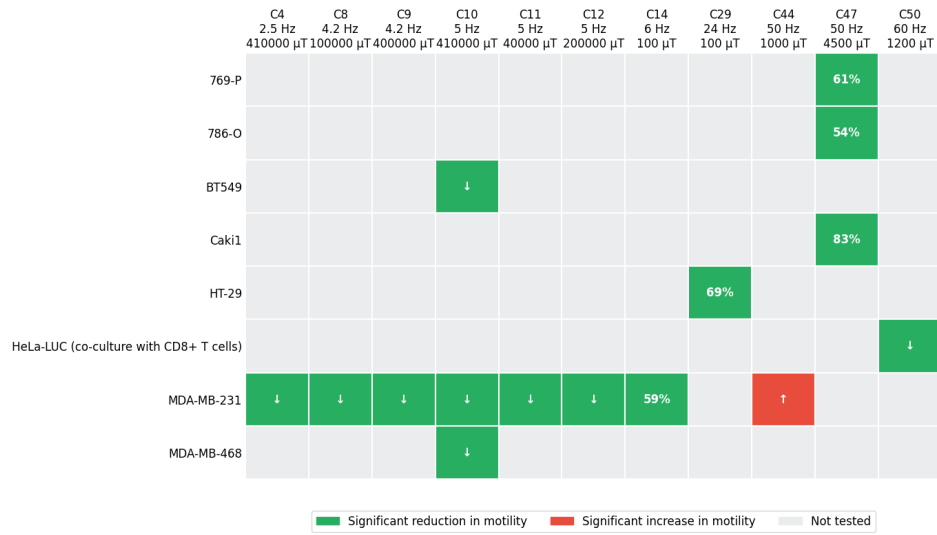


Figure 19: Motility heatmap for cancer cell lines across experimental conditions. Green cells indicate a significant decrease in motility, red cells indicate a significant increase. Where available, the percentage of motility relative to the control group is reported; otherwise, arrows indicate the direction of the effect. Grey cells denote untested combinations. Each condition is labeled with its C-ID, frequency, and intensity.

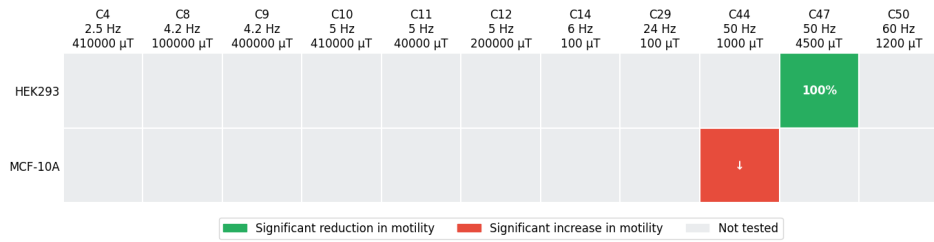


Figure 20: Motility heatmap for healthy cell lines across experimental conditions. Green cells indicate unchanged motility, red cells indicate a significant decrease. Each condition is labeled with its C-ID, frequency, and intensity.

Condition C10 (Gradient Rotating Magnetic Field, 5 Hz and 0.41 T, 48 hours of field exposure) significantly reduces migration, invasion, and wound healing capacity in three triple-negative breast cancer (TNBC) cell lines: MDA-MB-231, BT549, and MDA-MB-468. The same study shows that at lower frequencies (C4, 2.5 Hz) and lower intensities (C11, 0.04 T and C12, 0.2 T) the inhibition is less pronounced. This suggests a dose-response relationship in both frequency and intensity.

A significant inhibition of motility is documented under condition C47 (50 Hz and 4.5 mT, 30 minutes of field exposure per day for 5 consecutive days) on renal cell carcinoma lines: 786-O reaches 53.6% motility compared to control, 769-P 61%, and Caki1 83%. Healthy cells remain unchanged (100%), confirming the selectivity of treatment.

Condition C14 (6 Hz, 100 μ T + SMF 45 μ T, 72 hours of field exposure) significantly reduces wound closure in MDA-MB-231 (approximately 59% compared to the control), and condition C29 (24 Hz, 100 μ T + SMF 45 μ T, 72 hours of field exposure) has a similar effect on HT-29 (69%). In the latter case, field exposure reversed TGF- β -induced migration. TGF- β (Transforming Growth Factor beta) is a cytokine that promotes epithelial-mesenchymal transition and enhances cell motility by destabilising tubulin filaments; its reversal by ELF-EMF suggests an action on microtubule polymerization.

Inhibition of motility is confirmed for MDA-MB-231 exposed to LF-RMF (4.2 Hz) at both 0.1 T and 0.4 T (C8 and C9), with a more pronounced effect at the higher intensity.

Condition C44 shows that at 50 Hz and 1 mT, MDA-MB-231 exhibit a significant increase in migration and invasion after 4 hours of field exposure, accompanied by a decrease in adhesion. The

Mechanistic outcome	Cancer exp. conditions	Healthy exp. conditions	Total
ROS production	24	9	33
Antioxidant response	2	1	3
Membrane potential	19	6	25
Calcium signaling	14	3	17
Cellular morphology and cytoskeleton remodeling	28	10	38
Metabolic reprogramming	8	0	8
Differentiation	1	0	1
Cell cycle analysis	24	15	39

Table 6: Number of ELF-EMF macro-category experimental conditions that assess each mechanistic outcome, broken down by cancer and healthy cell lines.

authors interpret these results as an enhancement of the metastatic potential of the tumor. Interestingly, healthy MCF-10A cells show the opposite behavior, with a reduction in migration. This result is consistent with the viability findings discussed previously, where the same condition led to an increase in viability in MDA-MB-231, reinforcing the hypothesis that the 50 Hz and 1 mT window should be avoided in the treatment of this cell line.

Migration and invasion of HeLa-LUC cells are significantly reduced when co-cultured with CD8+ T cells exposed to PEMF at 60 Hz and 1.2 mT (C50, 72 hours of exposure), suggesting that the anti-migratory effect can also be indirectly mediated through activation of the immune response.

6.4 Mechanistic outcomes

Table 6 reports all the mechanistic outcomes analyzed within the ELF-EMF macro-category, divided into cancerous and healthy cell line experimental conditions.

6.4.1 ROS production

The production of reactive oxygen species was analyzed in 33 experimental conditions, 24 on cancer cell lines and 9 on healthy cell lines. Figures 21 and 22 show, for cancer and healthy cells respectively, the mechanistic flow from ROS direction to viability direction to apoptosis direction.

Cancer cells. In the majority of experimental conditions on cancer cells, field exposure induces a significant increase in ROS. The most marked increases are observed in [13] with DU-145 reaching 4400% compared to control (C1, 0.01 Hz and 100 mT, 120 hours) and MDA-MB-231 reaching 1266% (C2, 1 Hz and 100 mT, 120 hours); and in [28] (C47, 50 Hz and 4.5 mT) with Caki1 at 3100%, 786-O at 2800% and 769-P at 1900%. In [9] (C41, 50 Hz and 1.0–2.5 mT + SMF), G401 shows an increase of 310% and A549 of 140%, with a peak at 30–60 minutes and elevated values persisting for at least 24 hours after the end of exposure. NAC partially reduces but does not eliminate field-induced ROS. In [8] (spinning oscillating magnetic field), condition C55 (272 Hz, spinning magnet) produces a significant increase in ROS in both GBM115 (160%) and DIPG (155%), while condition C54 (137 Hz, Helmholtz coil) produces no increase. The authors state that an oscillating field alone is not sufficient to induce ROS: rotation of the field axis is required. However, these measurements were performed at different frequencies, and this must be taken into consideration.

In [19] (C46 – C38, 50 Hz, 0.1–1 mT), the behaviour of ROS depends on the cell line. MDA-MB-231 cells show an increase in ROS both at 0.1 mT (125%) and at 1 mT (120% at 96h after 24h of exposure). MCF-7 cells instead show no significant variations in ROS, despite a transient increase immediately after exposure.

From the flow in Fig. X, in most conditions where ROS increase, viability tends to decrease. The apoptotic response is not uniform: in some conditions the increase in ROS and the reduction in viability are accompanied by a significant increase in apoptosis (C41 G401 and A549, C2 MDA-MB-231, C3

MC4-L2, C47 786-O), while in others (C47 Caki1 and 769-P, C1 DU-145) viability decreases without a significant increase in apoptosis.

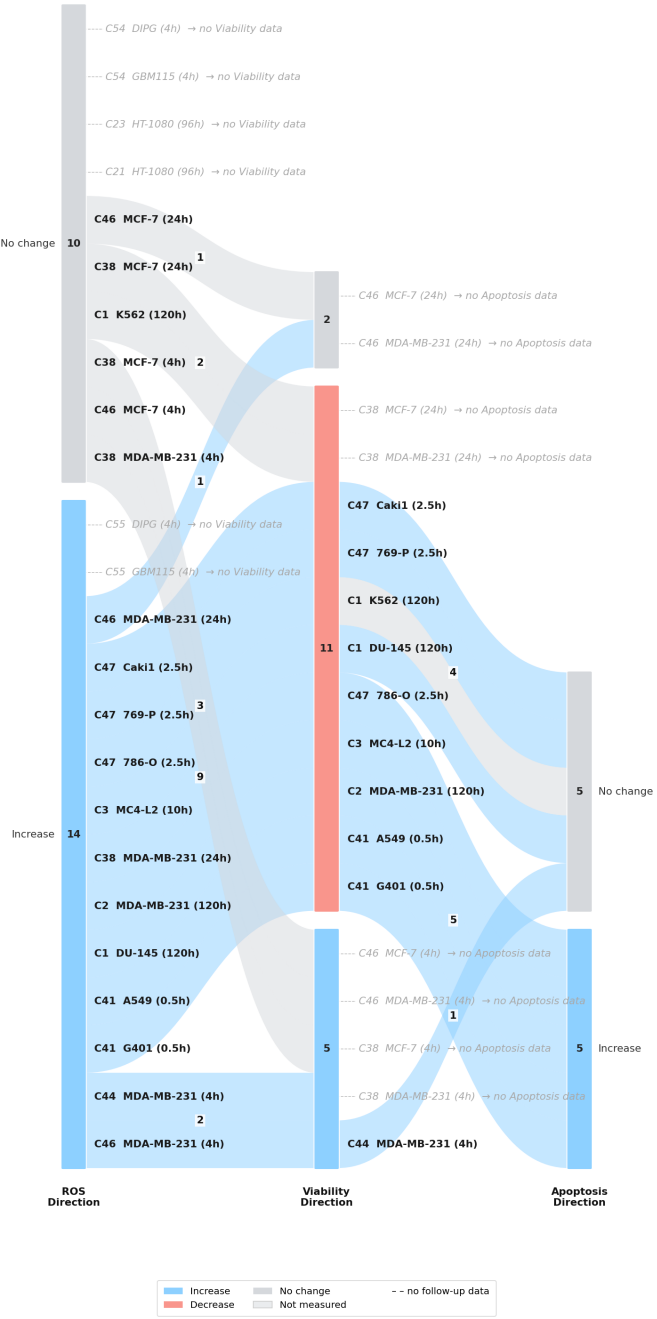


Figure 21: Mechanistic flow from ROS direction to viability direction to apoptosis direction in cancer cell lines. Each row represents one experimental condition and cell line. Color coding: blue = significant increase, red = significant decrease, grey = no significant change. Dashed lines indicate conditions where the downstream outcome was not measured.

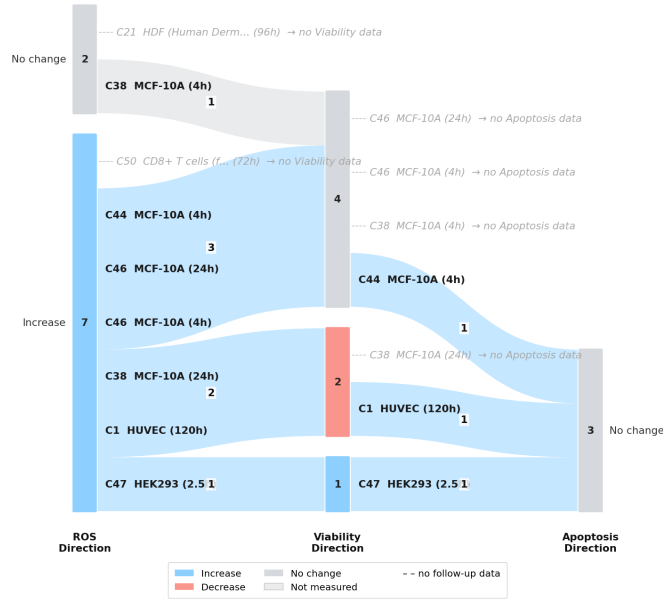


Figure 22: Mechanistic flow from ROS direction to viability direction to apoptosis direction in healthy cell lines. Colour coding and labeling conventions are the same as in the previous figure.

In K562 cells [13] (C1, 0.01 Hz and 100 mT), despite 120 hours of exposure, ROS do not increase. The authors attribute this absence to the fact that K562 are suspension cells, suggesting that the state of cell adhesion may influence the oxidative response to the field. Condition C44 (50 Hz and 1 mT) [41] on MDA-MB-231 shows an increase in mitochondrial ROS (145%), but as already discussed this does not translate into an antitumoral effect, but rather into an enhancement of the metastatic characteristics of the cell. This highlights that an increase in ROS is not automatically beneficial and the final outcome depends on the cellular context.

Healthy cells. In [13] (C1, 0.01 Hz and 100 mT), HUVEC cells show a 1900% increase in ROS, comparable to that of cancer cell lines, confirming the negative selectivity already observed for this condition. In [28] (C47, 50 Hz and 4.5 mT), HEK293 cells also show a significant increase in ROS (2100%), although the authors note that the cells were exposed outside the incubator, which may have contributed to the effect. In [19] (C46, 50 Hz and 0.1 mT, 4h and 24h of exposure), MCF-10A cells show an increase in ROS (125–130%) comparable to that of cancer MDA-MB-231 cells, indicating an absence of selectivity for this condition and intensity. In [41] (C44, 50 Hz and 1 mT), MCF-10A cells show an increase in mitochondrial ROS (117%). Despite these results, it is observed that cell viability does not decrease significantly, with the exception of C1 HUVEC, which however does not lead to significant changes in apoptosis, and C38 MCF-10A (24h).

In [18] (C21, 14.5 Hz and 10 μ T + SMF), healthy HDF cells show no significant variations (101.67%), unlike cancer HT-1080 cells, indicating selectivity for this condition at low intensity.

6.4.2 Antioxidant response

The antioxidant response was analyzed in 3 experimental conditions, 2 on cancer cell lines and 1 on immune cells.

In [9] (C41, 50 Hz and 1.0-2.5 mT + SMF 3-4 mT, 30 minutes of exposure), the level of NADPH, the main source of intracellular antioxidant defence, was measured. In G401 cells, NADPH decreases by approximately 50% compared to control after 30 minutes, accompanied by a 3.7-fold increase in malondialdehyde (MDA), a marker of lipid peroxidation. This combination indicates an exhaustion of the cell's antioxidant capacity leading to irreversible oxidative membrane damage. In A549 cells,

the NADPH reduction is less pronounced (approximately 15%) and levels are restored shortly after; MDA nevertheless increases 2.2-fold. The different intensity of the response between the two cancer cell lines suggests that the susceptibility to field-induced oxidative stress is cell-dependent.

In [48] (C50, PEMF, 60 Hz and 1.2 mT, 72 hours of exposure), the treatment was applied to CD8+ T cells derived from tumour tissues of cancer patients. In this specific experimental condition, the field does not act directly on cancer cells, but on immune cells. These exposed T cells show a significant reduction of glutathione (GSH), another intracellular antioxidant, and a significant increase in MDA. In this context, oxidative stress is not damage but is part of the immune activation mechanism, through which the enhanced T cells are able to exert greater antitumoral cytotoxicity.

In both studies, the reduction in antioxidant capacity is consistent with the increase in ROS observed under the same experimental conditions, confirming the existing link between the production of reactive oxygen species and the exhaustion of antioxidant defenses.

6.4.3 Membrane potential

C_ID	Cell Line	Target	Exposure	% vs Ctrl	Direction
C27	HepG2	CMP		80.0	↓
C27	A549	CMP		87.0	↓
C27	Raji	CMP		102.0	→
C27	Raji	CMP		109.0	↑
C29	HT-29	MMP	48h	–	↑
C38	MDA-MB-231	MMP	96h	96.0	↓
C38	MCF-7	MMP	96h	100.0	→
C46	MDA-MB-231	MMP	96h	100.0	→
C46	MCF-7	MMP	96h	96.0	↓

Table 7: Membrane potential changes in cancer cell lines. CMP = cell membrane potential; MMP = mitochondrial membrane potential. Arrows indicate the direction of change: ↑ = significant increase, ↓ = significant decrease, → = no significant change.

C_ID	Cell Line	Target	Exposure	% vs Ctrl	Direction
C27	293T	CMP		75.0	↓
C38	MCF-10A	MMP	4h	115.0	↑
C38	MCF-10A	MMP	96h	95.0	↓
C46	MCF-10A	MMP	96h	100.0	→

Table 8: Membrane potential changes in healthy cell lines. Labelling conventions are the same as in the previous table.

6.4.4 Calcium signaling

Calcium involvement was measured in 17 experimental conditions, 14 on cancer cell lines and 3 on healthy cell lines. Other studies ([9], [19], [20] and [24]) mention the role of calcium only at a hypothetical level. Direct measurements of intracellular or mitochondrial calcium come from 3 studies: [16], [18] and [25], to which [21] is added as it demonstrates involvement indirectly through pharmacological inhibitors.

Intracellular calcium. In [16] (C27, 20 Hz and 5 mT), the authors measured intracellular calcium concentration and observed that healthy 293T cells show a significant decrease, accompanied by membrane hyperpolarisation. The cancer cell lines HepG2, A549 and Raji instead show no significant variations in calcium; the authors hypothesise that in HepG2 this may be due to hypermethylation of the tumour-associated calcium signalling network. This means that, according to the authors, the

genes encoding the proteins of the calcium signalling network have been silenced, i.e. the cancer cells have shut down part of their calcium machinery. The field attempts to alter calcium fluxes in these cells but nothing happens because the proteins that should respond are no longer present or are present in reduced amounts. In the same study, intracellular sodium and potassium concentrations are not altered in any of the tested cell lines.

In [25] (C3, 1 Hz and 100 mT), field exposure induces a remarkable increase in intracellular calcium in MC4-L2 cells (from 205 to 9973 RFU, an approximately 50-fold increase). This calcium overload proved to be fundamental for ELF-EMF-induced necroptosis. Experiments with pharmacological inhibitors were conducted and showed that Verapamil, an L-type calcium channel blocker, completely suppresses calcium overload and abolishes both necroptosis and apoptosis; the addition of CaCl_2 enhances intracellular calcium and aggravates necroptosis. Treatment with NAC (N-acetylcysteine, a ROS scavenger) increases intracellular calcium but suppresses ROS and consequently necroptosis, demonstrating that calcium overload alone is not sufficient to induce necroptosis without the concomitant overproduction of ROS.

Mitochondrial calcium. In [18] (C21, 14.5 Hz and $10 \mu\text{T}$ + SMF $45 \mu\text{T}$), the cancer cell line HT-1080 shows a significant decrease in mitochondrial calcium of 14.05% compared to control. At 15.5 Hz (C22) the decrease is smaller (2.62%) and not significant. In healthy HDF cells, the decrease at 14.5 Hz is minimal (3.32%) and not significant, indicating selectivity of the treatment on cancer cells for this specific frequency.

Indirect involvement of calcium channels. In [21] (C56, ELF-EMF field modelled after the activation pattern of intercellular calcium waves), the use of an L-type voltage-gated calcium channel agonist (BAY K8644) abolishes the effect of the field on the viability of B16-BL6 cancer cells. This suggests that the field acts by perturbing T-type voltage-gated calcium channels, since when the drug that forces a massive and constant calcium influx through the L-type pathway is used, the effect of the field disappears as if the perturbation it causes on T-type channels were masked. Cancer cells have a much higher number of T-type channels compared to healthy cells. Indeed, healthy HEK293T cells, which express very low levels of T-type calcium channels, do not respond to field treatment, which supports the specificity of the proposed mechanism.

Some studies mention calcium involvement as a possible mechanism of action without however measuring it. In [20] (C15 and C16, 7.83 Hz), the authors hypothesise that the Schumann frequency approximates the resonance frequency of Ca^{2+} ions within certain cancer cells, and that the field may promote calcium entry through the modulation of calcium channels, causing an intracellular overload lethal to cancer cells. For healthy NIH/3T3 cells, the authors propose that these have adapted to the Schumann Resonance frequency through evolution, given the absence of negative effects. In [19] (C46 and C38, 50 Hz and 0.1–1 mT), it is mentioned that ELF-MF may alter mitochondrial membrane potential by interfering with Ca^{2+} homeostasis, but calcium was not measured.

6.4.5 Cellular morphology and cytoskeleton remodeling

Changes in cellular morphology and cytoskeleton were observed in 38 experimental conditions, 28 on cancer cell lines and 10 on healthy cell lines. As these data are predominantly qualitative in nature, they have been reported in two summary tables, Table X for cancer cell lines and Table Y for healthy cell lines.

C.ID	Cell Line	Notes
C1	DU-145	[Morphology] ELF-EMF induced morphological changes in DU145 cells and increased cell size.
C1	K562	[Morphology] K562 showed a minor increase in cell size compared to non-exposed cells. Smallest morphological change among all cell lines of the same study [13]. ELF-EMF also induced monocyte/macrophage differentiation: phagocytosis of latex particles was mostly observed after 24 h exposure; NBT reduction assay showed formazan deposits in about 40% of exposed K562 cells at 24 h.

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C_ID	Cell Line	Notes
C2	MDA-MB-231	[Morphology] ELF-EMF induced morphological changes in MDA-MB-231 cells and increased cell size. Most significant increase among cancerous adherent cells of the same study [13].
C3	MC4-L2	[Morphology] Transmission Electron Microscopy (TEM) showed ELF-EMF treated cells had translucent cytoplasm, disintegrated membrane, organelle swelling, and condensed chromatin; these are typical characteristics of necroptotic cell death. Some cells also showed typical apoptotic morphology. Numerous autophagosomes and autolysosomes were observed. AO/EtBr staining showed increased cells with orange nucleus (early apoptosis), homogenously orange cells (late apoptosis), homogenously red swollen round cells (necroptosis), and increased acidic vesicular organelles (AVOs). Co-treatment TEM observations: NAC+ELF showed predominantly apoptotic morphology; Verapamil+ELF showed normal physiologic morphology; NAC control, verapamil control, and verapamil+ELF groups showed normal morphology.
C3	MCF-7	[Morphology] MCF-7 cell size increased ~ 1.2 -fold vs. sham (TEM); not statistically significant ($p=0.1271$).
C3	U251	[Morphology] U251 cells changed to spherical shape by day 5 (optical microscopy). Cell size showed a slight, non-significant change vs. sham (TEM).
C3	U251-P53	[Morphology] U251-P53 cells changed to spherical shape by day 5 (optical microscopy). Cell size increased ~ 1.06 -fold vs. sham (TEM); not statistically significant ($p=0.4132$).
C3	U87	[Morphology] U87 cells changed to spherical shape by day 5 (optical microscopy). Cell size increased significantly ~ 1.7 -fold vs. sham (TEM, $p=0.0231$, Mann-Whitney U test). A reduction in cell population was also observed under optical microscope.
C4	MDA-MB-231	[Morphology] Morphology was altered, but spreading area and polarization increased less significantly than 5 Hz (see C10, MDA-MB-231). [Cytoskeleton] Rearrangement of F-actin was observed but less significantly than 5 Hz (see C10, MDA-MB-231).
C8	MCF-7	[Morphology] Cell area was significantly reduced ($P<0.05$). [Cytoskeleton] F-actin filament length was significantly reduced by 0.1 T LF-RMF in MCF-7 cells ($P<0.0001$). MCF-7 cells showed decreased attachment and spreading.
C8	MDA-MB-231	[Morphology] Cell area was significantly reduced. [Cytoskeleton] F-actin stress fibers were significantly disrupted by 0.1 T LF-RMF as early as 5 min. F-actin filament length was significantly reduced ($P<0.0001$). F-actin in the cell center was more affected than the peripheral area. Microtubules were not obviously affected. After 1.5 h treatment, F-actin changes in MDA-MB-231 did not recover even after 24 h. Cells also showed decreased attachment and spreading.
C9	MCF-7	[Morphology] Cell area was significantly reduced ($P<0.01$). [Cytoskeleton] F-actin filament length was significantly reduced by 0.4 T LF-RMF in MCF-7 cells ($P<0.0001$). MCF-7 cells showed decreased attachment and spreading, especially at 0.4 T.
C9	MDA-MB-231	[Morphology] After 0.4 T LF-RMF exposure, at 90 min, cells shrank and rounded up, similar to effects of actin drugs. Cell area was significantly reduced ($P<0.0001$). These morphological changes did not recover even 24 h after LF-RMF retrieval. [Cytoskeleton] F-actin stress fibers were significantly disrupted by 0.4 T LF-RMF as early as 5 min. F-actin filament length was significantly reduced ($P<0.0001$). MDA-MB-231 cells showed obvious disruption of stress fibers. Stress fibers switched to lamellipodia/ruffles from 10 to 60 min. F-actin changes did not recover even 24 h after retrieval. Microtubules were not obviously affected.
C10	BT549	[Cytoskeleton] F-actin underwent noticeable rearrangement. Cytoskeletal deformations and polarization were observed.
C10	MDA-MB-231	[Cytoskeleton] Rearrangement of F-actin with a tendency for parallel alignment. Spreading area and polarization coefficient increased significantly.
C10	MDA-MB-468	[Cytoskeleton] F-actin underwent noticeable rearrangement. Cytoskeletal deformations and polarization were observed.
C11	MDA-MB-231	[Morphology] Morphology altered but less significantly compared to 0.41 T. [Cytoskeleton] Rearrangement of F-actin was observed but less significantly than 0.41 T.
C12	MDA-MB-231	[Morphology] Morphology altered but less significantly compared to 0.41 T (C10). [Cytoskeleton] Rearrangement of F-actin was observed but less significantly than 0.41 T (C10).
C15	HeLa	[Morphology] No obvious morphological change after 2 days of SREMF exposure. After 4 days, some cells exhibited obvious morphological changes with a few showing signs of early-stage death. After 6 days, cells showed more dramatic morphological changes of shrinkage, blebbing, and fragmentation, while control cells formed healthy foci. These morphological changes are hallmarks of apoptosis.

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C.ID	Cell Line	Notes
C16	HeLa	[Morphology] No obvious morphological change after 2 days of SREMF exposure. After 4 days, some cells exhibited obvious morphological changes with a few showing signs of early-stage death. After 6 days, cells showed dramatic morphological changes of shrinkage, blebbing, and fragmentation, while control cells formed healthy foci. These morphological changes are hallmarks of apoptosis.
C32	U373MG	[Cytoskeleton] 2.3-fold drastic collapse of actin cytoskeleton area in U373 cells after 2 min MEMP. Actin collapsed around the nucleus. Nuclear envelope (DAPI staining) was not significantly impaired. Colony forming ability was also compromised at this exposure time.
C40	A549	[Morphology] Pulsed field exposure induced unique membrane ‘rippling’ or parallel-running ridging lines on the membrane surface of A549 cells. Nanopores <20 nm in diameter were significantly more frequent on magnet-exposed cells compared to controls. Larger 20–50 nm pores consistent with caveolae or endocytotic events were present on both control and exposed cells. Triton X-100 control showed extensive ‘moth-eaten’ conglomerate holes.
C44	MDA-MB-231	[Morphology] TEM revealed enlarged mitochondria with sparser cristae, less electron-dense matrix, dilation and loss of matrix. Cell density appeared higher in exposed cells. [Cytoskeleton] ELF-MF exposed MDA-MB-231 cells showed increased F-actin organization, F-actin marginalization from center toward plasma membrane, and greater number of pseudopodia in ELF-MF exposed MDA-MB-231 cells. Filopodia length (P=0.041) and density (P=0.043) were significantly increased. Increased filopodia suggest features of migrating/invading cancer cells.
C47	769-P	[Morphology] No visible changes in the size or shape of 769-P cancer cells were observed after ELF-EMF exposure.
C47	786-O	[Morphology] No visible changes in the size or shape of 786-O cancer cells were observed after ELF-EMF exposure.
C47	Caki1	[Morphology] No visible changes in the size or shape of Caki1 cancer cells were observed after ELF-EMF exposure.
C56	B16-BL6	[Morphology] Cells under Ca ²⁺ EMF appeared more uniformly distributed with less aggregation and greater spacing between cells compared to control. This is consistent with a decrease in viable cell number.
C56	B16-BL6	[Morphology] BAY K8644 + Ca ²⁺ EMF cells showed more variable distribution with cells more tightly packed forming dense clusters. The intensity of AO/EB stains was greater in this condition compared to all other conditions, indicative of nuclear condensation.

Table 9: Morphological and cytoskeletal effects of ELF-EMF exposure in cancer cell lines.

C.ID	Cell Line	Notes
C1	HUVEC	[Morphology] ELF-EMF induced morphological changes in HUVEC cells and increased cell size. Most significant size increase among all cell lines studied in the same study [13].
C3	Normal human fibroblast	[Morphology] No significant morphological changes were observed in fibroblasts under optical microscope or TEM after ELF-EMF exposure compared to sham exposure.
C9	MCF-10A	[Cytoskeleton] MCF10A noncancer cells were much more resistant to LF-RMF perturbations compared to MDA-MB231 cancer cells. Stress fibers were more robust. No stress fiber to lamellipodia/ruffles switch was observed. F-actin changes were subtle and could fully recover within 12-24 h after LF-RMF retrieval. This differential sensitivity is a key finding of the study.
C9	RPE1	[Cytoskeleton] RPE1 noncancer cells were much more resistant to LF-RMF perturbations compared to MDA-MB231 cancer cells. Stress fibers were more robust. No stress fiber to lamellipodia/ruffles switch was observed. F-actin changes could fully recover after LF-RMF retrieval. Cytochalasin D experiments also confirmed differential sensitivity: RPE1 cells fully recovered while cancer cells did not.
C15	NIH/3T3	[Morphology] All cells appeared healthy throughout the 6-day exposure period. No obvious difference in cell morphology was found between exposed and non-exposed groups at any time point (2, 4, or 6 days).
C16	NIH/3T3	[Morphology] All cells appeared healthy throughout the 6-day exposure period. No obvious difference in cell morphology was found between exposed and non-exposed groups at any time point (2, 4, or 6 days).

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C.ID	Cell Line	Notes
C32	A9	[Cytoskeleton] 1.4-fold collapse of actin cytoskeleton area in A9 cells after 2 min MEMP. Much less affected than U373. Actin cytoskeleton remained under a normal expanded configuration in the bulk of treated A9 cells. Nuclear envelope not impaired. Despite partial collapse, A9 cells recovered proliferative capacity and formed normal-sized colonies.
C44	MCF-10A	[Morphology] TEM revealed mitochondrial morphological changes: mitochondria were consistently larger, cristae sparser, matrix less electron dense, with dilation and loss of matrix, similar to MDA-MB-231. [Cytoskeleton] No substantial differences in actin microfilament distribution or morphology in ELF-MF exposed MCF-10A cells compared to unexposed. Filopodia length and density were not significantly affected.
C47	HEK293	[Morphology] No visible changes in size or shape were observed in HEK293 cells after ELF-EMF exposure.
C56	HEK293T	[Morphology] No notable changes in cell morphology were observed in HEK293T cells exposed to Ca ²⁺ EMF compared to control.

Table 10: Morphological and cytoskeletal effects of ELF-EMF exposure in healthy cell lines.

From the tables, several recurrent findings emerge. At the cytoskeletal level, the most consistent finding is the perturbation of F-actin observed in several independent studies. In [40] (C8 and C9, LF-RMF, 4.2 Hz and 0.1–0.4 T), the stress fibres of MDA-MB-231 cells are significantly disrupted as early as 5 minutes of exposure, with a transition from stress fibres to lamellipodia between 10 and 60 minutes, and at 90 minutes the cells contract and assume a rounded shape. These alterations do not recover even 24 hours after field removal. In contrast, healthy MCF-10A and RPE1 cells show a much greater resistance to the perturbation and a complete recovery within 12–24 hours. In [5] (C32, MEMP, >2 T), the actin cytoskeleton of U373 MG cells collapses drastically (2.3-fold reduction in area), while in healthy A9 cells the collapse is much more contained (1.4-fold) and the cells recover their proliferative capacity forming normal colonies.

At the morphological level, changes associated with cell death are observed under several conditions: in [25] (C3, 1 Hz and 100 mT), TEM of MC4-L2 cells shows features typical of necroptosis (translucent cytoplasm, disintegrated membrane, organelle swelling). In [20] (C15 and C16, 7.83 Hz), HeLa cells show a temporal progression from no visible change (2 days) to shrinkage, blebbing and fragmentation (6 days), typical of apoptosis. In [41] (C44, 50 Hz and 1 mT), MDA-MB-231 cells show enlarged mitochondria with sparser cristae and an increase in filopodia, indicative of a migratory/invasive phenotype, consistent with the pro-tumoral effects already discussed for this condition. Healthy MCF-10A cells show the same mitochondrial alterations but no significant changes in actin distribution or filopodia.

6.4.6 Metabolic reprogramming

Metabolic reprogramming was studied in 8 experimental conditions, all involving cancer cells.

In [2], following exposure to the frequencies calculated through the thermodynamic model (C6, C7, C13, C19 and C20), the cancer cell lines PANC-1, 1627 and MCF-7 undergo a metabolic switch toward mitochondrial respiration. The authors propose that cancer cells respond to the perturbation of ion fluxes induced by field exposure by reallocating energy resources to maintain cellular homeostasis, at the expense of proliferation. Even three-dimensional aggregates, despite the low basal hypoxia and reduced metabolic activity typical of these structures, show the same effect; this means that field exposure is able to force an increase in respiration.

In [5] (C32, MEMP, <30 Hz, >2 T), the authors argue that, according to their model, the membrane of cancer cells has a higher electronegativity due to the Warburg effect (aerobic glycolysis typical of tumours), and that this property would make them more susceptible to the electronic perturbation induced by MEMP. Healthy cells, according to the authors, would not be affected as they lack this metabolic reprogramming. They also observed that COS-1 cells, despite being immortalised through SV-40 (which inactivates the two tumour suppressor proteins p53 and pRb), respond to the treatment like healthy cells; this implies that what makes cancer cells vulnerable to the field must be something else, since p53 and pRb are not involved in field susceptibility. The authors propose metabolic reprogramming (Warburg effect) as the answer to this question.

In [8] (C53 and C55, 137–272 Hz and 5 mT, spinning oscillating magnetic field), the authors argue that field exposure may create a perturbation in electron transfer within the mitochondrial transport chain through the radical pair mechanism, causing an overproduction of superoxide.

In [36] (C14, 6 Hz and 100 μ T + SMF 45 μ T), following field exposure, MDA-MB-231 cells show an increased expression of mitochondrial respiratory chain components (COX2, COX4) and ATP synthase subunits (MT-ATP6, ATP5B), indicating an enhancement of oxidative phosphorylation. Under condition C29 (24 Hz), HT-29 cells show an increase in mitochondrial membrane potential.

In [41] (C44, 50 Hz and 1 mT), proteomic analysis shows an opposite trend between cancer and healthy cells. In MDA-MB-231, an upregulation of mitochondrial proteins (ECHS1, MAVS, OGDH, PCK2, PPA2, SFXN1, TIMM8A, TOMM22, TUFM) is observed, which are implicated in the ability of metastatic cells to alter their metabolism in response to stress and to induce epithelial-mesenchymal transition. In healthy MCF-10A cells, the same proteins are downregulated, accompanied by an increase in mitochondrial ROS, suggesting a metabolic alteration of a different nature and not comparable to the reprogramming observed in cancer cells.

6.4.7 Differentiation

Differentiation was studied only under experimental condition C1 (0.01 Hz, 100 mT) on the K562 cell line (chronic myeloid leukaemia), cultured in suspension. It consists of a monocyte/macrophage differentiation, which was confirmed by two assays. The first assay is latex particle phagocytosis: phagocytosis is the ability of cells to engulf and degrade foreign particles, and it is a function of mature macrophages; it was found to be significantly increased after 24 hours of exposure, indicating that the leukaemic cells are acquiring functional macrophage characteristics. The second assay is Nitro Blue Tetrazolium reduction: NBT reveals formazan deposits in approximately 40% of exposed cells compared to control, indicating that these cells have acquired the oxidative enzymatic activity characteristic of differentiated myeloid cells. The electromagnetic field alone was therefore able to induce differentiation through the inhibition of cell proliferation; the authors also suggest a possible involvement of the MAPK pathway.

In [41] (C44, 50 Hz and 1 mT), differentiation was not directly measured, but the analysis of pluripotency transcription factor expression shows a different trend between the cancer cell line and the healthy cell line analysed. In cancer MDA-MB-231 cells, an upregulation of KLF4, STAT3, BCLAF1 and NR2C2 is observed, suggesting a reprogramming toward stem-like properties associated with the transition to breast cancer stem cells. In contrast, in healthy MCF-10A cells, a downregulation of STAT3, BCLAF1 and NR2C2 is observed. This result is consistent with the previous findings for condition C44, which also showed pro-tumoral characteristics in viability and motility.

6.4.8 Cell cycle analysis



Figure 23: Cell cycle phase heatmap for cancer cell lines across experimental conditions. Each cell shows the direction of change for the corresponding phase: azure indicates a significant increase, soft red a significant decrease, grey no significant change, and white a phase not reported. Phases are ordered following the biological sequence of the cell cycle.

The cell cycle was analysed in 39 experimental conditions, 24 on cancer cell lines and 15 on healthy cell lines. From Figs. 12 and 13, two distinct patterns of cell cycle arrest emerge in cancer cells.

G2/M arrest. This is the most recurrent pattern, as it is observed in three independent studies. In [5] (C32, MEMP, <30 Hz and >2 T), the cell lines U373 MG, HeLa and MC-38 show an accumulation in G2/M phase and a reduction of the G0/G1 phase. As the exposure time increases from 2.5 minutes, where the G2/M increase and G0/G1 reduction are already evident, to 5 minutes the sub-G1 population also appears, indicating apoptosis. Healthy cells exposed to the same condition show no significant changes. The selectivity of this treatment is once again confirmed. In [13] (C1, 0.001 Hz and 100 mT; C2, 1 Hz and 100 mT), after 120 hours of exposure an accumulation in G2/M phase is observed in DU-145, MDA-MB-231 and K562. Healthy HUVEC cells also show the same pattern here, a result consistent with the negative selectivity shown for condition C1 in viability. In [6] (C3, 1 Hz and 100 mT, 10 hours of exposure), only U87 shows a significant increase in G2/M phase; the other cancer cell lines U251 and U251-P53 and normal fibroblasts show no significant variations.

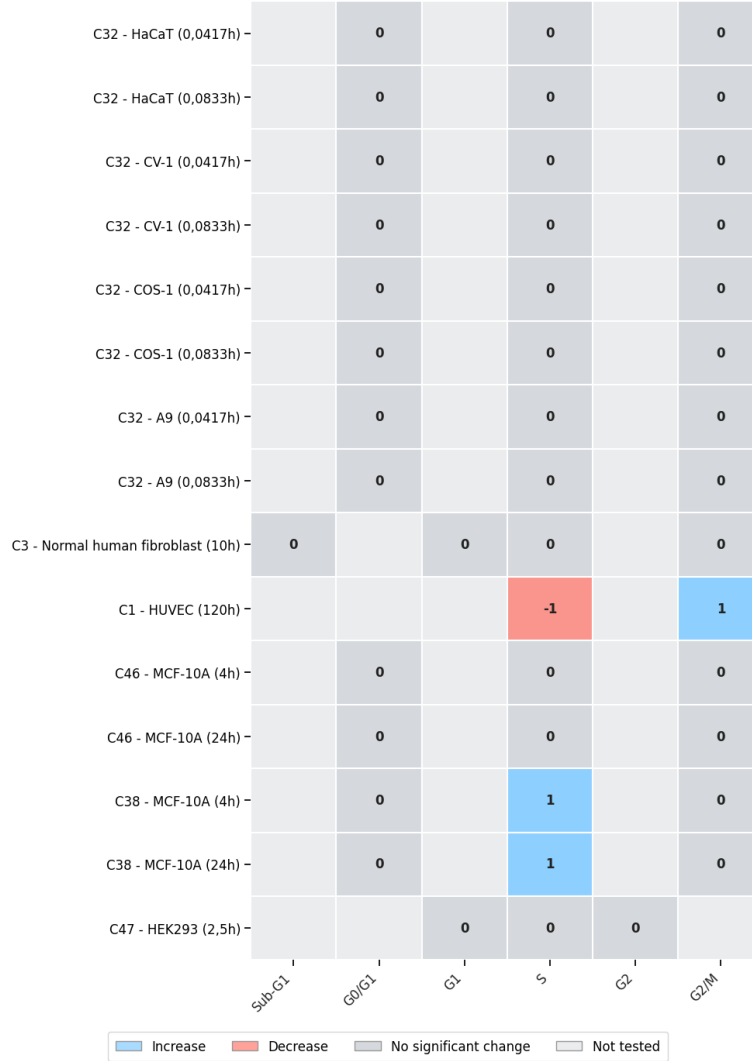


Figure 24: Cell cycle phase heatmap for healthy cell lines across experimental conditions. Color coding and labeling conventions are the same as in the previous figure.

G0/G1 arrest. In [28] (C47, 50 Hz and 4.5 mT, 2.5 hours of exposure), the cancer cell lines 786-O, 769-P and Caki1 show a significant increase in the G1 population and a decrease in G2, with no variations in S phase. Healthy HEK293 cells remain unchanged. In [40] (C8 and C9, LF-RMF, 4.2 Hz and 0.1–0.4 T, 30 hours of exposure), MDA-MB-231 cells show an accumulation in G0/G1 with a reduction of the S and G2/M phases. The effect is accompanied by the appearance of binucleated and multinucleated cells.

S phase increase. In [19], MDA-MB-231 cells show an increase in S phase that depends on both intensity and exposure time. Under condition C46 (50 Hz, 0.1 mT), the increase is present at 4 hours of exposure but disappears at 24 hours of exposure. Under condition C38 (50 Hz, 1 mT), the S phase increase persists at both 4 hours and 24 hours, and at 4 hours it is accompanied by a reduction in G2/M phase. This suggests an increase in replicative activity, consistent with the viability and proliferation data that showed values above control at short exposure times for these conditions. MCF-7 cells show no changes in any phase, at any intensity or exposure time. Regarding healthy cells, MCF-10A under condition C46 remain unchanged at both 4h and 24h, while under condition C38 they show an increase in S phase at both 4h and 24h, a finding worth noting as it indicates that an intensity of 1 mT at 50 Hz stimulates replication also in healthy cells.

In [10] (C18, 8 Hz and 0.011 T), the cell cycle was not directly measured, but the authors report senescence, an irreversible cell cycle arrest in the cancer cell lines MDA-MB-231 and MCF-7, confirmed by the downregulation of Ki67 and the upregulation of p21. Healthy FF95 cells do not show these effects.

6.5 Molecular outcomes

6.5.1 Genomic analysis

Genomic analysis was performed in a subset of experimental conditions using RT-qPCR or more comprehensive profiling approaches.

The most mechanistically informative genomic dataset comes from condition C3 (1 Hz and 100 mT) in glioblastoma lines and normal fibroblasts. In U87 cells (wild-type P53), field exposure induces a strong upregulation of P53 (14-fold) and P21 (2.5-fold), consistent with a P53-mediated checkpoint response, while CCNB1 and MCM6 are significantly downregulated, confirming G2/M arrest. MDM2 is also upregulated but proportionally less than P53, suggesting the P53 signal is not fully neutralized. In U251 cells (mutant P53), the same exposure downregulates P53, P21, and MCM6, reflecting the loss of functional P53 signaling; the attenuated apoptotic response in this line is directly consistent with this genomic pattern. In U251-P53 cells, in which wild-type P53 was reintroduced, P53 increases 3.6-fold and P21 1.6-fold, partially restoring the response. Critically, normal human fibroblasts exposed to C3 show the same qualitative pattern as U87 and U251-P53 (P53, P21, and MDM2 upregulation; MCM6 and CCNB1 downregulation), but do not undergo apoptosis. This dissociation, the same transcriptional signal leading to death in cancer cells but not in healthy cells, is one of the clearest mechanistic demonstrations of the context-dependence of the ELF-EMF response. Under conditions C6, C7, C13, C19 and C20 (thermodynamically-calculated frequencies, 70 μ T + 45 μ T DC), PANC-1, 1627 and MCF-7 cancer cells show significant upregulation of both mitochondrial-encoded (COX2, MT-ATP6) and nuclear-encoded (COX4, ATP5B) respiratory chain subunits, as well as uncoupling proteins UCP1 and UCP2. The same effect is observed in 3D spheroids, confirming that metabolic reprogramming toward oxidative phosphorylation occurs regardless of the structural organization of the culture model. Under condition C14 (6 Hz, 100 μ T + SMF 45 μ T), MDA-MB-231 cells show an analogous upregulation of COX2, COX4, MT-ATP6, and ATP5B, reinforcing the metabolic shift finding across independent studies and cell lines. Under condition C41 (50 Hz, 1.0–2.5 mT + SMF), G401 and A549 cells upregulate DNA damage repair genes LIG4, RAD9B, and BMI1, consistent with the extensive ROS-mediated DNA damage described in the ROS section. Under condition C44 (50 Hz, 1 mT), MDA-MB-231 cells upregulate pluripotency factors KLF4, STAT3, BCLAF1, and NR2C2, supporting the acquisition of stem-cell-like properties and epithelial-mesenchymal transition characteristics already inferred from viability, motility, and morphology data. Healthy MCF-10A cells show the opposite response, with significant downregulation of STAT3, BCLAF1, and NR2C2, while KLF4 is unchanged. Under condition C25 (C3 in the MC4-L2 context, 1 Hz and 100 mT), RIPK1, RIPK3, and MLKL are all significantly upregulated at the mRNA level, fully consistent with the protein-level necroptosis data already discussed. Under condition C50 (60 Hz, 1.2 mT, applied to CD8⁺ T cells), PEMF suppresses IL-37 expression in immune cells, disrupting the IL-37/SMAD3 interaction and activating the NF- κ B pathway, which leads to the enhanced expression of cytotoxic markers TNF- α , IFN- γ , granzyme B, and Ki67.

Finally, under condition C10 (5 Hz, 0.41 T) in MDA-MB-231 cells, the cytoskeleton-related gene *CCDC150* is significantly downregulated, consistent with the disruption of F-actin and suppression of TGF- β 1/SMAD3 signaling that underlies the anti-migratory effect reported for this condition.

6.5.2 Proteomic analysis

Proteomic analysis was carried out in a limited number of conditions and covers mechanistically distinct aspects of the ELF-EMF response.

Under condition C18 (8 Hz and 0.011 T), immunocytochemistry in MDA-MB-231 and MCF-7 cancer cells confirms downregulation of Ki67 and upregulation of p21WAF1/Cip1, providing molecular confirmation of the irreversible cell-cycle arrest and senescence phenotype already described. Healthy FF95 cells show no significant changes in either marker under the same condition, fully consistent with the viability and proliferation data. Under condition C3 (1 Hz and 100 mT), Western blot analysis of MC4-L2 cells confirms the necroptotic pathway at the protein level: p-RIPK1, p-RIPK3, and p-MLKL are all significantly increased (1.4-, 1.3-, and 1.6-fold respectively), alongside substantial increases in cleaved caspase-9, -3, and -8 (all approximately 11–13-fold). Co-treatment experiments confirm that ROS scavenging with NAC and calcium channel blockade with verapamil abolish the necroptotic protein signature, establishing the causal role of both ROS and calcium overload. Under condition C41 (50 Hz, 1.0–2.5 mT + SMF), gamma-H2AX and DNA-PKcs are both significantly increased in G401 and A549 cells, confirming ROS-mediated DNA double-strand breaks and the activation of repair machinery. Under conditions C8 and C9 (LF-RMF, 4.2 Hz, 0.1–0.4 T), a significant reduction in active GTP-Rho is observed in MDA-MB-231 and MCF-7 cells, alongside decreases in the focal adhesion proteins Talin and Tensin 2 in MCF-7 at the higher intensity. Rho GTPase inactivation is mechanistically consistent with the F-actin disruption and motility inhibition reported for these conditions, placing Rho signaling as a downstream mediator of cytoskeletal remodeling. Under condition C10 (5 Hz, 0.41 T), Western blot data in MDA-MB-231 cells confirm downregulation of CCDC150 protein and suppression of TGF- β 1/SMAD3 signaling: p-SMAD3 is strongly decreased while total SMAD3 remains unchanged, E-cadherin is upregulated, and N-cadherin is downregulated. This protein-level evidence provides direct mechanistic support for the anti-migratory and anti-EMT effect of this condition. The most comprehensive proteomic dataset in the subgroup is derived from condition C44 (50 Hz and 1 mT), where label-free quantitative mass spectrometry was performed in parallel on MDA-MB-231 cancer cells and MCF-10A healthy cells. In MDA-MB-231 cells, 328 differentially expressed proteins were identified; the upregulated set is enriched in focal adhesion, mitochondrion, and intracellular transport categories, with key proteins including ECHS1, MAVS, OGDH, PCK2, PPA2, SFXN1, TIMM8A, TOMM22, and TUFM. The simultaneous upregulation of proteins supporting metabolic plasticity and mitochondrial-membrane contact sites is consistent with a stress-adaptation response oriented toward motility and invasion. In MCF-10A healthy cells, 242 differentially expressed proteins were identified; the same set of mitochondrial proteins is downregulated rather than upregulated, indicating a qualitatively opposite metabolic response to the identical field. This cancer-vs-healthy proteomic divergence under C44 represents the most direct protein-level evidence that the 50 Hz / 1 mT window elicits fundamentally different cellular programs in cancer and healthy breast cells, and it reinforces the conclusion that this condition should be considered potentially detrimental in a therapeutic context.

6.6 Mechanisms of action and connections

The data collected across the ELF-EMF subgroup do not converge on a single mechanism of action, but they outline a coherent set of recurring biological events that are activated, in different combinations, by exposure to fields of this macro-category. These events form a connected chain whose entry point is the perturbation of the electrochemical state of the cell at the level of the plasma and mitochondrial membranes, and whose terminal outcomes are cell-cycle arrest, programmed cell death (in several distinct forms) or, in a smaller set of conditions, the opposite outcome of enhanced proliferation and acquisition of pro-metastatic features. The transition between entry point and outcome passes through three intertwined intermediate axes: oxidative stress, calcium signaling, and cytoskeletal/metabolic remodeling.

The selectivity for cancer cells over healthy cells is not a uniform property of ELF-EMF as a class; instead, it is a property of specific parameter windows and specific cell type pairings.

Mitochondrial perturbation and oxidative stress

The most reproducible mechanistic finding across the subgroup is the induction of oxidative stress. ROS were directly measured in 33 experimental conditions and increased significantly in the vast majority of cancer cells exposed, with magnitudes ranging from a modest 1.2-fold increase to extreme increases of more than 40-fold relative to the control group. The rise is documented across very different exposure regimes, from sub-Hertz frequencies at high intensity to 50 Hz fields at moderate intensity, and is observed in cell lines of distinct tissue origin (breast, prostate, glioma, renal, hepatic, sarcoma). The breadth of this finding suggests that ROS production is a general consequence of ELF-EMF exposure rather than a feature of a specific frequency.

Convergent evidence places mitochondria at the center of this oxidative response. When ROS were measured specifically as mitochondrial superoxide, an increase was observed; when mitochondrial membrane potential was measured in parallel, it was perturbed (either depolarized or, less frequently, hyperpolarized); when antioxidant defenses were measured, NADPH and reduced glutathione were depleted while malondialdehyde, a marker of lipid peroxidation, rose several-fold. The pharmacological dissection performed by some authors confirms ROS as a causal mediator and not a mere correlate: pre-treatment with the ROS scavenger NAC partially rescues viability and abolishes downstream cell death in the studies that tested it.

The hypothesis advanced by several authors to explain the biophysical primary event, that the field perturbs spin-correlated radical pairs in the mitochondrial electron transport chain, biasing the singlet/triplet equilibrium and increasing superoxide leakage, is consistent with data but is not directly demonstrated in any of the studies analyzed.

On the question of selectivity, the picture is mixed. In several conditions, healthy cells exposed to the same field show no significant ROS increase, or a markedly smaller one, than their cancer counterparts; in these cases, the oxidative response is genuinely cancer-selective. However, in other conditions, healthy cells show ROS increases of the same order as cancer cells, and the selectivity collapses. A useful interpretive key is the following: the oxidative perturbation itself is not the discriminating factor. What appears to discriminate between selective and non-selective conditions is the ability of the healthy cell to absorb that perturbation without crossing the threshold that triggers death programs. Cancer cells, which typically operate at a higher baseline oxidative load and with already partially exhausted antioxidant defenses, are pushed past this threshold more easily. This interpretation is suggested by the proteomic data of one of the studies, which show opposite directions of regulation of mitochondrial proteins in cancer and healthy cells exposed to the same field, indicating a fundamentally different metabolic response to identical input.

Calcium homeostasis and voltage-gated channels

The second recurring axis is the perturbation of calcium homeostasis. Direct measurements of intracellular or mitochondrial calcium are available in only a few studies, but where they are performed, they document substantial deviations from baseline: in one cancer line, the cytosolic calcium concentration rose approximately fifty-fold after exposure, while in others mitochondrial calcium was selectively reduced. Beyond direct measurement, calcium involvement is supported indirectly in several studies through the use of pharmacological inhibitors of voltage-gated calcium channels, both L-type blockers and L-type agonists, which respectively suppress or mask the effect of the field on cell viability. These results indicate that calcium fluxes through voltage-gated channels are part of the causal chain, not merely a downstream consequence of cell stress.

The relationship between the calcium axis and the oxidative axis is interesting and is not unidirectional. In the most thoroughly dissected study of the subgroup, blockade of L-type calcium channels suppressed both calcium overload and cell death without abolishing ROS rise, while ROS scavenging suppressed cell death without preventing the calcium rise.

This suggests that calcium overload and ROS overproduction are independent and both required: the field perturbs both pathways in parallel, and the death program is engaged only when both are simultaneously elevated. This dual-trigger logic is likely a reason for the marked dependence of the response on parameter windows: a condition that elevates one axis but not the other will not trigger the full response.

Selectivity at this level is documented in two ways. The first is intrinsic: cancer cells are reported to overexpress T-type voltage-gated calcium channels compared to healthy cells, and a selective response

in melanoma cells but not in HEK293T cells (which express low levels of these channels) is consistent with this differential. The hypothesis is plausible, but the differential expression is asserted from prior literature and not directly verified in the studies under analysis. The second is biophysical: when the field is tuned to a frequency that approximates a putative resonance of the calcium ion or of the dynamics of the calcium signaling network, cancer cells respond, and healthy cells do not.

Cytoskeletal remodeling and morphological changes

Across multiple independent works, ELF-EMF exposure on cancer cells produces disruption of stress fibers, reduction of filament length, transition of the actin organization from stress fibers to lamellipodia and membrane ruffles, contraction and rounding of the cell, and ultimately failure of cytokinesis with appearance of binucleated and multinucleated cells. In the most striking cases, the disruption is detectable within minutes of exposure and is not reversed even hours after the field is switched off.

Two important observations emerge specifically from the cytoskeletal data. The first concern selectivity: when the same field is applied to matched healthy lines, the F-actin perturbation is absent, much milder, or fully reversible within hours. Healthy cells thus appear to be intrinsically more resilient to the cytoskeletal stress imposed by ELF-EMF, even when the upstream perturbation (membrane potential changes, ROS rise) is comparable. This differential resilience is one of the cleanest demonstrations of selectivity in the entire dataset and is independent of P53 status, suggesting that it operates through cytoskeletal regulation rather than through the canonical tumor suppressor axis.

The second observation is that cytoskeletal remodeling is not always therapeutic. In a specific parameter window (50 Hz at moderate intensity, on triple-negative breast cancer cells), exposure reinforces rather than disrupts the F-actin network, with marginalisation of actin towards the plasma membrane and a significant increase in filopodia length and density, features of an invasive, migrating phenotype. Under the same condition, healthy breast epithelial cells do not show detectable cytoskeletal change. This is one of several signals across the dataset that the same exposure may be cytotoxic, neutral, or even pro tumoral depending on parameters and cell line, and it is an argument against considering ELF-EMF a uniformly anti-tumoral modality.

Metabolic reprogramming

Metabolic reprogramming is the fourth recurring axis. The picture that emerges from the studies that addressed it, by transcriptional profiling of mitochondrial respiratory-chain components, by measurement of mitochondrial membrane potential, or by proteomic analysis, is that ELF-EMF exposure tends to push cancer cells away from glycolytic dependence and towards oxidative phosphorylation. Several studies report upregulation of mitochondrial-encoded and nuclear-encoded subunits of the respiratory chain and of ATP synthase, accompanied by changes in mitochondrial membrane potential consistent with enhanced respiratory activity. The interpretation offered by the original authors is that the field forces a homeostatic reaction in which energy is reallocated to maintain ionic balance at the expense of biosynthesis and proliferation; this is consistent with the parallel observation that proliferation declines under the same conditions.

The relevance of metabolic reprogramming for selectivity is articulated most clearly by one author group, which proposes that the higher membrane electronegativity of cancer cells, itself a consequence of the Warburg effect, is the upstream determinant of differential vulnerability to electromagnetic perturbation. Healthy cells, lacking this metabolic phenotype, would have a less negative membrane and therefore a smaller response to the same field. A particularly observation in support of this view is that immortalized but non-tumorigenic cells with inactive p53 and pRb behave like healthy cells under this exposure, showing no loss of viability or cell-cycle deregulation: this excludes p53/pRb status as the discriminating factor. The metabolic hypothesis is consistent but not directly proven, since the relevant studies do not measure membrane electronegativity or metabolic flux in parallel.

When direct cancer-vs-healthy comparison is available at the proteomic level, the result is informative: the same set of mitochondrial proteins is regulated in opposite directions in the two cell types under identical exposure, with cancer cells upregulating proteins that support metabolic plasticity and stress response, and healthy cells downregulating the same proteins.

Cell-cycle arrest and the role of P53

Two arrest patterns dominate the dataset. The first, observed in three independent studies, is a G2/M arrest associated with the upregulation of p53 and p21 and the downregulation of CCNB1 and MCM6. The second, observed in two further studies, is a G0/G1 arrest with concurrent reduction of S and G2/M phases, sometimes accompanied by the appearance of binucleated cells consistent with cytokinesis failure. A third, distinct pattern, an increase in S phase without arrest, occurs in the same parameter window that produces the pro-tumoral effects mentioned above and reflects accelerated replicative activity rather than a block.

The cell-cycle data demonstrate that the apoptotic and cell-cycle response to ELF-EMF is conditional on functional P53. In one study, glioblastoma cells with wild-type P53 underwent strong G2/M arrest and apoptosis after exposure; isogenic cells carrying a P53 mutation did not respond; and reintroduction of wild-type P53 in the mutant background restored the response. Healthy fibroblasts exposed to the same field showed the same qualitative gene-expression pattern (P53 and p21 upregulation, MCM6 and CCNB1 downregulation) but did not undergo apoptosis. This dissociation is mechanistically informative: P53 functionality is necessary for the response, but not sufficient on its own. The cancerous cellular context, most plausibly the metabolic and redox context discussed above, is required to convert the P53 signal into a death program. From the selectivity perspective, this means that ELF-EMF can engage stress signaling in healthy cells without committing them to die; the same signal in cancer cells, on a different metabolic background, becomes lethal.

Cell-death programs

Classical caspase-dependent intrinsic apoptosis is the most frequent program engaged and is observed across many cell lines and exposure regimes; in one of the studies, the apoptotic markers (cleaved caspase-3, -8 and -9) increase more than tenfold. A subset of conditions engages, alongside or instead of apoptosis, a regulated necroptotic program, biochemically characterized by the phosphorylation of RIPK1, RIPK3 and MLKL; this program requires both calcium overload and ROS overproduction, and is suppressed by inhibitors of either. A single study documents ferroptosis (the iron-dependent, lipid-peroxidation-driven form of regulated cell death) as a contributor to the antitumoral effect, with the specific ferroptosis inhibitor Ferrostatin-1 partially rescuing growth inhibition; this represents the first report of magnetic-field-induced ferroptosis and is, at present, isolated. Senescence is documented in one further condition, with classical molecular markers (Ki67 down, p21 up).

In conditions where cancer cells undergo apoptosis or necroptosis, healthy cells exposed to the same field most often do not, even when they show the same upstream gene-expression changes. The rare exceptions are concentrated in two specific parameter regions of the dataset: very low frequency combined with very high intensity (where the effect on healthy cells becomes substantial and the selectivity collapses or even inverts), and the 50 Hz / 1 mT window already discussed, where healthy breast epithelial cells undergo necrosis while cancer cells become more aggressive. Outside these regions, the cell-death response of ELF-EMF can be considered selective for cancer cells under most of the conditions examined.

A separate cell-death route is the indirect and immune-mediated one. In one study, the field was not applied directly to cancer cells, but to $CD8^+$ T cells. Exposed T cells acquired enhanced cytotoxic activity (increased TNF- α , IFN- γ , granzyme B, Ki67) and, when co-cultured with HeLa cancer cells, induced apoptosis in cancer cells while leaving the T cells themselves activated rather than damaged. This indirect mechanism is mechanistically distinct from the direct one discussed in the rest of this section and suggests that ELF-EMF may operate not only as a direct cytotoxic agent but also as an immune modulator. Within the dataset, this is documented in a single study and should be read as a hypothesis-generating observation rather than as an established mechanism.

The selectivity question and the role of parameter windows

Selectivity of ELF-EMF for cancer cells over healthy ones is real and documented at every level of the mechanistic chain and engagement of death programs, but it is conditional on the parameter window. Under specific frequency–intensity–waveform combinations, cancer cells are pushed over the threshold that triggers cell death, while healthy cells absorb the perturbation without consequence. Outside these windows, the same fields can produce no effect, equal effects on both cell types, or, in a small but well-documented subset of conditions, the opposite effect of stimulating cancer cell

proliferation and metastatic potential. There is no single ELF-EMF effect on cancer; instead, there are parameter-defined regimes within which the effect can be antitumoral and selective.

The biophysical basis for the existence of these narrow windows is debated within the dataset itself. Three distinct hypotheses are advanced by different authors: a radical-pair mechanism operating on mitochondrial electron transport, a nuclear-spin / J-coupling resonance, and a cell-line-specific thermodynamic resonance defined by a characteristic time τ . Each is internally consistent with the data of the study that proposes it, but none has been tested in cross-comparison, and the three are not mutually exclusive.

Two cellular determinants of selectivity that emerge consistently across the dataset deserve to be highlighted. The first is the metabolic phenotype: cells that depend on aerobic glycolysis (Warburg effect) appear to be more vulnerable to the same exposure that leaves oxidative-phosphorylation-competent cells unaffected. The second is functional P53: when present, P53 enables the conversion of stress signaling into a death program; in cancer cells, P53 functionality combines with the altered metabolic and redox context to produce death; in healthy cells, the same P53 signaling is engaged but does not lead to death, presumably because the cellular context can dissipate the stress. These two determinants are not independent (the metabolic phenotype shapes the redox context, and the redox context modulates P53 activity) and together describe a cellular state in which ELF-EMF perturbation is preferentially lethal. The corollary is that ELF-EMF selectivity is unlikely to be uniform across cancer types: tumors that retain a more glycolytic phenotype and a functional P53 are predicted to be more responsive than those that have already lost one or both of these features. The dataset is too small to test this prediction, but the pattern is consistent with the available observations.

An integrated picture

The mechanistic picture that emerges from the dataset can be summarized as a connected sequence of events whose individual steps are supported by direct measurement in multiple independent studies, even when the causal links between adjacent steps are sometimes inferred rather than proven. The field interacts with the cell at the level of the membrane, perturbing ion fluxes and the mitochondrial electron transport chain; the perturbation produces in parallel an increase in reactive oxygen species and an alteration of calcium homeostasis; these two parallel signals act together to remodel the cytoskeleton, shift cellular metabolism towards oxidative phosphorylation, and engage cell-cycle checkpoints; depending on the parameter window and on the cellular context, the integrated stress is then resolved either through adaptation (in healthy cells, and in cancer cells under sub-threshold exposures) or through commitment to one of cell-death programs (apoptosis, necroptosis, ferroptosis, or senescence).

Selectivity for cancer cells, where it is observed, is not located at any single step of this chain but emerges from the cumulative effect of several differences between cancer and healthy cells: a higher baseline oxidative load, an altered metabolic phenotype, a more responsive (or differently responsive) calcium-signaling network, a less robust cytoskeleton, and, when present, a functional P53 that can convert stress into a commitment to die. This makes ELF-EMF mechanistically interesting as a candidate cancer-selective modality, but it also makes the response intrinsically context-dependent, which is reflected in the wide variability of outcomes observed across the dataset.

6.7 Limitations

The analysis above is constrained by a set of structural limitations of the dataset that must be made explicit before any conclusion is drawn from it. These limitations restrict the depth of the analysis and define the boundary between what the data support and what remains speculative.

Heterogeneity of exposure parameters across studies

Although the ELF-EMF subgroup is the largest in the dataset (178 experimental conditions, 24 studies), its parameter space is extremely heterogeneous. Frequencies span more than four orders of magnitude (0.01 Hz to 385 Hz), AC intensities span more than seven (1 μ T to >2 T), and the waveforms include sinusoidal, square, square-pulse, periodic gradient, multifrequency, and complex pattern signals modeled on physiological calcium waves. With the single exception of condition C3 (1 Hz, 100 mT), used in two studies, each experimental condition can be traced back to a single study, and conversely

the same study often explores many conditions. As a result, the dataset is dense in parameter space but extremely sparse in independent replication: there are essentially no two studies that test the same combination of frequency, intensity, waveform, exposure duration and cell line. This makes the formal cross-study comparison impossible in almost every case, and any conclusion about the effectiveness of one parameter combination over another is based on indirect inference rather than direct head-to-head comparison.

Heterogeneity of cell lines and absence of matched cancer/healthy pairs

A total of 38 cancer cell lines and 19 healthy cell lines are tested within the subgroup, but the matching between the two is uneven. 17 of the 24 studies include both a cancer and a healthy cell line, which is the strength of the subgroup; however, when the comparison is made within a single study, the healthy line chosen is not always biologically appropriate as a non-tumoral counterpart of the cancer line under test. For example, in [13] HUVEC (vascular endothelium) is the only healthy reference for DU-145 (prostate), MDA-MB-231 (breast) and K562 (haematological); in [6] normal human fibroblasts are the reference for U87, U251 and U251-P53 (glioblastoma); in [10] FF95 dermal fibroblasts are the reference for MDA-MB-231 and MCF-7 (breast). In several other studies ([3], [8], [15], [23], [25], [36], [40], [41], [48]) the choice is more biologically appropriate (matched epithelial breast lines, matched glial lines). The implication is that, when assessing selectivity, part of the differential response between cancer and healthy lines may reflect tissue-of-origin rather than the cancer/non-cancer dichotomy itself, and this confounding factor cannot be resolved within the present dataset.

Sparse and uneven coverage of the outcomes

The coverage of the seven cell fate outcomes and the eight mechanistic outcomes is markedly uneven between conditions. Within the ELF-EMF subgroup, viability is measured in 115 conditions and proliferation in 112, but apoptosis only in 44, necrosis in 18, motility in 17, senescence in 3 and ferroptosis in only 2 (both from a single study). Among the mechanistic outcomes, cell cycle (39 conditions), cellular morphology and cytoskeleton (38), and ROS (33) are reasonably well represented; antioxidant response is measured in only 3 conditions, and differentiation in 1. The proteomic and genomic outcomes are concentrated in a few studies and address different protein/gene panels (necroptosis pathway in [25]; P53/P21/MDM2/MCM6/CCNB1 in [6]; mitochondrial OXPHOS subunits in [2] and [36]; pluripotency factors in [41]; cytotoxic T-cell markers in [48]). The consequence is that, for any given experimental condition, only a fraction of the mechanistic chain is observed, and the connections between adjacent steps must be reconstructed from different conditions in different studies, with the obvious risk of conflating distinct mechanisms.

Missing parameters

Several studies do not report all the parameters needed to reproduce the exposure exactly. The waveform is sometimes implicit; in [5] the device is explicitly described as proprietary (industrial secret); in some cases (e.g. condition C3 used by [6] and [25]) the same nominal parameters are reported, but the precise device geometry, coil configuration and induced electric field at the cell layer are not always specified. Exposure duration is reported but the time of measurement relative to exposure end is not always consistent across studies, complicating the comparison of acute vs delayed effects.

Taken together, these limitations imply that the mechanistic framework reconstructed in Section 6.6 should be read as the most coherent picture compatible with the available data, not as a validated mechanistic model. The translational implication is that ELF-EMF mono-therapy in oncology, on the basis of available in vitro evidence, cannot yet be associated with a single, predictable mechanism of action, and the reproducibility of even the most promising parameter windows requires independent validation under harmonized exposure conditions before any clinical inference can be drawn.

7 Low Frequency Electromagnetic Fields

According to the general frequency classification adopted by the IEEE (Institute of Electrical and Electronics Engineers), Low Frequency electromagnetic fields span the range from 30 kHz to 300 kHz. Compared to ELF-EMFs, LF-EMFs therefore operate at considerably higher frequencies. In this report, conditions up to 320 kHz are included in this macro-category, as the studies grouped here share comparable frequency ranges.

7.1 Dataset overview and cell line distribution

The LF-EMF macro-category comprises 38 experimental conditions drawn from 3 independent studies. Of these, 27 (71%) involve cancer cell lines (Table 11) and 11 (29%) involve healthy cell lines (Table 12), all of human origin.

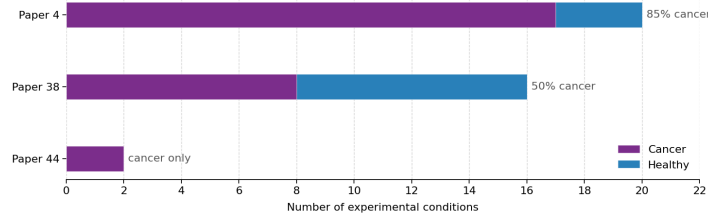


Figure 25: Cancer vs healthy cell line distribution across the LF-EMF studies.

Cell line	Type	Cancer Category	Origin	Exp. cond.
U87	Glioblastoma	Brain & CNS	Human	5
LN229	Glioblastoma	Brain & CNS	Human	5
SH-SY5Y	Neuroblastoma	Brain & CNS	Human	4
T98G	Glioblastoma multiforme	Brain & CNS	Human	4
MDA-MB-231	Triple-negative breast cancer (TNBC)	Breast	Human	3
U251	Glioblastoma	Brain & CNS	Human	1
T98	Glioblastoma	Brain & CNS	Human	1
A172	Glioblastoma	Brain & CNS	Human	1
PANC-1	Pancreatic cancer	Pancreatic	Human	1
MCF-7	Breast cancer	Breast	Human	1
MDA-MB-453	Breast cancer	Breast	Human	1

Table 11: Cancer cell lines used in the LF-EMF macro-category: type, cancer category, origin, and number of experimental conditions.

Cell line	Type	Category	Origin	Exp. cond.
NHA	Normal astrocytes	Healthy – Brain & CNS	Human	5
HDFa	Dermal fibroblasts	Healthy – Skin	Human	4
HCF	Cardiac fibroblast	Healthy – Heart	Human	1
HUVEC	Umbilical vein endothelial cells	Healthy – Vascular	Human	1

Table 12: Healthy cell lines used in the LF-EMF macro-category: type, cancer category, origin, and number of experimental conditions.

7.2 Exposure parameters

The parameters characterizing LF-EMF exposures are the same as those described for the ELF-EMF macro-category (waveform, frequency, and intensity) in Section 6.2, and their definitions are unchanged. One difference with respect to the ELF-EMF macro-category is that no static (DC) magnetic field component was superimposed in any of the LF-EMF conditions.

C_ID	Paper	Field type	Wave type	Freq. (kHz)	Max AC intensity (mT)
C1	38	EMF	Sine	87	1.6
C2	44	AMF	Sine	100	10
C3	4	AMF	Sine	160	18
C4	4	AMF	Sine	198	18
C5	38	EMF	Sine	207	1.4
C6	4	AMF	Sine	227	18
C7	4	AMF	Sine	280	18
C8	4	AMF	Sine	320	18

Table 13: LF-EMF exposure conditions lookup table.

Exposure duration varied across conditions. Cells tested under conditions C3, C4, C6, C7, and C8 were exposed for 30 minutes, while those under conditions C1 and C5 received a 60 minute exposure. Condition C2 was the only one in which two different durations were compared: 60 and 90 minutes.

7.3 Frequency distribution and rationales

The two frequencies of 87 kHz and 207 kHz were selected because they correspond to the resonant operating parameters of a wireless power transfer (WPT) prototype system [38].

The frequency of 100 kHz was chosen based on the capabilities of the exposure system used and on a previous study that had already demonstrated significant biological effects under the same experimental setup [44].

The remaining frequencies of 160, 198, 227, 280, and 320 kHz were selected through a systematic screening of what the authors describe as a black box frequency range, broadly located between induction heaters (low kHz) and microwave devices (MHz/GHz range). Preliminary experiments showed that frequencies below 160 kHz or above 320 kHz did not produce significant inhibitory effects in human glioblastoma cell lines (U87 and LN229); the frequency of 227 kHz emerged as the most effective and was therefore the focus of further investigation [4].

7.4 Intensity distribution and rationales

The magnetic flux density values of 1.4 mT and 1.6 mT were chosen to exceed the reference level ICNIRP 2010 of 27 μ T by a factor of 48–63, to explore biological effects at field strengths well above current regulatory limits [38].

The intensity of 10 mT was selected on the basis of a previous study demonstrating significant biological effects at 5, 10 and 15 mT using the same exposure system. Intensities of 5 mT and 15 mT were subsequently excluded from further analysis because they did not produce statistically significant effects on cell viability [44].

The intensity of 18 mT reflects the maximum output achievable by the equipment used and was therefore adopted as the upper limit of the tested range [4].

7.5 Cell fate outcomes

As shown in Table 14, viability is the most frequently assessed cell fate outcome in the LF-EMF macro-category, being evaluated in all 38 experimental conditions.

Proliferation was measured in only 2 conditions, both on cancer cell lines.

Apoptosis and necrosis were assessed in 8 conditions, equally distributed between cancer (4) and healthy (4) cell lines. Data for ferroptosis, senescence, or motility were not available within this macro-category.

Mechanistic outcome	Cancer exp. conditions	Healthy exp. conditions	Total
Viability	27	11	38
Proliferation	2	0	2
Apoptosis	4	4	8
Necrosis	4	4	8
Ferroptosis	0	0	0
Senescence	0	0	0
Motility	0	0	0

Table 14: Number of LF-EMF macro-category experimental conditions assessing each cell fate outcome, broken down by cancer and healthy cell lines.

7.5.1 Viability

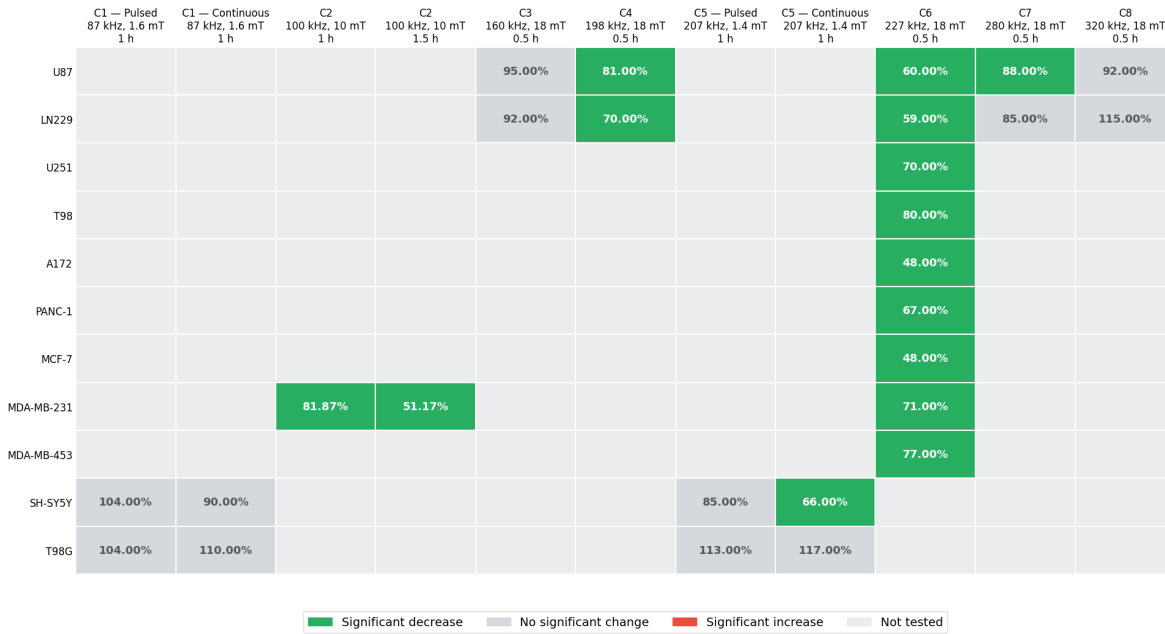


Figure 26: Residual viability (% of control) of cancer cell lines across LF-EMF exposure conditions. Green cells indicate a significant decrease in viability; dark gray cells indicate no significant change; red cells indicate a significant increase in viability; light gray cells indicate conditions not tested for that line.

Figures 26 and 27 report the residual viability (% of control) of cancer and healthy cell lines across LF-EMF exposure conditions.

Condition C6 (227 kHz, 18 mT) is the most promising in the entire LF-EMF dataset. It was tested for nine distinct cancer cell lines (U87, LN229, U251, T98, A172, PANC-1, MCF-7, MDA-MB-231, MDA-MB-453) and produced a statistically significant reduction in viability in each one. Residual viability ranged from 48% in A172 and MCF-7 to 80% in T98, with intermediate values recorded for LN229 (59%), U87 (60%), PANC-1 (67%), U251 (70%), MDA-MB-231 (71%) and MDA-MB-453 (77%), all relative to the untreated control.

Conditions C3, C4, C7 and C8 (160, 198, 280 and 320 kHz respectively, all at 18 mT) were tested exclusively for U87 and LN229 cells and together delineate a narrow window of efficacy centered between 198 and 227 kHz. Within this range, 227 kHz produced the strongest inhibition, while 198 kHz (C4) ranked second, reducing viability to 81% in U87 and 70% in LN229. Efficacy decreased gradually

at 280 kHz, and the boundary frequencies of 160 and 320 kHz did not show a significant effect on either line (U87: 95% and 92%; LN229: 92% and 115%, respectively). The value of 115% observed in LN229 at 320 kHz is particularly noteworthy, as it suggests a growth-stimulatory rather than inhibitory response, that is the opposite of the desired therapeutic effect.

Turning to the healthy cell lines tested under C6, none of the three showed any significant reduction in viability: normal astrocytes (NHA) reached 90%, cardiac fibroblasts (HCF) 105%, and umbilical vein endothelial cells (HUVEC) 96% compared to the control group. This is particularly meaningful because these healthy cells were exposed to the exact same frequency, intensity, and duration that reduced cancer cell viability.

	C1 — Pulsed 87 kHz, 1.6 mT 1 h	C1 — Continuous 87 kHz, 1.6 mT 1 h	C5 — Pulsed 207 kHz, 1.4 mT 1 h	C5 — Continuous 207 kHz, 1.4 mT 1 h	C6 227 kHz, 18 mT 0.5 h
NHA	104.00%	100.00%	101.00%	92.00%	90.00%
HCF					105.00%
HUVEC					96.00%
HDFa	104.00%	103.00%	101.00%	96.00%	

Significant increase / No significant change
Significant decrease
Not tested

Figure 27: Residual viability (% of control) of healthy cell lines across LF-EMF exposure conditions. Green cells indicate values at or above control level; dark gray cells indicate no significant change; red cells indicate a significant decrease in viability; light gray cells indicate conditions not tested for that line.

Under condition C2 (100 kHz and 10 mT), MDA-MB-231 cells were exposed for either 60 or 90 minutes. At 60 minutes, viability dropped to 81.87% of control; at 90 minutes, it fell further to 51.17%. These results point to a clear time-dependent relationship: exposures shorter than 60 minutes (5, 10, 20, and 30 minutes) produced no significant effect, indicating the existence of a temporal threshold somewhere between 30 and 60 minutes, below which the field is insufficient to trigger measurable cellular damage.

At condition C1 (87 kHz and 1.6 mT), none of the four tested lines (HDFa, NHA, SH-SY5Y, and T98G) showed any significant change in viability, regardless of whether exposure was delivered in pulsed mode (six 10-minute sessions separated by 10-minute intervals) or continuously (a single 60-minute session). At condition C5 (207 kHz and 1.4 mT) the situation is almost identical, with one exception: SH-SY5Y neuroblastoma cells showed a significant viability reduction to 66% of control under continuous exposure, but not under pulsed exposure (85%). This divergence suggests that SH-SY5Y cells may be particularly sensitive to sustained field exposure, pointing to a cell-line specific vulnerability. Notably, T98G glioblastoma multiforme cells exhibited the opposite trend under C5, reaching 117% of control viability under continuous exposure, an increase that the authors themselves consider non-significant, but which raises the possibility that prolonged exposure at this frequency could promote rather than inhibit tumor growth in certain glioblastoma subtypes.

Comparing pulsed and continuous delivery across C1 and C5 (Figure 28), at 87 kHz the two modalities produce virtually identical outcomes across all cell lines, consistent with the inefficacy of this condition. At 207 kHz, differences emerge specifically in SH-SY5Y (continuous: 66% vs. pulsed: 85%) and T98G (continuous: 117% vs. pulsed: 113%), suggesting that exposure mode may interact with tumor type in ways that warrant further investigation. However, the available data are too limited to draw conclusions.

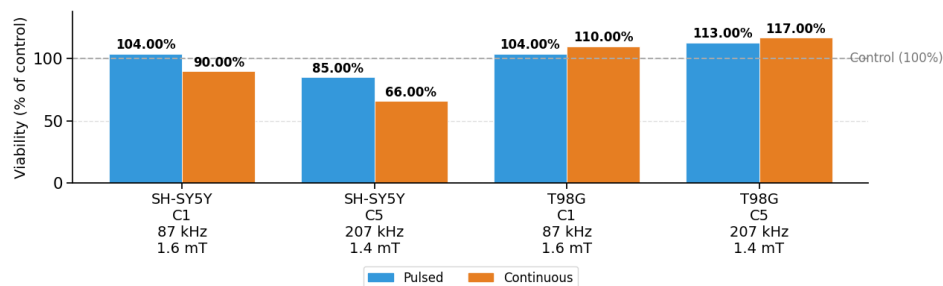


Figure 28: Comparison of residual viability (% of control) between pulsed and continuous exposure in SH-SY5Y and T98G cancer cell lines at conditions C1 (87 kHz and 1.6 mT) and C5 (207 kHz and 1.4 mT).

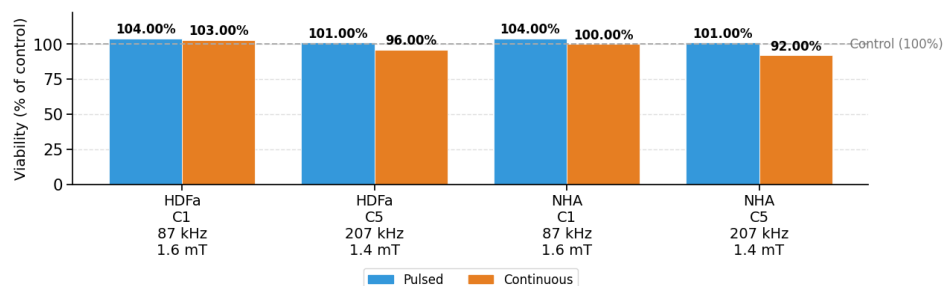


Figure 29: Comparison of residual viability (% of control) between pulsed and continuous exposure in healthy cell lines HDFa and NHA at conditions C1 (87 kHz and 1.6 mT) and C5 (207 kHz and 1.4 mT).

7.5.2 Proliferation

Proliferation was assessed only under condition C6 (227 kHz and 18 mT), in the two glioblastoma lines U87 and LN229. Cell counts performed 96 hours after the end of field exposure revealed a reduction in proliferation of 65% compared to control group in U87 and 70% in LN229. Taken together with the viability data recorded 24 hours post exposure, these results indicate that the antiproliferative effect of a single 30-minute exposure persists for at least four days.

7.5.3 Apoptosis

Apoptosis was measured exclusively under condition C1 (87 kHz and 1.6 mT) across all four cell lines (HDFa, NHA, SH-SY5Y, T98G), under pulsed and continuous exposure modes, for a total of eight experimental conditions. No significant increase in apoptosis was detected in any of them. This result is fully consistent with the absence of viability effects observed under the same condition.

7.5.4 Necrosis

The same considerations reported for apoptosis apply here: no significant increase in necrosis was observed under condition C1 in any of the eight conditions tested.

7.6 Mechanistic outcomes

7.6.1 ROS production

Under condition C6 (227 kHz and 18 mT), intracellular ROS levels rose to 150% of control in U87 cells and to 145% in LN229 cells, both statistically significant increases. In healthy NHA astrocytes exposed under the same condition, ROS increased by only 10% (110% of control), a change that did

Mechanistic outcome	Cancer exp. conditions	Healthy exp. conditions	Total
ROS production	2	1	3
Antioxidant response	5	4	9
Membrane potential	1	0	1
Calcium signaling	0	0	0
Cellular morphology and cytoskeleton remodeling	5	4	9
Metabolic reprogramming	2	1	3
Differentiation	0	0	0
Cell cycle analysis	2	1	3

Table 15: Number of experimental conditions in the LF-EMF macro-category that assess each mechanistic outcome, broken down by cancer and healthy cell lines.

not reach significance. The authors demonstrated that both KCN (potassium cyanide, an inhibitor of mitochondrial electron transport chain complex IV) and NAC (N-acetylcysteine, a ROS scavenger) suppressed the field-induced ROS elevation, identifying the mitochondrial respiratory chain as the primary source of radical production. This implies that the field acts on mitochondria, triggering an uncontrolled burst of free radical generation. Cancer cells, owing to their altered metabolism, already operate at higher baseline ROS levels than healthy cells; field exposure appears to push them past a critical oxidative stress threshold from which they cannot recover. Healthy cells, starting from a lower baseline, remain below that threshold even under the same exposure conditions.

7.6.2 Antioxidant response

Reduced glutathione (GSH) was measured under condition C1 (87 kHz and 1.6 mT) in HDFa, NHA, SH-SY5Y, and T98G cells. No significant changes were detected in any of the four lines, a result that is entirely consistent with the absence of viability effects at this frequency–intensity combination: if the field is insufficient to generate oxidative stress, there is no reason to expect depletion of antioxidant defences.

Total antioxidant capacity was assessed in MDA-MB-231 cells under condition C2 (100 kHz and 10 mT), where it showed a significant reduction to 89.97% of control. This finding complements the ROS data from C6 and suggests that, at this frequency–intensity combination, the field both increases radical production and simultaneously erodes the cellular defences available to neutralise them — a two-pronged mechanism that would amplify oxidative damage beyond what either effect alone could produce.

7.6.3 Membrane potential

Mitochondrial membrane potential was measured in U87 cells under condition C6 (227 kHz and 18 mT) and 60 minutes exposure, where it increased significantly to 180% of control. This hyperpolarization is mechanistically linked to the ROS data described above: when the inner mitochondrial membrane becomes hyperpolarized, electron flow through the respiratory chain slows, leading to electron accumulation and a consequent rise in superoxide and other reactive species. The authors argue that this hyperpolarization event acts as the upstream trigger for the excessive ROS production and the mitochondrial dysfunction that follows.

7.6.4 Cellular morphology and cytoskeleton remodelling

Under condition C1 (87 kHz and 1.6 mT), none of the four cell lines (HDFa, NHA, SH-SY5Y, and T98G) exhibited morphological alterations, consistent with the absence of significant viability changes at this exposure level.

Under condition C2 (100 kHz and 10 mT) and 60 minutes exposure, MDA-MB-231 cells displayed a constellation of morphological changes including nuclear fragmentation, nuclear condensation, mem-

brane blebbing and protrusions, cytoplasmic vacuolisation, cell shrinkage, and apoptotic nuclear alterations. This profile is characteristic of late-stage apoptosis and provides morphological confirmation that the field induces programmed cell death under these exposure parameters. The authors also suggest that the field may destabilize the cytoskeleton, though no direct measurements were performed to support this hypothesis.

Regarding the cytoskeleton specifically, direct quantification of filamentous actin (F-actin) under condition C1 (87 kHz and 1.6 mT) revealed no significant reorganisation in any of the tested lines.

7.6.5 Metabolic reprogramming

In U87 and LN229 cancer cells exposed under condition C6 (227 kHz and 18 mT), the field induced a profound shift in cellular energy metabolism. The extracellular acidification rate (ECAR), a proxy for glycolytic activity, decreased significantly, while the oxygen consumption rate (OCR), indicative of oxidative phosphorylation, increased simultaneously. The authors describe this as an anti-Warburg shift: a reversal of the aerobic glycolysis dependency that characterizes cancer cell metabolism. Tumor cells, which rely heavily on glycolysis for proliferation and survival, appear to be forced into an oxidative metabolic programme to which they are not adapted. In healthy NHA cells, the same exposure produced no significant changes in either ECAR or OCR, suggesting that normal cells possess sufficient metabolic flexibility to accommodate the perturbation without functional consequences, whereas cancer cells, locked into glycolytic dependency, cannot.

This functional observation is corroborated at the genomic level. Microarray analysis of U87 cells exposed at 227 kHz revealed a broad down-regulation of glycolytic gene expression, consistent with the anti-Warburg shift observed functionally. Interestingly, the same glycolytic gene down-regulation was also detected in NHA cells, yet without any accompanying functional changes in ECAR or OCR. This dissociation between transcriptional and functional responses suggests that the field acts on gene regulation similarly in both cell types, but that the functional vulnerability is specific to cancer cells, most likely because healthy cells can compensate at the metabolic level in ways that tumor cells cannot.

7.6.6 Cell cycle analysis

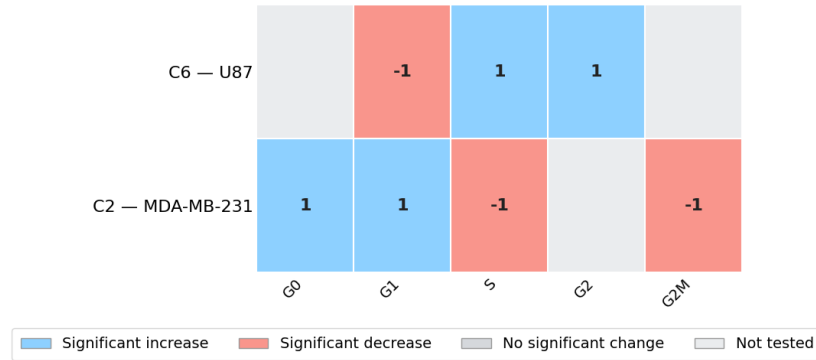


Figure 30: Cell cycle phase distribution in cancer cell lines U87 (condition C6, 227 kHz and 18 mT) and MDA-MB-231 (condition C2, 100 kHz and 10 mT). Values of +1, 0, and -1 indicate a significant increase, no significant change, and a significant decrease in the fraction of cells in each phase respectively.

Figures 30 and 31 show the cell cycle phase distribution in cancer and healthy cell lines, respectively.

In U87 cells under condition C6 (227 kHz and 18 mT), cell cycle profiling revealed a reduction of the G1 population and an accumulation of cells in both the S and G2 phases. This pattern is consistent with the observed down-regulation of Cyclin A (an S/G2 regulator) and Cyclin B1 (required for mitotic entry): cells progress into S phase but fail to complete it, stalling before mitosis. In healthy NHA cells exposed under the same condition, cell cycle phase distribution remained unchanged, further supporting the selectivity of the effect for cancer cells.

In MDA-MB-231 cells under condition C2 (100 kHz and 10 mT), a different arrest profile emerged. An increase in the G0 and G1 populations was accompanied by a simultaneous decrease in S phase

and G2/M fractions, indicating a G0/G1 arrest with inhibition of both DNA synthesis and mitotic entry. The expansion of the G0 fraction is particularly notable, as it suggests that a subset of cells is actively exiting the proliferative cycle and entering a state of quiescence or senescence.

Taken together, the two conditions induce cell cycle arrest at distinct checkpoints: 227 kHz stalls cells in S/G2, while 100 kHz arrests them in G0/G1. Whether this difference reflects the distinct frequencies, the different field intensities (18 vs. 10 mT), the different exposure durations (30 vs. 60–90 minutes), or the intrinsically different biology of the cell lines tested (glioblastoma vs. triple-negative breast carcinoma) remains an open question. What is clear is that both arrests ultimately reduce viability, suggesting that the downstream cytotoxic outcome is achieved regardless of the specific phase at which the cell cycle is blocked.

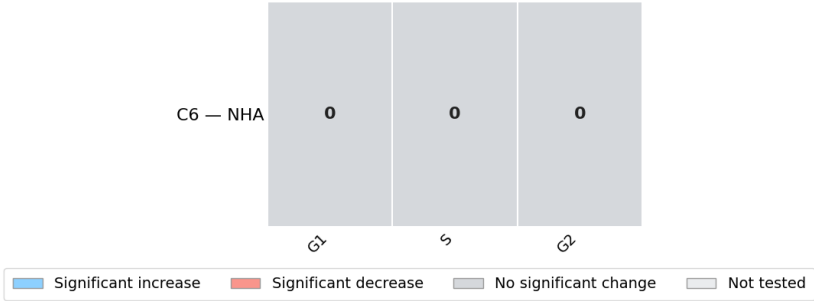


Figure 31: Cell cycle phase distribution in healthy NHA astrocytes under condition C6 (227 kHz and 18 mT). Values of +1, 0, and –1 indicate a significant increase, no significant change, and a significant decrease in the fraction of cells in each phase respectively.

7.7 Molecular outcomes

7.7.1 Genomic analysis

Microarray analysis was performed under condition C6 (227 kHz and 18 mT) in U87 cells to validate at the transcriptional level the anti-Warburg metabolic shift described in Section 7.6.5. The results are discussed there.

7.7.2 Proteomic analysis

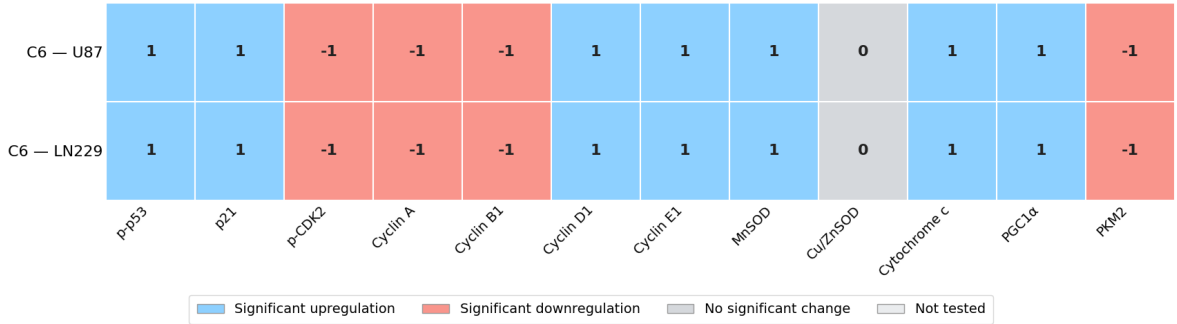


Figure 32: Proteomic profile of U87 and LN229 cancer cell lines under condition C6 (227 kHz and 18 mT). Twelve proteins spanning cell cycle regulation, mitochondrial function, and glycolytic metabolism were quantified. Values of +1, 0, and –1 denote significant up-regulation, no significant change, and significant down-regulation respectively.

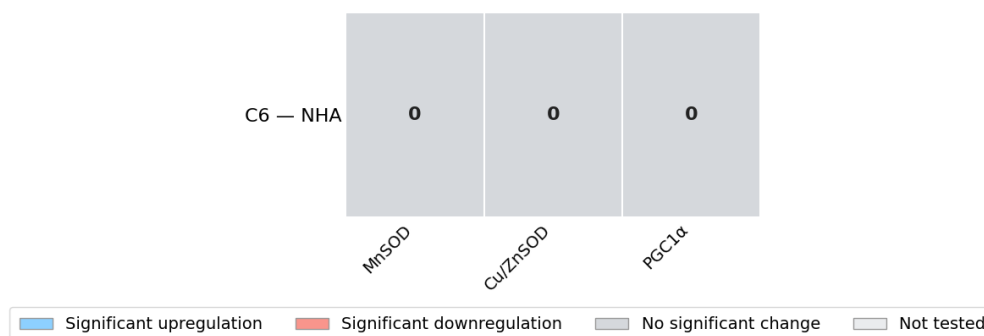


Figure 33: Proteomic profile of healthy NHA astrocytes under condition C6 (227 kHz and 18 mT). Three proteins from the antioxidant defence and mitochondrial biogenesis pathways were quantified. Values of +1, 0, and -1 denote significant up-regulation, no significant change, and significant down-regulation respectively.

Condition C6 (227 kHz and 18 mT) was the only one for which a detailed proteomic profile was assembled, covering twelve proteins across three functional categories: cell cycle regulation, mitochondrial biogenesis, and glycolytic metabolism. As shown in Figure 32 and 33 the analysis was carried out in U87 and LN229 cancer lines and in healthy NHA astrocytes. U87 and LN229 showed an identical expression profile across all twelve proteins, suggesting that the field acts through a common mechanism in these two glioblastoma lines despite their distinct genetic backgrounds.

Up-regulated proteins in cancer cell lines

- **p-p53.** Phosphorylated p53 is the active form of the principal tumor suppressor, often referred to as the guardian of the genome. It is mobilized in response to cellular stress and can either halt the cell cycle to allow damage repair or, when damage is irreparable, commit the cell to apoptosis. Its up-regulation in both lines is consistent with the elevated ROS levels observed under the same condition and points to p53-mediated stress signaling as a central component of the field's mechanism of action.
- **p21.** p21 (CDKN1A) is a cyclin-dependent kinase inhibitor and a direct transcriptional target of activated p53. By binding cyclin-CDK complexes, it prevents retinoblastoma protein (Rb) phosphorylation and thereby blocks entry into S phase. Its up-regulation is a direct downstream consequence of p-p53 activation and confirms that the p53-mediated cell cycle arrest cascade is engaged.
- **Cyclin D1 and Cyclin E1.** These two proteins regulate G1 progression: Cyclin D1 partners with CDK4/6 in early G1, while Cyclin E1 partners with CDK2 at the G1/S transition. Their accumulation under field exposure is best explained as a build-up of G1-phase proteins in cells that are arrested at this checkpoint and therefore unable to consume them through normal cycle progression.
- **MnSOD.** Manganese superoxide dismutase is the primary antioxidant enzyme of the mitochondrial matrix, where it converts superoxide into hydrogen peroxide. Its up-regulation reflects a compensatory response to mitochondrial oxidative stress: cells attempt to counteract excess ROS by increasing their enzymatic defences. The fact that this response fails to prevent viability loss in cancer cells indicates that ROS production exceeds the cell's capacity to neutralize it.
- **Cytochrome c.** Under normal conditions, cytochrome c resides in the mitochondrial intermembrane space where it shuttles electrons along the respiratory chain. Its release into the cytoplasm is a defining event in the intrinsic apoptotic pathway: once cytosolic, it binds Apaf-1 to form the apoptosome, which activates caspase-9 and triggers the executioner caspase cascade. The up-regulation detected here, reflecting cytoplasmic release, constitutes the strongest direct evidence of apoptotic commitment in the LF-EMF dataset, all the more notable given that no Annexin V assay was performed at 227 kHz.

- **PGC1 α .** PGC1 α is the master regulator of mitochondrial biogenesis and oxidative metabolism. Its up-regulation is consistent with the anti-Warburg metabolic shift described in the previous section: the field appears to drive cancer cells towards increased mitochondrial activity at the expense of glycolysis. In tumour cells, however, this forced shift towards oxidative phosphorylation may be inherently damaging, as dysfunctional mitochondria operating beyond their normal load generate disproportionate amounts of ROS.

Down-regulated proteins in cancer cell lines

- **PKM2.** Pyruvate kinase M2 is a key glycolytic enzyme and widely regarded as the molecular engine of the Warburg effect. Its down-regulation is the most direct proteomic confirmation of the anti-Warburg shift: the field is actively suppressing the glycolytic machinery that cancer cells depend on.
- **p-CDK2.** CDK2, in its phosphorylated (active) form, is required for the G1-to-S transition and for progression through S phase, where it operates in complex with Cyclin A or Cyclin E. Its down-regulation is fully coherent with the p53/p21-mediated arrest described above.
- **Cyclin A.** Cyclin A governs the S/G2 transition and S phase progression. Its reduction indicates that cells are failing to advance through DNA replication and prepare for mitosis.
- **Cyclin B1.** Cyclin B1 is indispensable for mitotic entry: in complex with CDK1 it forms the maturation promoting factor (MPF), which initiates chromosome condensation, nuclear envelope breakdown, and mitotic spindle assembly. Its down-regulation signals that cells cannot enter mitosis, effectively blocking cell division at the last gate before it occurs.

Unchanged proteins in cancer cell lines

- **Cu/ZnSOD.** Unlike MnSOD, which is mitochondrial, Cu/ZnSOD operates in the cytoplasm. The fact that it remains unchanged while its mitochondrial counterpart is up-regulated supports the interpretation that oxidative stress under condition C6 is compartmentalised within the mitochondria rather than being a generalised cytoplasmic phenomenon, further reinforcing the mitochondrion as the field's primary target.

Healthy NHA astrocytes In healthy NHA cells, only three of the twelve proteins were quantified (MnSOD, Cu/ZnSOD, and PGC1 α) and none showed a significant change. While the limited panel prevents a full comparison with the cancer lines, the absence of any detectable proteomic response in these three markers is consistent with the overall picture: the same exposure that triggers extensive molecular remodelling in glioblastoma cells leaves normal astrocytes functionally undisturbed.

7.8 Mechanisms of action and connections

Taken together, the data collected across the LF-EMF macro-category outline a mechanistic framework for the action of LF-EMFs on cancer cells, most completely defined for the C6 (227 kHz and 18 mT) condition.

The sequence of events appears to begin inside the mitochondria. Field exposure induces hyperpolarization of the inner mitochondrial membrane, slowing electron flow through the respiratory chain, causing electron accumulation, and ultimately driving superoxide generation. The exact biophysical mechanism by which the electromagnetic field interacts with the mitochondrial membrane remains unclear; the authors of the studies analyzed here suggest that voltage-dependent ion channels or field-induced redistribution of charges across the inner membrane may be involved. Cells attempt to compensate by up-regulating MnSOD, but the antioxidant response proves insufficient to contain the oxidative burden.

Simultaneously, the field suppresses glycolysis, evidenced by PKM2 down-regulation and broad glycolytic gene down-regulation in the microarray data, while enhancing oxidative phosphorylation, as reflected by PGC1 α up-regulation and increased OCR. Cancer cells, which depend heavily on glycolysis for biosynthesis and proliferation, are ill-equipped to handle this forced metabolic shift. Healthy cells, being metabolically more flexible, adapt without apparent functional consequences.

The combined burden of ROS accumulation and metabolic stress activates p53, which in turn induces p21 and triggers cell cycle arrest. Downstream, mitochondrial damage leads to cytochrome c release, committing the cell to the intrinsic apoptotic pathway. Cells that cannot repair the accumulated damage ultimately undergo programmed cell death.

For the C2 (100 kHz and 10 mT) condition the mechanistic picture is less complete, but several parallel elements are nonetheless present: a significant reduction in total antioxidant capacity, cell cycle arrest at G0/G1 (as opposed to the S/G2 arrest seen at 227 kHz), and morphological changes characteristic of apoptosis, including nuclear fragmentation, membrane blebbing, and chromatin condensation.

7.9 Limitations

The most compelling viability data obtained at 227 kHz across nine cancer cell lines derive from a single study [4]. Independent replication remains an essential next step before these findings can be considered robust.

The C2 (100 kHz and 10 mT) condition was tested on only one cancer cell line (MDA-MB-231) and on no healthy cell lines. In the absence of a normal cell control, it is not possible to make any claim about the selectivity of this frequency–intensity combination.

Almost all of the mechanistic evidence in this macro-category (ROS quantification, mitochondrial membrane potential, metabolic reprogramming, proteomic profiling, and cell cycle analysis) comes from a single study [4]. A mechanistic model resting on one source is fragile and should be treated as hypothesis-generating rather than conclusive.

Apoptosis and necrosis were measured exclusively under condition C1 (87 kHz and 1.6 mT), which produced no significant effects on viability. These endpoints were never assessed under the conditions that did reduce viability (227 kHz and 100 kHz), leaving a gap in the understanding of the cell death pathway actually responsible for the observed losses.

The field intensities used across the three studies (18 mT, 10 mT, and 1.4–1.6 mT) differ by more than an order of magnitude and were never tested in a common experimental framework. This makes cross-study comparisons of biological effects unreliable.

8 Radio Frequency Electromagnetic Fields

RF-EMF is the term used to describe the part of the electromagnetic spectrum comprising the frequency range from 100 kHz to 300 GHz. Within this frequency range, the electric and magnetic fields, which together make up the electromagnetic fields, are interrelated and considered jointly for measurements. RF-EMF exposure is conventionally quantified either as power density (W/m^2) or as specific absorption rate (W/kg), depending on the frequency range and the context of exposure [ICNIRP].

In practice, the boundary between RF-EMFs and lower-frequency fields is not always sharp, and the allocation of borderline cases to one category or another inevitably involves a degree of judgment. In this report, the LF-EMF section (Section 7) covers frequencies up to 320 kHz, which falls within the lower portion of the RF band as defined by ICNIRP, but was kept separate because the studies grouped there share comparable frequency ranges, making them more informative when analyzed together. The present RF-EMF subgroup, by contrast, comprises conditions operating in the MHz range and above.

8.1 Dataset overview and cell line distribution

Cell line	Type	Cancer category	Cate-	Origin	Exp. cond.	Subgroup
HT-1080	Fibrosarcoma	Sarcoma		Human	21	RF-MF + SMF
H1299	Non-small cell lung cancer (NSCLC)	Lung		Human	13	MMW + THz
Huh7	Hepatocellular carcinoma	Liver		Human	8	LTE
HeLa	Cervical adenocarcinoma	Cervical		Human	6	LTE
MCF-7	Breast cancer	Breast		Human	5	MMW + THz, RF-EMF
MCF-7	Invasive breast adenocarcinoma (wild-type TP53)	Breast		Human	5	MMW + THz, RF-EMF
A375	Melanotic melanoma	Melanoma		Human	4	MMW + THz
B103	Neuroblastoma	Brain & CNS		Rat	4	LTE
Hep3B	Liver cancer stem cells (hepatocellular carcinoma)	Liver		Human	2	LTE
SH-SY5Y	Neuroblastoma	Brain & CNS		Human	2	LTE
Ramos	Burkitt lymphoma (B-cell origin, EBV-negative, homozygous...)	Hematological		Human	2	MMW + THz
Jurkat	T-cell acute lymphoblastic leukemia	Hematological		Human	2	MMW + THz
HT29	Colorectal carcinoma (primary tumor)	Colorectal		Human	2	RF-EMF
SW480	Colorectal adenocarcinoma (primary, Duke's B)	Colorectal		Human	2	RF-EMF
LoVo	Colorectal adenocarcinoma (left supraclavicular metastasis)	Colorectal		Human	2	RF-EMF
LNCaP	Prostate carcinoma (wild-type p53)	Prostate		Human	1	MMW + THz

Table 16: Cancer cell lines used in the RF-EMF macro-category: type, cancer category, origin, number of experimental conditions, and the specific subgroups within the RF-EMF macro-category (RF-MF + SMF, RF-EMF, LTE, MMW + THz) in which each cell line was tested.

The RF-EMF macro-category comprises 99 experimental conditions taken from 10 independent studies. Four subgroups were identified on the basis of the field type and the frequency range used: RF-MF + SMF (0.9–7.2 MHz, 21 conditions, 21.2%), RF-EMF (13.56 MHz and unspecified frequency,

9 conditions, 9.1%), LTE (~ 1.7 GHz, 29 conditions, 29.3%), and MMW + THz (35–140 GHz, 40 conditions, 40.4%). Given the heterogeneity between subgroups, each is treated separately in the following subsections.

Cell line	Type	Category	Origin	Exp. cond.	Subgroup
MCF-10A	Breast epithelial cells	Healthy – Breast	Human	15	MMW + THz
ASC	Adipose tissue-derived stem cells	Healthy – Stem Cells	Human	5	LTE
IMR-90	Normal fibroblast	Healthy – Connective Tissue	Human	2	LTE
PBMCs	Primary mononuclear blood cells (monocytes, B-lymphocytes...)	Healthy – Blood/Immune	Human	1	MMW + THz

Table 17: Healthy cell lines used in the RF-EMF macro-category: type, category, origin, number of experimental conditions, and the specific subgroups within the RF-EMF macro-category (RF-MF + SMF, RF-EMF, LTE, MMW + THz) in which each cell line was tested.

As shown in Tables 16 and 17, experimental conditions involving cancer cell lines substantially outnumber those involving healthy cell lines across the macro-category. Moreover, healthy cell lines were included in only 4 of the 10 studies, and exclusively within the LTE and MMW + THz subgroups. This uneven distribution limits the extent to which selectivity can be evaluated across all four subgroups: for RF-MF + SMF and RF-EMF, no healthy cell data are available and any conclusions about differential effects between cancer and normal cells are restricted to the two subgroups where such comparisons are possible.

8.2 Exposure parameters

The four subgroups that constitute the RF-EMF macro-category differ not only in frequency range, but also in the physical quantities used to characterize the field. Each subgroup is therefore described by its own lookup table.

RF-MF + SMF (Table 18). These twenty conditions, taken from Papers 7 and 29, combine an oscillating radiofrequency magnetic field (AC component) with a superimposed static magnetic field (DC component). Three parameters define each condition: frequency (MHz), maximum AC intensity (nT), and DC intensity (μ T).

In both studies the frequency range of 0.9–7.2 MHz was chosen based on the known hyperfine resonance frequencies of iron-sulfur (Fe-S) clusters in the mitochondrial electron transport chain, particularly Complex I. These clusters exhibit hyperfine interactions with ^{14}N nuclei in the protein scaffold at frequencies between approximately 2.15 and 3.85 MHz, and the broader range tested was designed to scan around these resonances to investigate frequency-dependent biological effects via the radical pair mechanism [7], [29]. Condition C16 is the only one for which the waveform is explicitly described (a 4 MHz carrier frequency-modulated at 50 kHz); the waveform type of the remaining conditions is not reported in the original studies.

AC intensities range from 1 nT to 1500 nT were selected to investigate biological effects at levels well below current safety standards. The 20 nT AC intensity was chosen to explore whether biologically significant effects could occur at intensities of orders of magnitude below the regulatory limits [7]. Paper 29 extended this approach by varying the AC intensity from 1 nT to 1.5 μ T.

DC intensities range from 10 to 300 μ T. The baseline value of 45 μ T was chosen to approximate the geomagnetic field in Boulder, Colorado, where the experiments were conducted [7]. Paper 29 then varied the DC component across 10, 45, 90, and 300 μ T to study how the Zeeman effect, which governs the energy gap between spin states, modulates the biological response at different strengths of the background field [29].

C_ID	Paper	Wave Type	Freq. (MHz)	Max AC intensity (nT)	DC intensity (μ T)
C1	29		0.9	100	45
C2	29		1.8	1	10
C3	29		1.8	1	45
C4	29		1.8	1	300
C5	29		1.8	100	90
C6	29		1.8	1500	10
C7	29		1.8	1500	45
C8	29		1.8	1500	300
C9	7		2	20	45
C10	7		2.5	20	45
C11	7		3	20	45
C12	7		3.5	20	45
C13	29		3.6	1	45
C14	29		3.6	1500	45
C15	7, 29		4	20	45
C16	29	Frequency modulated at 50 kHz	4	20	45
C17	7		4.5	20	45
C18	7		5	20	45
C19	29		7.2	1	45
C20	29		7.2	1500	45

Table 18: RF-MF + SMF exposure conditions lookup table.

RF-EMF (Table 19). This subgroup comprises three conditions from two studies. Conditions C21 and C22 (Paper 37) operate at 13.56 MHz. This frequency was selected because it is the operating frequency of clinical capacitive hyperthermia systems (e.g., EHY-2000+), which allows direct translational comparison with established clinical devices [37]. C21 delivers a continuous sinusoidal waveform; C22 applies amplitude modulation with a pink noise spectrum ($100/f$, $f \geq 100$ Hz) at a modulation index of 50%, enabling a direct comparison between the two waveform types. Researchers hypothesize that cancer cell membranes possess natural mechanical vibration frequencies. Using this type of modulation (covering frequencies from 100 Hz and above), they aim to resonate with the cell membrane to induce mechanical damage. Absolute field intensity values are not reported for these conditions; the adjusted input power was 10 W [37].

Condition C23 uses a different approach. The device used (BENE 110; ASMED, Italy) operates through radioelectric asymmetric conveyance (REAC), a proprietary technology designed to restore endogenous bioelectric activity using weak radioelectric fields. The carrier frequency and field intensity values are not reported in the original study [30], which limits direct comparison with the remaining RF-EMF conditions.

C_ID	Paper	Freq. (MHz)	Wave Type
C21	37	13.56	Continuous sine
C22	37	13.56	Amplitude-modulated sine
C23	30		

Table 19: RF-EMF exposure conditions lookup table.

LTE (Table 20). The six LTE conditions replicate the signal profile of 4G mobile telecommunications using WCDMA modulation at 1700 and 1760 MHz. These frequencies were selected because they represent the most widely used bands in 4G-LTE mobile communications, making their biological characterization a direct public health concern [31], [35]. The studies in this subgroup were not designed with cancer mono-therapy in mind: the primary objective was to assess whether LTE-frequency fields cause cellular damage, and both cancer and healthy cell lines were included to evaluate potential adverse effects on normal tissue. The results are reported here for completeness.

Field intensity is expressed as specific absorption rate (SAR, W/kg), the standard dosimetric quantity for radiofrequency fields absorbed by biological tissue. The tested values (0.4, 1, 2, 4, and 8 W/kg) were chosen with reference to regulatory limits: the general public reference level of the ICNIRP is 2 W/kg averaged over 10 g of tissue, and the US FCC limit is 1.6 W/kg. Both studies included values below and above these thresholds to map the dose-response relationship across the relevant regulatory boundary [31], [35].

C.ID	Paper	Freq. (MHz)	SAR (W/kg)
C24	31	1700	1
C25	31	1700	2
C26	35	1760	0.4
C27	35	1760	1
C28	35	1760	4
C29	35	1760	8

Table 20: LTE exposure conditions lookup table.

MMW and THz (Table 21). This subgroup covers the highest frequencies in the entire dataset, from 35.2 GHz to 140 GHz. At these frequencies, the field intensity is expressed as power density (mW/cm²).

The frequency of 35.2 GHz (C30, Paper 32) corresponds to the operating frequency of the KFA-100A, a millimetre wave therapeutic instrument used clinically in China. It was selected because MMW penetration depth is limited to the epidermis and dermis (up to approximately 1 mm), which makes it directly relevant for melanoma cells, which originate from skin structures [32]. The W-band range of 75–110 GHz (C31) and 75–105 GHz (C32) was chosen because the torsional eigenmodes of biomolecules such as carbohydrates, proteins, and DNA are estimated to resonate around 70 GHz; sweeping across the W-band provides an integrated effect covering multiple resonant frequencies without requiring precise tuning to any single discrete frequency [11], [12].

The terahertz frequency of 140 GHz (C33) corresponds to the upper limit of THz exposure defined by the IEEE standard for human radiofrequency safety, where intensities between 1 and 10 mW/cm² are considered safe [33].

C.ID	Paper	Field Type	Wave Type	Freq. (GHz)	Power density (mW/cm ²)
C30	32	MMW	Continuous sine	35.2	0.16
C31	11	MMW	Sine regime (frequency skipping across W-band 75-110 GHz) - CW	75-110	0.2
C32	12	MMW	Sine regime (frequency skipping across W-band 75-105 GHz) - CW	75-105	0.2
C33	33	THz radiation	Continuous	140	7.4

Table 21: MMW and THz exposure conditions lookup table.

The power density values are low: 0.16 mW/cm² for C30, 0.2 mW/cm² for C31 and C32 (below the FCC maximum permissible exposure of 1 mW/cm²) and 7.4 mW/cm² for C33, within the IEEE

safety window [11], [12], [32], [33].

8.3 Cell fate outcomes

Since no experimental conditions within the RF-EMF macro-category assessed necrosis, ferroptosis, or motility, these outcomes are not discussed in this section.

8.3.1 Viability

RF-MF + SMF. As shown in Figure 34, viability was assessed in HT-1080 cells at seven frequencies with a continuous 96-hour exposure. The results show a clear non-linear frequency-dependent pattern. C9 (2 MHz, 20 nT, 45 μ T) produced a small but significant decrease to 96% of control, while C10 (2.5 MHz, 20 nT, 45 μ T) caused a significant increase to 102.3%, suggesting a stimulatory effect at this frequency. C11 (3 MHz, 20 nT, 45 μ T) did not show a significant change (104%) and C12 (3.5 MHz, 20 nT, 45 μ T) produced the strongest inhibition in the subgroup at 91%, correlating with increased oxidative stress and GSH depletion. C15 (4 MHz, 20 nT, 45 μ T) and C17 (4.5 MHz, 20 nT, 45 μ T) yielded significant increases to 103% and 102.5% respectively, while C18 (5 MHz, 20 nT, 45 μ T) returned to a slight but significant decrease (97%). The alternation between inhibitory and stimulatory effects across closely spaced frequencies is a distinctive feature of this subgroup, consistent with a resonance-based mechanism rather than a monotonic dose-response.

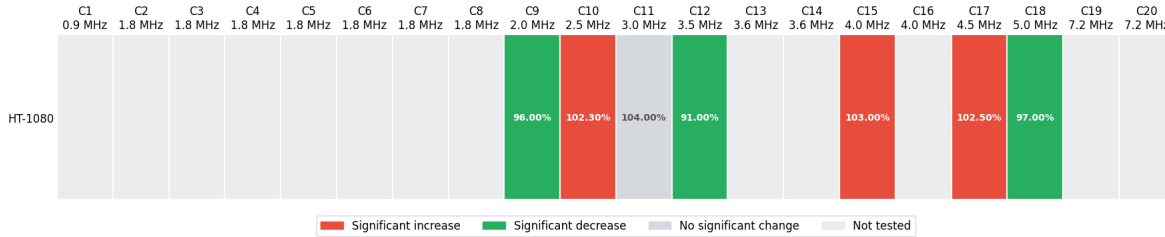


Figure 34: Residual viability (% of control) of HT-1080 fibrosarcoma cells across RF-MF + SMF exposure conditions (C1–C20).

RF-EMF. Viability was assessed in MCF-7 at condition C23 at three time points. A progressive reduction was observed: 95% of control at 96 h, 96.7% at 168 h, and 84.53% at 336 h, indicating that the effect of REAC exposure accumulates with the prolonged duration of treatment.

LTE. Viability was assessed after 72 h of continuous exposure in Huh7, HeLa, B103, and ASC. Conditions C26–C29 (1760 MHz, 0.4–8 W/kg) did not produce a significant change in viability in any of the four cell lines at any SAR value. This contrasts to the proliferation results of Paper 31, which used a similar but technically distinct exposure system at 1700 MHz and found marked anti-proliferative effects; the authors of Paper 35 attribute this discrepancy to differences between the two exposure systems rather than a true absence of biological effect.

MMW and THz. Under condition C30 (35.2 GHz, 0.16 mW/cm²), viability of A375 decreased in a time-dependent manner: 76% at 15 min, 73% at 30 min, 62% at 60 min, and 36.7% at 90 min of exposure.

THz irradiation at condition C33 (140 GHz, 7.4 mW/cm²) affected suspension cancer cell lines after 15 minutes of surface exposure: Ramos Burkitt lymphoma cells reached approximately 75% viability and Jurkat T-cell leukaemia cells similarly 75%, with bottom irradiation for 20 minutes yielding comparable results (73% in both). LNCaP prostate carcinoma cells were more sensitive, dropping to 63% after 15 minutes. MCF-7 breast cancer adherent cells were largely unaffected by surface irradiation (100%) but showed a marginal reduction to 90% under bottom irradiation. Healthy PBMCs showed an increased death rate compared to non-irradiated controls, though baseline cell loss under culture conditions limits direct quantitative comparison.

8.3.2 Proliferation

RF-MF + SMF. Proliferation was measured in HT-1080 cells at two 4 MHz conditions. C15 (4 MHz, 20 nT, 45 μ T) produced a significant increase to 108.5% of control, while C16 (4 MHz modulated at 50 kHz, 20 nT, 45 μ T) reached 133%. The greater effect of the frequency-modulated signal at the same carrier frequency and intensity suggests that the modulation envelope itself plays a biologically active role beyond the carrier wave alone.

LTE. In Paper 31, proliferation was the primary outcome assessed after 72 h of continuous exposure. At C24 (1700 MHz, 1 W/kg), all cancer cell lines showed significant reductions: Huh7 to 79%, Hep3B to 64%, HeLa to 66% and SH-SY5Y to 64%. At C25 (1700 MHz, 2 W/kg), the effects were considerably stronger: Huh7 to 27%, Hep3B to 36%, HeLa to 28%, and SH-SY5Y to 12%. Among healthy cell lines, IMR-90 reached 70% at C24 and 51% at C25, and ASC reached 88% at C24 and 46% at C25, making them the least sensitive cell type overall. NAC pre-treatment rescued proliferation by 49% in Huh7 and 42% in ASC, confirming ROS as a primary functional mediator.

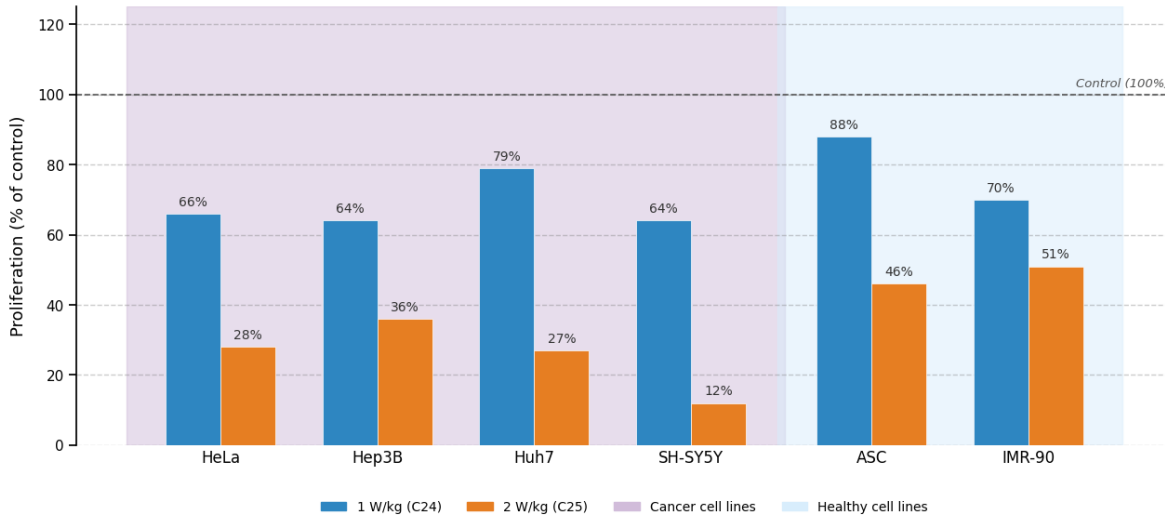


Figure 35: Residual proliferation (% of control) in cancer and healthy cell lines exposed to LTE signals at 1700 MHz under conditions C24 (1 W/kg) and C25 (2 W/kg) for 72 h [31].

In Paper 35, proliferation was assessed in Huh7 and ASC cells at C26 (1760 MHz, 0.4 W/kg) and C28 (1760 MHz, 4 W/kg). No significant change was detected in either cell line at either SAR value, in direct contrast to the results of Paper 31.

RF-EMF, MMW and THz. Proliferation was not assessed.

8.3.3 Apoptosis

RF-MF + SMF. Apoptosis was not assessed.

RF-EMF. In Paper 37, apoptosis was the primary outcome, assessed in HT29, SW480, and LoVo. Under C21 (13.56 MHz, continuous sine), SW480 cells showed a significant increase to 145% of the control group and LoVo to 130%, but HT29 reached only 108% (n.s.). Under C22 (13.56 MHz, amplitude-modulated), HT29 showed a significant increase to 134%, while SW480 remained at 145% and LoVo at 130%, unchanged relative to C21. The selective additional effect of amplitude modulation in HT29 but not in the other two lines is interpreted in terms of cell membrane mechanical resonance: HT29 has the highest membrane stiffness and an estimated resonance frequency above 100 Hz, which falls within the applied AM spectrum ($100/f$, $f \geq 100$ Hz); instead, LoVo has the lowest stiffness and a resonance frequency below this threshold.

LTE. In Paper 31, cleaved PARP was not detected in Huh7 or ASC cells at C24 or C25, confirming that the anti-proliferative effects observed in this study are not mediated by apoptosis. In Paper 35, at C29 (1760 MHz, 8 W/kg) apoptotic population was not detected in HeLa, ASC, or B103.

MMW and THz. Under condition C30 (35.2 GHz, 0.16 mW/cm²), A375 melanoma cells showed an increase in apoptosis that was dependent on time: 214.5%, 224.6%, 288.7%, and 330.6% of sham at 15, 30, 60, and 90 minutes respectively. Western blot confirmed up-regulation of both cleaved caspase-8 and cleaved caspase-3, establishing a caspase-dependent pathway.

8.3.4 Senescence

RF-MF + SMF. Senescence was not assessed.

RF-EMF. Senescence was assessed in MCF-7 cells at C23 at three time points. A significant increase was detected at 96 h (116.6% of control), no significant change at 168 h (111%) and a marked significant increase at 336 h (133%), coinciding with the strongest reduction in viability and Bcl-2 down-regulation. Senescence appears to become the dominant cellular response at the longest exposure duration tested.

LTE. At C24 (1700 MHz, 1 W/kg), neither Huh7 nor ASC showed significant changes. At C25 (1700 MHz, 2 W/kg), Huh7 cells showed a dramatic increase to 142% of the control, confirmed by the molecular markers p21, p53-Ser15 phosphorylation, and decreased pRb-Ser780, establishing senescence as the primary fate under this SAR. ASC cells showed only an increase of 103%, indicating that healthy stem cells are substantially less susceptible to LTE-induced senescence than cancer cells.

MMW and THz. Under condition C32 (75–105 GHz), senescence in H1299 cells was assessed during Days 1, 7, and 14. Under 2-min exposure, senescence was already elevated on Day 1 (182.8%), not significant on Day 7 (119.1%) and significantly increased on Day 14 (159.7%). Under 4-min exposure, the trajectory was 198.4% on Day 1, 176.5% on Day 7 and 430.5% on Day 14.

Healthy MCF-10A cells did not show a significant increase in senescence at any exposure duration or observation time point, providing direct evidence of cancer-selective senescence induction.

8.4 Mechanistic outcomes

8.4.1 ROS production

RF-MF + SMF. Mitochondrial superoxide was measured in HT-1080 cells using MitoSOX Red at seven frequencies. A clear inverse relationship with viability emerged at several conditions: C9 (2 MHz, 20 nT, 45 μ T) showed a significant superoxide increase to 127% while viability dropped to 96%, and C11 (3 MHz) reached 121% superoxide despite no significant viability change, suggesting oxidative stress without sufficient magnitude to reduce survival. Conversely, C15 (4 MHz) and C17 (4.5 MHz) showed significant superoxide reductions to 84% and 85% respectively, matching their stimulatory viability effects. At C12 (3.5 MHz), superoxide was not significantly changed (109%) despite the strongest viability inhibition, suggesting that at this frequency the cytotoxic mechanism may involve GSH depletion rather than mitochondrial superoxide directly.

Hydrogen peroxide was measured in HT-1080 cells. H₂O₂ was consistently reduced across most conditions: C2 (1.8 MHz, 1.5 μ T, 10 μ T: 87%), C6 (1.8 MHz, 1 nT, 45 μ T: 90%), C13 (3.6 MHz, 1.5 μ T, 45 μ T: 93%), C14 (3.6 MHz, 1 nT, 45 μ T: 90%), C19 (7.2 MHz, 1.5 μ T, 45 μ T: 91%), and C1 (0.9 MHz, 0.1 μ T, 45 μ T: 91%) and C5 (1.8 MHz, 0.1 μ T, 90 μ T: 94%). Peroxynitrite was measured at C15 and C16 (4 MHz unmodulated and modulated), showing its lowest levels at C15 (92% of the control). The authors interpret the reduction in H₂O₂ as evidence of RF-induced redox modulation of Fe-S clusters in the electron transport chain.

LTE. In Paper 31, intracellular and mitochondrial ROS were assessed by carboxy-H2DCFDA and MitoSOX Red staining in Huh7 and ASC cells. Positive staining was detected at C25 (1700 MHz, 2 W/kg) in both cell lines, with Huh7 showing brighter staining than ASCs, consistent with the greater anti-proliferative effect in cancer cells. At C24 (1700 MHz, 1 W/kg), only mitochondrial ROS

was detected in Huh7 cells at lower intensity. NAC rescue confirmed ROS as a functional mediator. No quantitative values were reported.

RF-EMF, MMW and THz. ROS production was not assessed.

8.4.2 Antioxidant response

RF-MF + SMF. Three antioxidant markers were measured in HT-1080 cells: total SOD activity, GSH/GSSG ratio, and H_2O_2 . The responses were frequency-dependent and sometimes internally inconsistent. At C15 (4 MHz), SOD increased to 137% and GSH to 160% with H_2O_2 unchanged, suggesting active antioxidant compensation under stimulatory conditions. At C12 (3.5 MHz), GSH was significantly depleted to 79% and H_2O_2 slightly reduced (96%), indicating oxidative stress exceeding antioxidant capacity at this inhibitory frequency. At C10 (2.5 MHz), GSH dropped markedly to 41% despite a stimulatory viability effect, a dissociation the authors do not directly address. At C18 (5 MHz), SOD (118%), GSH (119%), and H_2O_2 (111%) all increased together, pointing to an oxidative stress response without a corresponding loss in viability.

RF-EMF, LTE, MMW and THz. Antioxidant response was not assessed in RF-EMF, LTE and MMW + THz subgroups.

8.4.3 Membrane potential

RF-MF + SMF. Cell membrane potential was measured in HT-1080 cells. At 1.8 MHz, C3 (1.5 μT , 45 μT) there was no significant change (101%), while C2 (1.5 μT , 10 μT) showed a significant increase to 130% and C4 (1.5 μT , 300 μT) returned to baseline (98%). With the lower AC intensity in C6 (1 nT, 45 μT) the increase was 118%, and in C5 (1 nT, 10 μT) the largest effect in the subgroup was recorded (140%). Increasing DC to 300 μT at C8 (1 nT, 300 μT) abolished the effect (101%). At 3.6 MHz, C13 (1.5 μT , 45 μT) increased to 118% while C14 (1 nT, 45 μT) showed a significant decrease to 83%. At 7.2 MHz, both C19 (1.5 μT) and C20 (1 nT) showed no significant change or slight decreases.

RF-EMF, LTE, MMW and THz. Membrane potential was not measured.

8.4.4 Calcium signaling

RF-MF + SMF. Mitochondrial calcium was measured in HT-1080 cells under two conditions: C1 (0.9 MHz, 0.1 μT , 45 μT) and C5 (1.8 MHz, 0.1 μT , 90 μT) both showed significant decreases to approximately 87% and 85% of control respectively. At all other conditions of the same study (1 nT or 1.5 μT AC, 45 μT DC), mitochondrial calcium variation was less than 3% and was not significant. The authors note that C1 and C5 satisfy the Zeeman equation when both the RF frequency and the DC field are doubled simultaneously (0.9 MHz/45 μT and 1.8 MHz/90 μT), producing equivalent biological responses, a pattern they interpret as evidence for the radical pair mechanism.

RF-EMF, LTE, MMW and THz. Calcium signaling was not assessed.

8.4.5 Cellular morphology and cytoskeleton remodeling

RF-MF + SMF. Cellular morphology and cytoskeleton remodeling were not investigated.

RF-EMF. At C23, MCF-7 cells were assessed at all three time points (96, 168, 336 h). A progressive reduction in cell dimensions was observed at each time point, suggesting cumulative cell shrinkage as a consistent morphological correlate of REAC exposure.

MMW and THz. For conditions C31 and C32, morphological analysis of H1299 revealed progressive and exposure duration dependent changes over 14 days. Shorter exposures (2 min) produced delayed effects, with significant morphological alterations appearing from Day 7; longer exposures (4 min) caused immediate apoptotic shrinkage at Day 1 that evolved into a mixed apoptotic and senescent phenotype by Day 14. The 10-min exposure caused complete cell death within two days. Healthy MCF-10A breast epithelial cells did not show morphological changes under any condition in either study.

For C33 (140 GHz, 7.4 mW/cm²), dead and dying cells were identified in all mononuclear blood cell subtypes (monocytes, B-cells, T-cells) in the PBMC population.

8.4.6 Cell cycle analysis

RF-MF + SMF. Cell cycle was not assessed.

RF-EMF. Cell cycle was not directly measured but was indirectly investigated through RT-qPCR of p19ARF, p21, and p53 in MCF-7 cells at C23. No significant modulation was found at 96 h. p19ARF was significantly up-regulated at 168 h, suggesting an early onset of cycle arrest. By 336 h, all three inhibitors were significantly up-regulated, consistent with strong cell cycle arrest and senescence induction at the longest duration of exposure.

LTE. At C24 (1700 MHz, 1 W/kg), Huh7 showed a slight increase in G1 and a decrease in S phase without sub-G1 accumulation, consistent with early arrest in G1/S. At C25 (1700 MHz, 2 W/kg), G1 increased, S decreased and G2/M also decreased, confirmed by up-regulation of p21 and p53-Ser15 phosphorylation and by down-regulation of pRb-Ser780. ASC cells at C24 showed no change; at C25, a slight increase in G1, a decrease in S, and a decrease in G2/M were detected without sub-G1 accumulation, indicating a mild arrest that does not culminate in cell death.

Cell cycle was also assessed in HeLa, ASC, and B103 cells at C29 (1760 MHz, 8 W/kg). No significant changes in phase distribution were detected in any cell line, consistent with the absence of viability and apoptosis effects.

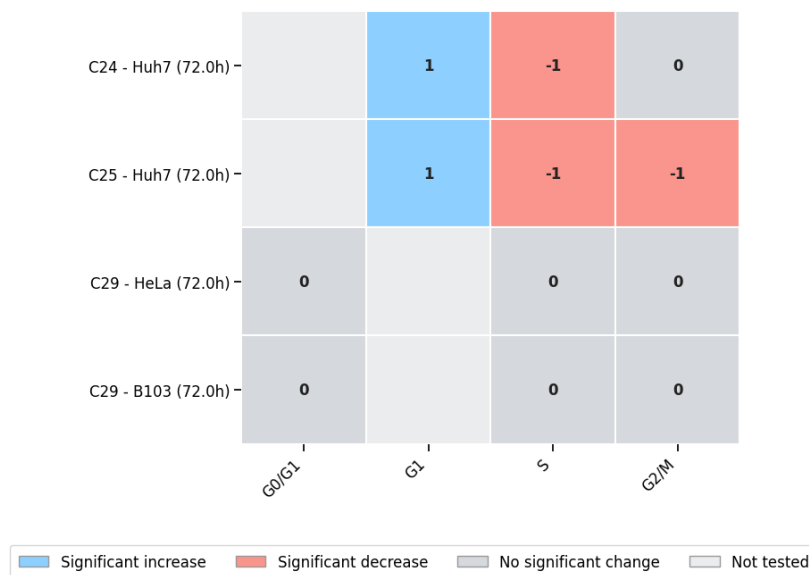


Figure 36: Cell cycle phase heatmap for cancer cell lines across experimental conditions in LTE subgroup. Each cell shows the direction of change for the corresponding phase: azure indicates a significant increase, soft red a significant decrease, gray no significant change. Phases are ordered following the biological sequence of the cell cycle.

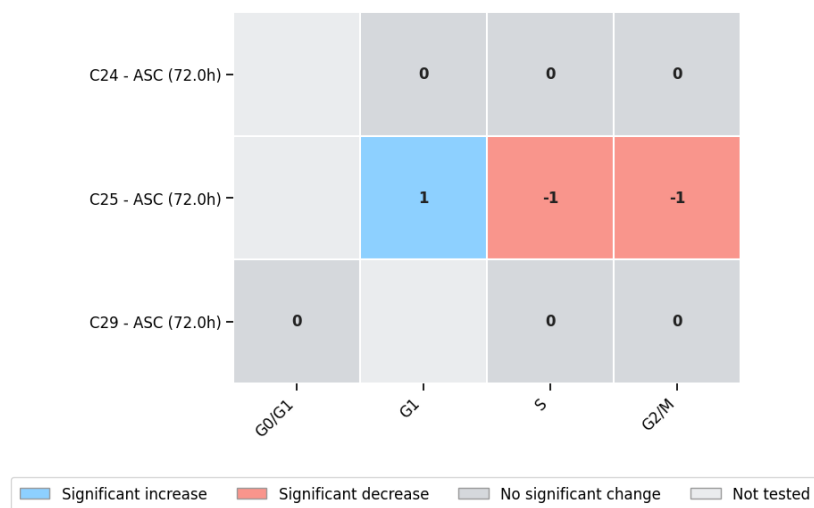


Figure 37: Cell cycle phase heatmap for healthy cell lines across experimental conditions in LTE subgroup. Each cell shows the direction of change for the corresponding phase: azure indicates a significant increase, soft red a significant decrease, gray no significant change. Phases are ordered following the biological sequence of the cell cycle.

MMW and THz. Cell cycle phases were not measured directly here. In Papers 11 and 12, nuclear morphology subgroups were used to infer approximate cell cycle phases, and the accumulation of enlarged polyploid forms was interpreted as indirect evidence of the arrest in G1/G2.

8.5 Molecular outcomes

8.5.1 Genomic analysis

RF-MF + SMF, LTE, MMW and THz Genomic analysis was not performed.

RF-EMF. Gene expression was assessed in MCF-7 cells at C23 at three time points for nine targets: stemness markers Oct-4, Sox2, NANOG, and c-Myc; cell cycle and senescence regulators p19ARF, p21, and p53; and apoptosis and stress response markers Bcl-2 and HSP70. Stemness markers and HSP70 did not show significant changes at any time. p19ARF, p21, and p53 were progressively up-regulated, reaching significance for all three at 336 h. Bcl-2 showed a transient up-regulation at 168 h followed by a significant down-regulation at 336 h, suggesting a late shift from anti-apoptotic to pro-apoptotic transcriptional programming.

8.5.2 Proteomic analysis

RF-EMF. Protein expression was assessed in MCF-7 cells at C23 for DKK1, SFRP1, Beclin-1, and LC3-I at all three time points. DKK1, a promoter of tumor progression, was significantly reduced at all time points (most strongly at 336 h), representing the most consistent molecular response to REAC exposure. SFRP1, a Wnt pathway suppressor, was significantly increased only at 336 h. Beclin-1 was significantly increased at 96 h and gradually declined thereafter, while LC3-I increased at 336 h. The combined Beclin-1 and LC3-I trajectory suggests progressive autophagy induction, with Beclin-1 initiating the process early and LC3-I marking later autophagosome formation.

LTE. At C25 (1700 MHz, 2 W/kg) in Huh7 cells, p21 was up-regulated, p53-Ser15 phosphorylation increased, and pRb-Ser780 significantly decreased, while total p53, total Rb, γ -H2AX, and cleaved PARP were unchanged. This profile leads to p53-p21-driven G1/S arrest and senescence without DNA damage or apoptosis. In ASC cells at C24 (1 W/kg) and C25 (2 W/kg), γ -H2AX and cleaved PARP were also absent. p-ERK1/2, ERK1/2, and γ -H2AX were assessed in HeLa, ASC, and B103 cells at C29 (1760 MHz, 8 W/kg) and showed no significant changes.

MMW and THz. In C30 (35.2 GHz), western blot in A375 melanoma cells quantified cleaved caspase-8 and cleaved caspase-3 at all four exposure durations (15, 30, 60, 90 min). Both proteins were significantly up-regulated at all time points, and functional inhibition of caspase-3 with AC-DEVD-fmk partially rescued viability at 90 min, providing direct evidence of caspase-dependent apoptosis via the extrinsic pathway.

8.6 Mechanisms of action and connections

Given the extreme heterogeneity of the RF-EMF macro-category, no single mechanistic model can characterize all four subgroups. Each is addressed separately in the following sections, followed by a set of cross-cutting observations.

RF-MF + SMF

The mechanistic framework proposed for this subgroup is based on the radical pair mechanism, which suggests that weak oscillating magnetic fields can influence the spin dynamics of radical pairs generated during biochemical reactions, altering the reaction in a frequency and intensity dependent manner [7], [29]. The primary target proposed is the iron-sulfur (Fe-S) cluster system of the mitochondrial Complex I, whose hyperfine interactions with ^{14}N nuclei in the surrounding protein scaffold are predicted to resonate in the 2–4 MHz range [7], [29]. The non-linear and frequency dependent pattern of viability and ROS observed in HT-1080 cells is consistent with a resonance-based interaction rather than a simple thermal or monotonic dose response [7].

RF-EMF

Cancer cell membranes are mechanically softer than healthy cells, a difference that corresponds to lower natural resonance frequencies. When a 13.56 MHz carrier is amplitude-modulated with a pink noise spectrum that covers frequencies above 100 Hz, the modulation envelope overlaps with the resonance frequencies of stiffer cancer membranes, inducing mechanical disturbance and triggering apoptosis through a caspase-dependent pathway [37]. The selective additional apoptotic effect of AM observed specifically in the cell line whose estimated resonance frequency falls within the AM spectrum provides indirect mechanistic support for this model [37].

A fundamentally different mechanism is proposed for REAC-based field exposure. Progressive senescence, down-regulation of the tumor promoting factor DKK1, up-regulation of the Wnt suppressor SFRP1, autophagy induction, and late down-regulation of Bcl-2 suggest a multi-path response involving Wnt signaling modulation and eventual apoptotic commitment at prolonged durations of exposure [30]. However, the absence of reported exposure parameters and the proprietary nature of the device make it impossible to identify the upstream biophysical trigger responsible for these downstream events.

LTE

ROS generation was confirmed as a causal mediator rather than a mere correlate: pre-treatment with ROS scavenger NAC partially rescued cell numbers in both cancer and healthy cells, with mitochondria identified as the primary source of reactive species [31]. The path p53–p21 drives the arrest of G1/S and senescence, with DNA damage markers and apoptosis markers consistently absent, excluding these pathways as primary endpoints. The proposed mechanistic chain is: field exposure, mitochondrial ROS, activation of p53–p21, dephosphorylation of Rb, arrest of G1/S, senescence [31].

MMW and THz

At 35.2 GHz, a caspase-8/caspase-3 apoptotic cascade was biochemically confirmed in melanoma cells, and partial rescue with a caspase-3 inhibitor established its role [32]. The trigger that initiates caspase-8 activation is unknown.

At W-band frequencies (75–110 GHz), exposure is proposed to affect the membrane fluidity of cancer cells, a mechanism analogous to the membrane resonance model described above but operating through direct electromagnetic interaction rather than mechanical resonance [11], [12]. The persistence

of morphological effects for up to 14 days following a single brief exposure at very low delivered energy implies a self-sustaining downstream cascade rather than a continuous field effect [11], [12].

Cross-subgroup observations

Several themes recur across subgroups despite their mechanistic diversity. Mitochondria emerge as a common functional target in the MHz and GHz ranges, with ROS as the key downstream mediator [7], [31]. The axis p53–p21–Rb appears as a shared convergence point for cell cycle arrest in two independent subgroups [30], [31]. Cell membrane properties, such as stiffness, fluidity, and resonance frequency, are proposed as primary determinants of selective sensitivity at both RF and MMW frequencies [11], [12], [37], pointing to the membrane as a potential frequency-independent target.

8.7 Limitations

The most fundamental limitation of the RF-EMF macro-category is its internal heterogeneity. Within each subgroup sample sizes are small. The MHz-range subgroup tested only one cancer cell line with no healthy cell controls, making selectivity impossible to evaluate [7], [29]. The 13.56 MHz subgroup contains only three conditions from two studies with different technologies and no shared endpoints [30], [37]. The MMW + THz subgroup is based on a single cancer cell line per study. The LTE subgroup is affected by a direct contradiction between two studies that tested overlapping cell lines and similar exposure systems but obtained opposite results on proliferation and viability [31], [35]. Without a controlled replication study, it is not possible to determine whether LTE signals at these SAR values affect cancer cell biology.

The mechanistic characterization is fragmentary.

Where healthy cells were included, the selectivity picture is mixed. These inconsistencies cannot be understood with the current data. No study in this macro-category simultaneously exposed matched cancer and healthy cells under sham-controlled conditions.

9 Static Magnetic Fields

SMFs are magnetic fields characterized by a constant magnitude and direction that do not vary over time. Unlike alternating or pulsed fields, SMFs deliver a continuous, unidirectional magnetic force on the biological system throughout the entire exposure period.

The SMFs are classified by intensity into four ranges: weak (< 1 mT), moderate (1 mT – 1 T), strong (1 – 5 T) and ultrastrong (> 5 T) [44]. Conditions analyzed in the subgroup fall within the weak to moderate range.

9.1 Dataset overview and cell line distribution

The SMF subgroup includes data from two independent studies, each using a different cell line (as shown in Table 22), with a total of only four experimental conditions.

Cell line	Type	Cancer category	Origin	Exp. cond.
MDA-MB-231	Triple-negative breast cancer (TNBC)	Breast	Human	2
HT-1080	Fibrosarcoma	Sarcoma	Human	2

Table 22: Cancer cell lines used in the SMF macro-category: type, cancer category, origin, and number of experimental conditions.

9.2 Exposure parameters

Given the static nature of these fields, the parameter that characterizes the SMF exposure is the DC intensity, expressed here in microTesla (μ T). It represents the magnitude of the applied magnetic flux density and is constant throughout the entire exposure period. The exposure conditions are listed in Table 23.

C_ID	Paper	Field type	DC intensity (μ T)
C1	18	SMF	-45
C2	44	SMF	15000

Table 23: SMF exposure conditions lookup table.

9.2.1 Intensity rationales

In study [18], a field of -45μ T was applied to specifically mimic the magnitude of the Earth’s geomagnetic field but with a reversed polarity. This was done to test the hypothesis that fibrosarcoma cells possess a permanent magnetic dipole moment, evaluating whether reversing the field direction would influence cell growth, membrane potential, or mitochondrial calcium levels.

The study [44] explored higher intensities. 5, 10 and 15 mT were initially selected based on previous literature and the researchers found that 5 and 10 mT did not have a significant impact on cell viability for any of the exposure durations investigated (5, 10, 20, 30, 60 and 90 minutes). As a result, these lower intensities were discarded and the study focused exclusively on the 15 mT condition, which was the only one to produce measurable biological effects. These data suggest the presence of an intensity threshold between 10 mT and 15 mT below which this static field does not produce biological effects on the viability of MDA-MB-231 cells.

9.3 Cell fate outcomes

Table 24 summarizes the cell fate outcomes investigated for this subgroup. The following analysis focuses only on viability and proliferation data, as other outcomes were not investigated by the studies.

Mechanism	Number of cancer exp. conditions
Viability	2
Proliferation	2
Apoptosis	0
Necrosis	0
Ferroptosis	0
Senescence	0
Motility	0

Table 24: Number of SMF macro-category experimental conditions assessing each cell fate outcome. No data were available for healthy cell lines in this macro-category.

9.3.1 Viability

For condition C2 (15 mT), the effect on viability depends on the duration of exposure, as shown in Figure 38. After 60 minutes of field exposure, viability is significantly reduced to 87.03% compared to the control group; after 90 minutes, the reduction is more pronounced, with viability of 58.14% compared to the control. An increase of only 30 minutes in exposure duration at a constant intensity (15 mT) produces a more than proportional cytotoxic effect on cells, suggesting a non-linear response as a function of time within the 60-90 minute interval.

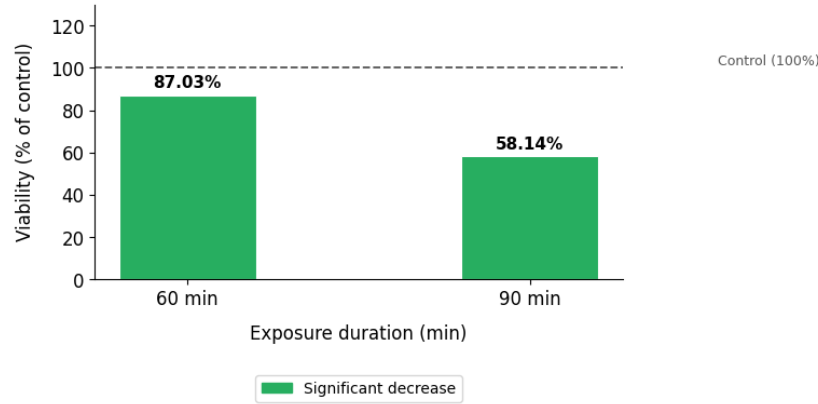


Figure 38: Viability of MDA-MB-231 cells exposed to condition C2 at two exposure durations (60 min and 90 min), expressed as percentage of the control group (dashed line at 100%).

9.3.2 Proliferation

Condition C1 highlighted the biphasic nature of the response over time. Figure 39 shows that after 48 h of exposure to the reversed field, HT-1080 cell proliferation is significantly inhibited (85.48% compared to the control group). However, after 96 h of exposure to the reversed field, the effect reverses and proliferation increases relative to the control group (123.05%). This behavior suggests that fibrosarcoma cells develop an adaptive response to the perturbation of the geomagnetic field during prolonged exposure, transitioning from an initial phase of inhibition to a subsequent phase of proliferation stimulation. This reversal can be explained by mechanisms of cellular adaptation, homeostatic regulation, or desensitization of field-sensitive receptors, as the authors suggest.

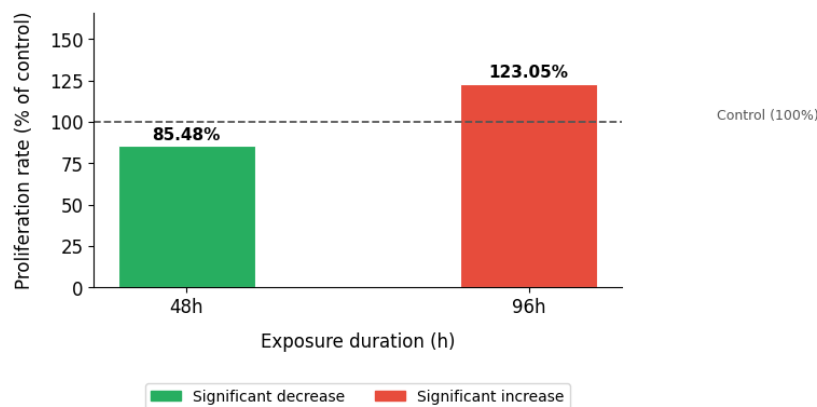


Figure 39: Proliferation rate of HT-1080 fibrosarcoma cells exposed to a reversed static magnetic field ($-45 \mu\text{T}$) at two time points (48 h and 96 h), expressed as percentage of the control group (dashed line at 100%). Green bar: proliferation decreased relative to control; red bar: proliferation increased relative to control.

9.4 Mechanistic outcomes

Mechanistic outcome	Number of cancer exp. conditions
ROS production	0
Antioxidant response	1
Membrane potential	1
Calcium signaling	1
Cellular morphology and cytoskeleton remodeling	1
Metabolic reprogramming	0
Differentiation	0
Cell cycle analysis	1

Table 25: Number of SMF macro-category experimental conditions assessing each mechanistic outcome. No data were available for healthy cell lines.

9.4.1 Antioxidant response

In condition C2 and an exposure time of 60 min, the total antioxidant capacity was significantly reduced to 74.87% compared to the control group. This reduction is not accompanied by a direct measurement of ROS levels, which are mentioned by the authors as a probable mechanism involved, without measuring it. The reduction in TAC suggests that exposure to condition C2 induces a state of oxidative stress in MDA-MB-231 cells, consuming cellular antioxidant reserves.

9.4.2 Membrane potential

Plasma membrane potential showed a significant reduction in condition C1 and an exposure time of 96 h, reaching 91.81% compared to the control group. This depolarization corresponds to an increase in proliferation, as described above. The authors suggest that depolarization of the cytoplasmic membrane may reflect a reorganization of cellular ionic balance as part of the adaptive response described above; however, this interpretation is speculative given the absence of direct measurements of ion channels.

9.4.3 Calcium signaling

In condition C1 and an exposure time of 96 h, mitochondrial calcium showed an increase of only 4.54% compared to the control group, a result that was not statistically significant.

9.4.4 Cellular morphology and cytoskeleton remodeling

Morphological analysis was conducted 24 h after 60-minute exposure to condition C2. Significant morphological alterations were identified in the exposed cells: nuclear fragmentation, chromatin condensation, plasma membrane blebbing, cytoplasmic protrusions, cytoplasmic vacuolations, and reduction of cell volume. These morphological features are typically associated with the apoptotic process, which the authors mention without performing a direct assay. Apoptosis was therefore hypothesized based on morphological analyzes and on the cell cycle analyzes reported below. The authors also mention a possible involvement of the cytoskeleton: they attribute the observed morphological changes to a destabilization of the cytoskeletal network induced by field exposure.

9.4.5 Cell cycle analysis

Cell cycle analysis was conducted 24 h after 60-minutes of exposure to condition C2. Significant arrest was measured in the G0/G1 phases: the G0 phase increased from 0.0% to 4.3% relative to the control group and the G1 phase from 0.2% to 15.73% relative to the control group. The G2/M phase decreased from 47.87% to 30.13%. In contrast, phase S did not show a significant variation (49.8% vs 49.47% of the control group). These results indicate that MDA-MB-231 cells exposed to a 15 mT field accumulate in the G0/G1 phase and do not enter mitosis. The arrest of G0/G1 is consistent with the observed reduction in viability and with the morphological alterations suggestive of apoptosis; it also suggests that the static field at 15 mT may interfere with the mechanisms controlling the cell cycle, possibly through oxidative stress pathways, as evidenced by the reduction in TAC.

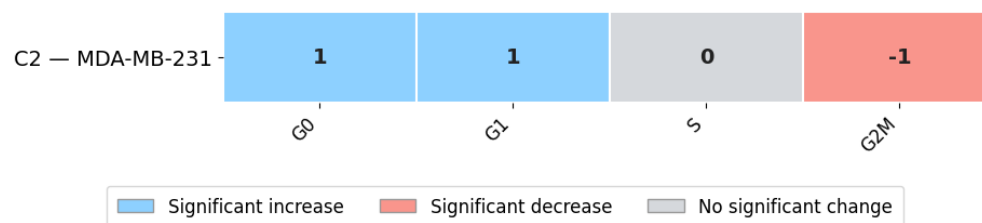


Figure 40: Cell cycle phase heatmap for cancer cell lines across experimental conditions. Each cell shows the direction of change for the corresponding phase: azure indicates a significant increase, soft red a significant decrease, grey no significant change. Phases are ordered following the biological sequence of the cell cycle.

9.5 Molecular outcomes

In the SMF macro-category, no analyses were conducted on genomic and proteomic results.

9.6 Mechanisms of action and connections

The two studies show that static magnetic fields can produce relevant biological effects on tumor cells even at very different intensities (C1, -45 μ T; C2, 15 mT). However, the biphasic nature of the response observed in the study [18] and the dependence on the duration of field exposure in the study [44] highlight that the effects of SMFs depend strongly on experimental parameters and are not easily generalizable.

9.7 Limitations

The most evident limitation is the small number of experimental conditions available, which alone is insufficient to draw any robust conclusion about the optimal field parameters for a possible therapeutic application of SMFs on cancer cells.

A second limitation concerns the absence of healthy cell line controls under all conditions. Without a paired non-tumor reference, it is not possible to assess whether the effects observed on cancer cells are selective or whether they would equally affect normal tissue.

In addition, the two studies investigated different results. The absence of a shared set of measured outcomes makes any direct comparison between the two studies impossible, and the picture that emerges from combining them is fragmented. Even within each individual study, the mechanistic outcomes were not measured in all experimental conditions but only for a selected subset. This means that comparisons are limited not only between studies, but also between conditions within the same study, further restricting the depth of the analysis that can be conducted on this subgroup.

10 Pulsed Magnetic Fields

PMFs are time-varying magnetic fields characterized by a non-continuous waveform, in which the field is delivered in discrete pulses rather than as a sustained sinusoidal or static exposure.

10.1 Dataset overview and cell line distribution

The analyzes involve only the study [22], which evaluates the effects on the viability of a rat glioma cell line (see Table 26) using two types of PMF: nsPMF and μ sPMF.

Cell line	Type	Cancer category	Origin	Exp. cond.
C6	Glioma	Brain & CNS	Rat	24

Table 26: Cancer cell line used in the PMF macro-category: type, cancer category, origin, and number of experimental conditions.

10.2 Exposure parameters

The parameters involved in PMF exposure are the following:

- **Waveform.** The temporal profile of each individual pulse. The waveforms used in this study are the square wave, in which the field rises abruptly to its peak value and is maintained for a defined duration before returning to zero, and the exponential wave, in which the field rises steeply and then decays progressively. The waveform determines the high-frequency spectral content of the pulse and, in particular, governs the rise rate dB/dt , the rate at which the magnetic field increases from zero to its maximum value.
- **Max AC intensity.** The maximum magnetic flux density reached during each pulse (mT). It represents the amplitude of the applied field and determines the magnitude of the induced electric field inside the biological system at the moment of maximum field strength.
- **Pulse duration.** The length of time during which the magnetic field remains active within a single pulse, expressed in nanoseconds (ns) or microseconds (μ s) depending on the temporal scale of the exposure. Together with the peak intensity, it determines the integral of the magnetic field action $\Sigma B\Delta t$ (mT·s), which quantifies the total magnetic field dose delivered per pulse.
- **Pulse repetition rate.** The number of pulses delivered per unit time (Hz). Determines how frequently the biological system is stimulated and, together with the duration of exposure, governs the cumulative dose received during the full treatment session.

C.ID	Paper	Field type	Waveform	Max AC intensity (mT)	Pulse duration	Pulse rep. rate (Hz)
C1	22	nsPMF	Square	0.24	600 ns	1
C2	22	nsPMF	Square	0.24	600 ns	5
C3	22	nsPMF	Square	0.24	600 ns	10
C4	22	nsPMF	Square	0.24	600 ns	20
C5	22	nsPMF	Square	0.353	600 ns	1
C6	22	nsPMF	Square	0.353	600 ns	5
C7	22	nsPMF	Square	0.353	600 ns	10
C8	22	nsPMF	Square	0.353	600 ns	20
C9	22	nsPMF	Square	0.598	600 ns	1
C10	22	nsPMF	Square	0.598	600 ns	5

Continued on next page

C.ID	Paper(s)	Field type	Waveform	Max AC intensity (mT)	Pulse duration	Pulse rep. rate (Hz)
C11	22	nsPMF	Square	0.598	600 ns	10
C12	22	nsPMF	Square	0.598	600 ns	20
C13	22	nsPMF	Square	0.787	600 ns	1
C14	22	nsPMF	Square	0.787	600 ns	5
C15	22	nsPMF	Square	0.787	600 ns	10
C16	22	nsPMF	Square	0.787	600 ns	20
C17	22	μ sPMF	Exponential	16.4	205 μ s	
C18	22	μ sPMF	Exponential	35.9	205 μ s	
C19	22	μ sPMF	Square	45.4	100 μ s	
C20	22	μ sPMF	Square	122	100 μ s	
C21	22	μ sPMF	Exponential	151	205 μ s	
C22	22	μ sPMF	Exponential	311	205 μ s	
C23	22	μ sPMF	Square	363	100 μ s	
C24	22	μ sPMF	Square	854	100 μ s	

Table 27: PMF exposure conditions lookup table.

10.3 Cell fate outcomes

Mechanism	Number of cancer exp. conditions
Viability	24
Proliferation	0
Apoptosis	0
Necrosis	0
Ferroptosis	0
Senescence	0
Motility	0

Table 28: Number of PMF macro-category experimental conditions assessing each cell fate outcome. No data were available for healthy cell lines in this macro-category.

Table 28 summarizes the cell fate outcomes investigated for this macro-category. The following analysis focuses exclusively on viability data, as other outcomes were not addressed in the study.

10.3.1 Viability

nsPMF. For the nsPMF subgroup, viability was tested immediately after 2 hours of cell exposure to the field. The analysis of experimental conditions using the nsPMF suggests a relationship between pulse repetition frequency and magnetic field intensity. From Figure 41 the pulse repetition frequency of 1 Hz, all four intensities tested in the study (C1, 0.24 mT; C5, 0.353 mT; C9, 0.598 mT; C22, 0.787 mT) led to a significant reduction in cell viability, with values ranging between 84.97% and 91.86% compared to the control group. The most effective experimental condition among nsPMF is C1 (1 Hz and 0.24 mT) (84.97% with respect to the control group), which confirms that the best parameters for inhibiting viability in C6 cells using nsPMF are those at low pulse repetition frequency and low intensity.

At higher frequencies, the situation changes. At the pulse repetition frequency of 5 Hz, only the experimental condition C2 (0.24 mT) reports a significant decrease (89.31%), while higher field intensities do not report significant changes, with condition C14 (0.787 mT) showing viability above the control group (100.7%). At 10 Hz, an inverse trend is observed: field intensities at the extremes (C3, 0.24 mT; C14, 0.787 mT) are effective, while intermediate ones (C7, 0.353 mT; C11, 0.598 mT)

do not report significant effects. At 20 Hz, the two lowest intensities (C4, 0.24 mT; C8, 0.353 mT) report a significant reduction, while the two highest field intensities (C12, 0.598 mT; C16, 0.787 mT) do not report significant effects.

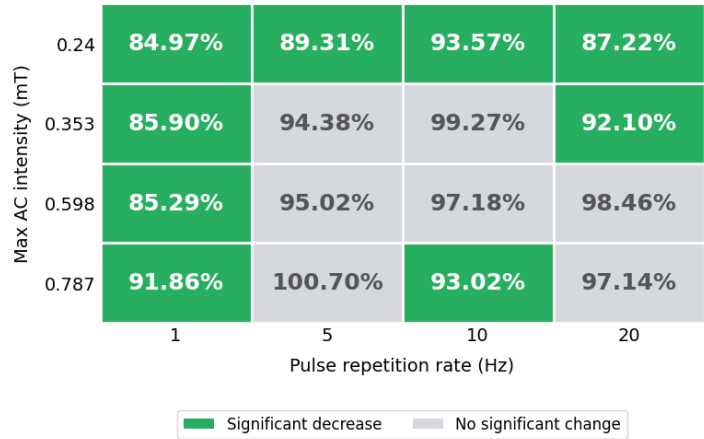


Figure 41: nsPMF cell viability percentages across varying magnetic field intensities (mT) and pulse repetition rates (Hz). Green cells highlight conditions that resulted in a significant decrease in viability, while grey cells indicate no significant change.

μsPMF. For the μsPMF subgroup, cells were treated 4 hours per day over 4 consecutive days. To monitor the progression of the phenomenon, analyzes were performed daily: from day 0 (pre-experimental control), through day 1 after the initial 4 hour session, days 2 and 3, and finally day 4 after completion of the full treatment cycle.

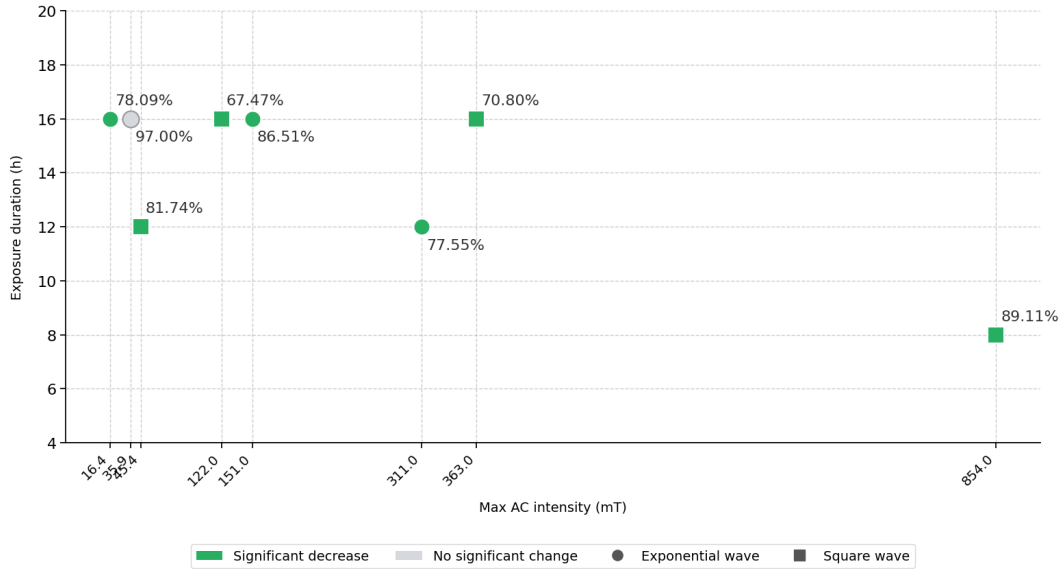


Figure 42: Viability outcomes for the μsPMF subgroup as a function of maximum AC intensity (mT) and total daily exposure duration (h). Each marker represents one experimental condition; shape encodes the pulse waveform and color encodes the outcome. Percentages indicate the residual viability relative to the control group.

In Figure 42 the results show similar effects: 7 out of 8 reported conditions led to a significant reduction in viability. The exception concerns condition C18 in which the pulse presents an exponential waveform with intensity 35.9 mT and a field exposure duration of 16 h, which did not report significant effects on any of the four measurement days (97% compared to the control group during all observation

days). In contrast, condition C17 at 16.4 mT with the same exponential wave produced a reduction of 21.91% on the fourth measurement day: the effect was observed with prolonged exposures; authors suggest that in some conditions the cumulative action of the field over time can be more effective than peak intensity. The most evident reductions in viability are observed when the pulse presents a square waveform. Condition C20 (122 mT) of exposure showed the greatest reduction in the study, with an effect that grew over time, highest on the fourth observation day (67.47%). Condition C23 (363 mT) showed a similar trend, with significant inhibition of viability only on the fourth and last observation day (70.8%). These two results suggest that the effect of μ sPMF has a cumulative character and requires greater exposures to obtain significant results.

However, the condition at 854 mT and 8 h of exposure produced the greatest reduction (10.89% less compared to the control group) in viability on the second observation day, without significant effects on the following days. This also highlights the fact that the highest intensity did not lead to the most lasting effect, indicating that beyond a certain threshold, increasing intensity does not lead to a proportional benefit and that exposure duration assumes a fundamental role. In support of this, condition C19 (45.4 mT) with 12 h of exposure and square waveform pulse, despite being the least intense, showed the most constant inhibition of viability over time of the entire subgroup, with significant reductions in all four measurement days with values ranging between 11.32% and 18.26% lower compared to the control group.

10.4 Mechanistic outcomes

Mechanistic outcome	Number of cancer exp. conditions
ROS production	0
Antioxidant response	0
Membrane potential	0
Calcium signaling	0
Cellular morphology and cytoskeleton remodeling	24
Metabolic reprogramming	0
Differentiation	0
Cell cycle analysis	0

Table 29: Number of PMF macro-category experimental conditions assessing each mechanistic outcome. No data were available for healthy cell lines.

This section reports only the analyzes on cell morphology, as other mechanistic outcomes were not investigated within the study (see Figure 29).

10.4.1 Cellular morphology and cytoskeleton remodeling

Cell morphology remained unchanged under all conditions tested.

nsPMF. The authors report that no cell fragments, micronuclei, or growth alterations were found. The absence of these suggests that the inhibitory effect on viability is not due to damage related to cell morphology.

μ sPMF. No significant morphological alterations were observed. The absence of morphological damage further supports the fact that the triggered outcome is not related to them.

10.5 Molecular outcomes

In PMF macro-category, no analyzes were conducted on genomic and proteomic results.

10.6 Mechanisms of action and connections

The authors hypothesized the involvement of the membrane potential as the main mechanism of action, although they did not directly measure it, and argue that this is influenced by the rate of change of the field dB/dt . The authors argue that PMFs with a higher rate of field change would generate a stronger induced electric field within the cell, thus capable of altering the potential of the plasma membrane and organelle membranes. The cell membrane behaves like a low-pass filter due to its capacitance: the low-frequency spectral components contained within each pulse would act on the plasma membrane, while the high-frequency ones would penetrate the nucleus and organelles [22]. This mechanism concerns the structure of a single pulse, not the frequency at which pulses repeat over time.

Instead, the pulse repetition frequencies (1 Hz, 5 Hz, 10 Hz, 20 Hz) are about the biological response that is triggered. This pulse repetition frequency does not change the depth of penetration (which is determined by the shape of the single pulse), but decides whether the cell has time to rest between one hit and the next. The best results observed at 1 Hz are explained by the authors through cell response and recovery dynamics between one pulse and the next: at low pulse repetition frequencies, cells are stimulated with longer intervals; this could prevent the development of adaptation mechanisms that could instead develop at higher frequencies.

The hypothesis is consistent with the non-linear trend observed in conditions at higher frequencies, where the effect does not increase with field intensity but depends on the specific combination of the two parameters. This suggests the existence of therapeutic windows, precise combinations of these parameters that can maximize cell damage. Outside of these therapeutic windows, the cell is capable of managing its own response.

In addition, for μ sPMF, the role of exposure duration and its effect that accumulates over time is observed. Viability inhibition progressively intensifies with the passing of days, independently of peak intensity; therefore, the accumulation of biological action over time could be an important parameter to take into account.

10.7 Limitations

Comparing the two subgroups is difficult because the parameters used are different. The intensities for μ sPMF are one to three orders of magnitude higher than those of nsPMF, and there are significant discrepancies in pulse durations and exposure patterns.

Beyond these technical differences, the evidence is extremely limited: all available data come from a single study [22] and focus on a single tumor cell line. The total lack of data on healthy cell lines, combined with the fact that the pulse repetition frequency for μ sPMF was never reported, makes it impossible to weigh the actual effectiveness of treatment or the role of specific parameters.

It is impossible to formulate any reliable hypotheses or draw conclusions regarding the potential of Pulsed Magnetic Fields as a mono-therapy for cancer.

11 Electric Fields

Electric fields differ from the electromagnetic fields discussed in the preceding sections. In all previous macro-categories, the field propagated through space and interacting with cells at a distance. In the Electric Fields macro-category, the field is applied directly to the cell culture via electrodes or conductive devices in physical contact with the biological sample. From a clinical perspective, this distinction is significant.

A further difference from the other macro-categories concerns the nature of the field itself. All previous groups involved oscillating magnetic or electromagnetic fields; here, the primary physical stimulus is the electric field component, applied either as a low-intensity alternating field (TTFields) or as ultra-short, high-intensity voltage pulses (nsPEF). The two subgroups differ in physical parameters and mechanisms of action, so they are treated separately in the following sections.

11.1 Data overview and cell line distribution

The Electric Fields macro-category comprises 89 experimental conditions from 11 independent studies. It is divided into two subgroups: TTFields (52 conditions, 58.4%), covering alternating electric field technologies including TTFields proper, a single TTIF (Tumor Treating Induced Field) condition, and one triboelectric nanogenerator condition; and nsPEF (37 conditions, 41.6%), covering nanosecond pulsed electric field applications. The macro-category spans 17 cancer cell lines and 2 healthy cell lines.

The representation of healthy cells is limited, accounting for 14 out of 89 experimental conditions (15.7%).

Cell line	Type	Cancer Category	Origin	Exp. cond.
MDA-MB-231	Triple-negative breast cancer (TNBC)	Breast	Human	34
U251	Glioblastoma	Brain & CNS	Human	9
HCC38	Triple-negative breast cancer (TNBC)	Breast	Human	5
RBE	Intrahepatic cholangiocarcinoma	Liver	Human	5
HCCC-9810	Intrahepatic cholangiocarcinoma	Liver	Human	5
MCF-7	Breast cancer	Breast	Human	3
U87	Glioblastoma	Brain & CNS	Human	3
A375	Melanotic melanoma	Melanoma	Human	2
B16F10	Melanoma	Melanoma	Mouse	1
Hep3B	Hepatocellular carcinoma	Liver	Human	1
C6	Glioma	Brain & CNS	Rat	1
GL261	Glioma	Brain & CNS	Mouse	1
C32	Amelanotic melanoma	Melanoma	Human	1
MeWo	Malignant melanoma (metastatic)	Melanoma	Human	1
Colo-829	Metastatic melanoma	Melanoma	Human	1
Me45	Melanoma (metastatic)	Melanoma	Human	1
U-87 MG	Glioblastoma	Brain & CNS	Human	1

Table 30: Cancer cell line used in the Electric Fields macro-category: type, cancer category, origin, and number of experimental conditions.

Cell line	Type	Category	Origin	Exp. cond.
MCF-12	Breast epithelial cells	Breast	Human	11
MCF-10A	Breast epithelial cells	Breast	Human	3

Table 31: Healthy cell line used in the Electric Fields macro-category: type, cancer category, origin, and number of experimental conditions.

11.2 Exposure parameters

The two subgroups differ in the physical quantities used to characterize the field and are therefore described by separate lookup tables.

TTFields (Table 32). TTFields are sinusoidal alternating electric fields delivered at low intensities (1–6 V/cm) and intermediate frequencies (100–300 kHz).

Larger cells respond to lower frequencies, since the optimal frequency is inversely proportional to the size of the cell [49]. 150 kHz is the most tested in the subgroup and corresponds to the parameter of the Optune device established for breast cancer treatment in previous clinical and preclinical work [26],[34]. The range 100–300 kHz was tested in biliary tract cancer cells to identify the frequency with the strongest anti-proliferative effect [17], while U251 glioblastoma cells were tested at 100, 200 and 300 kHz to confirm the cell-size dependence [49].

C_ID	Paper	Field Type	Wave Type	Freq. (Hz)	Electric field intensity (V/cm)
C1	17	TTFields	Sine	100000	2.1
C2	49	TTFields	Sine	100000	2.2
C3	17	TTFields	Sine	150000	1.3
C4	26	TTFields	Sine (UM)	150000	1.5
C5	26	TTFields	Sine (FM)	150000	1.5
C6	34	TTFields	Sine	150000	1.5
C7	17	TTFields	Sine	150000	1.7
C8	17	TTFields	Sine	150000	2.1
C9	34	TTFields	Sine	150000	3
C10	34	TTFields	Sine	150000	3.75
C11	34	TTFields	Sine	150000	4.4
C12	34	TTFields	Sine	150000	6
C13	49	TTFields	Sine	200000	1
C14	49	TTFields	Sine	200000	1.5
C15	17	TTFields	Sine	200000	2.1
C16	49	TTFields	Sine	200000	2.2
C17	27	Tumor Treating Induced Fields	Sine	200000	
C18	49	TTFields	Sine	300000	2.2
C19	39	AEF			0.26

Table 32: TTFields exposure conditions lookup table.

The intensity was investigated from 1 to 6 V/cm. The 1.5–3 V/cm range was selected as the therapeutic window for breast cancer based on the prior literature that showed optimal selectivity between cancer and healthy cells in this range [26], [34]. Higher intensities (4.4–6 V/cm) were specifically tested to determine the upper limit of selectivity, such as the point at which normal cells begin to be significantly affected [34].

Two waveform variants were tested under breast cancer conditions: unmodulated (UM) sinusoidal fields and frequency-modulated (FM) fields in which the direction of the field rotates periodically [26]. The rationale for FM is that tumor cells divide in multiple orientations and rotating the field maximizes the probability that dividing cells are exposed to the optimal field direction relative to their mitotic spindle.

Condition C17 (TTIF) is methodologically distinct: the field is delivered by electromagnetic induction rather than direct electrode contact, using a magnetic ring with an LC resonance circuit at 200 kHz. The induced current density at the cell level has not been reported [27].

Condition C19 (AEF) uses a triboelectric nanogenerator (TENG) to deliver a very low intensity field (0.26 V/cm) without a specified carrier frequency. This condition is included for completeness, but is not directly comparable to the other TTFields conditions [39].

nsPEF (Table 33). nsPEF are characterized by three parameters: electric field intensity (kV/cm), pulse duration (ns), and number of pulses. The determinant of the biological effect is the combination of these three parameters.

C.ID	Paper(s)	Field Type	Wave Type	Electric field intensity (kV/cm)	Pulse duration (ns)	Pulse rep. rate (Hz)	N pulses
C20	47	nsPEF	Square	8	100	10000	100
C21	43	nsPEF		10	300	2	800
C22	45	nsPEF		10	50	2000	
C23	45	nsPEF		15	50	2000	
C24	45	nsPEF		20	50	2000	
C25	46	nsPEF		40	100	200000	60
C26	42	nsPEF		70	5	100	2000
C27	42	nsPEF		70	5	100	4000
C28	42	nsPEF		70	5	100	6000
C29	42	nsPEF		70	5	100	8000
C30	42	nsPEF		70	5	100	10000
C31	42	nsPEF		80	5		1000
C32	42	nsPEF		80	5	100	2000
C33	42	nsPEF		80	5	100	4000
C34	42	nsPEF		80	5	100	6000
C35	42	nsPEF		80	5	100	8000
C36	42	nsPEF		80	5	100	10000
C37	42	nsPEF		90	5	100	2000
C38	42	nsPEF		90	5	100	4000
C39	42	nsPEF		90	5	100	6000
C40	42	nsPEF		90	5	100	8000
C41	42	nsPEF		90	5	100	10000
C42	42	nsPEF		100	5	100	2000
C43	42	nsPEF		100	5	100	4000
C44	42	nsPEF		100	5	100	6000
C45	42	nsPEF		100	5	100	8000
C46	42	nsPEF		100	5	100	10000

Table 33: nsPEF exposure conditions lookup table.

Pulse durations range from 5 ns [42] to 300 ns [43]. Very short pulses (5 ns) have high-frequency spectral components that allow the field to penetrate the plasma membrane and act directly on intracel-

lular membranes; they were tested at high intensities (70–100 kV/cm) to ensure effective intracellular delivery [42]. Longer pulses (50–300 ns) overlap with the plasma membrane charging time constant and act through both intracellular and outer membrane mechanisms; they were tested at lower intensities (8–40 kV/cm) [43], [45], [46]. Condition C20 (8 kV/cm, 100 ns, 100 pulses) was deliberately designed below the electroporation threshold to study sub-cytotoxic effects on membrane exocytosis without inducing cell death [47].

11.3 Cell fate outcomes

Cell fate outcome	Cancer exp. conditions	Healthy exp. conditions	Total
Viability	24	11	35
Proliferation	19	0	19
Apoptosis	3	2	5
Necrosis	4	4	8
Ferroptosis	0	0	0
Senescence	0	0	0
Motility	5	3	8

Table 34: Number of Electric Fields macro-category TTFields subgroup experimental conditions assessing each cell fate outcome, broken down by cancer and healthy cell lines.

Cell fate outcome	Cancer exp. conditions	Healthy exp. conditions	Total
Viability	12	3	15
Proliferation	5	0	5
Apoptosis	4	1	5
Necrosis	0	0	0
Ferroptosis	0	0	0
Senescence	0	0	0
Motility	2	0	2

Table 35: Number of Electric Fields macro-category nsPEF subgroup experimental conditions assessing each cell fate outcome, broken down by cancer and healthy cell lines.

11.3.1 Viability

TTFields. Viability was the primary endpoint in several breast cancer conditions. At C4 (150 kHz, 1.5 V/cm, UM) and C5 (150 kHz, 1.5 V/cm, FM), MDA-MB-231 cells were reduced to approximately 29% of the control after 24 h under both conditions. Under UM treatment (C4), cells began recovering at 48 h (74%) and fully recovered at 72 h (100%), indicating that the cytotoxic effect is reversible after field removal. FM treatment (C5) prevented this recovery: the cells continued to decline to 51% at 48 h, with no viable cells detectable at 72 h. This difference highlights the importance of the field modality: FM treatment appears to overcome the adaptive resistance that cancer cells develop under continuous UM exposure. The same C6 condition (150 kHz, 1.5 V/cm) confirmed the initial drop to 30% at 24 h in MDA-MB-231 cells, with complete recovery at 72 h under unmodulated conditions.

In healthy MCF-12 breast epithelial cells, C4 and C5 produced modest and inconsistent effects: viability ranged from 86% to 102% across time points and conditions, with only some values reaching statistical significance and none showing the dramatic reduction observed in cancer cells. At the intensity limit, C12 (150 kHz, 6 V/cm) showed that MCF-12 cells experienced greater death than MDA-MB-231 or HCC38, confirming that at higher intensities the selectivity inverts.

Under C17 (TTIF, 200 kHz), U87 density decreased to 80% at 24 h, 65% at 48 h, and 48% at 72 h of exposure.

At C19 (AEF, 0.26 V/cm), B16F10 melanoma cells dropped to 18% of the control after 2 h of TENG stimulation.

nsPEF. In Hep3B, C21 (10 kV/cm, 300 ns, 800 pulses) reduced the viability to 24.94% of the control group. In glioblastoma multicellular tumor spheroids, C25 (40 kV/cm, 100 ns, 60 pulses) reduced viability to 9% in U-87 MG, 7% in C6 cells and 27% in GL261 [46].

11.3.2 Proliferation

TTFields. At C8 (150 kHz, 2.1 V/cm), proliferation was reduced to 58.2% of the control in HCCC-9810 and 48% in RBE, this is the lowest proliferation values in the TTFields subgroup. Across the intensity series at 150 kHz, a dose-dependent relationship was observed: C3 (1.3 V/cm) did not produce significant effect in HCCC-9810 (90% of control), while C7 (1.7 V/cm) reduced it to 72% and C8 (2.1 V/cm) to 58.2%. Colony formation capacity was also reduced to 32.6% in HCCC-9810 and 50.4% in RBE at C8. In U251 glioblastoma, proliferation was significantly reduced across all conditions tested (C2, C13–C16, C18) at 200 kHz, with 200 kHz identified as the most effective frequency [49].

nsPEF. In Hep3B cells, C21 reduced proliferation to 48.74% of the control. In glioblastoma spheroids at C25 (40 kV/cm, 100 ns), Ki67 expression was reduced by 46.83% in U-87 MG, 53.19% in C6 cells and 54.44% in GL261. Ki67 is a proliferation marker present during active phases of the cell cycle.

11.3.3 Apoptosis

TTFields. Apoptosis was confirmed in MDA-MB-231 cells at C4 and C5 (150 kHz, 1.5 V/cm) with both UM and FM conditions showing a significant increase. In healthy MCF-12 cells, no significant increase in apoptosis was detected under the same conditions. In B16F10 melanoma cells at C19, significant apoptosis was confirmed along with the viability reduction.

nsPEF. In glioblastoma spheroids at C25 (40 kV/cm, 100 ns), apoptosis rates were elevated: 320% of control in U-87 MG, 330% in C6 and 340% in GL261. In MCF-7 at C24 (20 kV/cm, 50 ns), significant apoptosis was confirmed by Annexin V and corroborated by caspase-3/7 activation; in MCF-10A neither Annexin V-positive population nor caspase-3/7 activation was detected, providing direct evidence of cancer-selective apoptosis induction.

11.3.4 Necrosis

TTFields. Necrosis was assessed in cancer MDA-MB-231 cells and in healthy MCF-12 cells at C4 and C5 (150 kHz, 1.5 V/cm, UM and FM). No significant necrotic response was detected in either cell line under either condition, indicating that the cell death induced by TTFields at this frequency and intensity occurs primarily through apoptosis rather than necrosis.

nsPEF. Necrosis was not assessed in the nsPEF subgroup.

11.3.5 Motility

TTFields. Migration was assessed by wound-healing assay in three breast cancer conditions. At C8 (150 kHz, 2.1 V/cm), migration rate was reduced to 53.4% of the control in HCCC-9810 and 27.6% in RBE after 24–72 h of exposure. At C4 (150 kHz, 1.5 V/cm, UM) and C5 (FM), wound closure in MDA-MB-231 cells was reduced to 30.4% and 19.6% of the control, respectively. In MCF-12 healthy epithelial cells, wound closure was not significantly different from control under UM and FM conditions (100% and 93% respectively, both n.s.), providing further evidence of cancer-selective suppression of cell motility.

nsPEF. Motility was assessed in melanoma cells at C20 (8 kV/cm, 100 ns, 100 pulses). A375 and MeWo cells showed a significant reduction in migration capacity following nsPEF exposure, consistent with the cytoskeletal remodeling and membrane vesicle release observed under the same condition.

11.4 Mechanistic outcomes

Mechanistic outcome	Cancer exp. conditions	Healthy exp. conditions	Total
ROS production	1	0	1
Antioxidant response	0	0	0
Membrane potential	1	0	1
Calcium signaling	1	0	1
Cellular morphology and cytoskeleton remodeling	15	6	21
Metabolic reprogramming	0	0	0
Differentiation	0	0	0
Cell cycle analysis	0	0	0

Table 36: Number of experimental conditions in the Electric Fields macro-category TTfields subgroup that assess each mechanistic outcome, broken down by cancer and healthy cell lines.

Mechanistic outcome	Cancer exp. conditions	Healthy exp. conditions	Total
ROS production	1	1	2
Antioxidant response	0	0	0
Membrane potential	21	1	22
Calcium signaling	1	1	2
Cellular morphology and cytoskeleton remodeling	28	1	29
Metabolic reprogramming	6	1	7
Differentiation	0	0	0
Cell cycle analysis	0	0	0

Table 37: Number of experimental conditions in the Electric Fields macro-category nsPEF subgroup that assess each mechanistic outcome, broken down by cancer and healthy cell lines.

11.4.1 ROS production

TTFields. ROS was assessed only at C19 (AEF, 0.26 V/cm) in B16F10 melanoma cells, where a significant increase in ROS was detected.

nsPEF. At C24 (20 kV/cm, 50 ns) in MCF-7 cells, ROS increased to 150% of the control, while in MCF-10A normal cells exposed to the same condition ROS was below the control (50%), indicating opposite responses in cancer vs normal cells under identical nsPEF parameters.

11.4.2 Membrane potential

TTFields. Mitochondrial membrane potential was assessed in B16F10 melanoma cells at C19 (AEF, 0.26 V/cm), where a significant decrease was detected after TENG stimulation.

nsPEF. Mitochondrial membrane potential was assessed in study 42 in MDA-MB-231 cells in all combinations of intensity and pulse number using a dye whose uptake depends on the mitochondrial membrane potential. Fluorescence intensity decreased with increasing intensity and pulse number: at 70 kV/cm it dropped from 85% at 2000 pulses to 40% at 10000 pulses; at 90 kV/cm from 78% to 25%; at 100 kV/cm from 75% to 25%. This progressive mitochondrial depolarization indicates dose-dependent mitochondrial damage as one of the primary intracellular consequences of nsPEF exposure.

In MCF-7 cells at C24 (20 kV/cm, 50 ns), the mitochondrial membrane potential dropped drastically to approximately 20% of baseline within 300 seconds of nsPEF application, while in MCF-10A healthy cells the decrease was modest (remaining around 70% of baseline), which confirms differential mitochondrial sensitivity between cancer and normal cells.

11.4.3 Calcium signaling

TTFIELDS. Calcium signaling was assessed in B16F10 melanoma cells at C19 (AEF, 0.26 V/cm), where a significant increase in intracellular Ca^{2+} levels was detected after TENG stimulation.

nsPEF. At C24 (20 kV/cm, 50 ns), cytosolic Ca^{2+} levels in MCF-7 cells increased dramatically after nsPEF application. In MCF-10A cells under the same condition, the Ca^{2+} elevation was not pronounced.

11.4.4 Cellular morphology and cytoskeleton remodelling

TTFIELDS. At C8 (150 kHz, 2.1 V/cm), HCCC-9810 and RBE cells revealed hallmarks of TTFIELDS-induced mitotic disruption: disorganized spindle filament alignment, multipolar spindle formation, and asymmetric chromosome distribution [17]. These morphological findings are consistent with the established anti-mitotic mechanism of TTFIELDS. At the cellular level, C8 also produced increased cell size, indistinct cell boundaries, and significantly reduced cell density [17]. In U251 cells, loss of tentacle-shaped projections and significant enlargement of cell bodies were observed, along with multi-nucleated cells and abnormal chromosome segregation patterns [49].

In MDA-MB-231 cells at C4 and C5 (150 kHz, 1.5 V/cm), cell diameter decreased significantly (by 1.45 μm under UM at 24 h, $p < 0.0001$), consistent with early apoptotic cell shrinkage, while FM treatment at 72 h produced extensive membrane blebbing and complete loss of membrane integrity [26]. MCF-12 healthy cells did not show any significant morphological changes under UM and FM conditions at any time [26].

nsPEF. In MDA-MB-231 cells, ultra-short 5 ns pulses at 70–100 kV/cm caused progressive pulse number dependent disruption of intracellular organelles [42]. At C21 (10 kV/cm, 300 ns), Hep3B cells revealed nuclear shrinkage, chromatin condensation and margination, and enlarged dysfunctional mitochondria 2 h after pulsing [43]. In glioblastoma MCTSs at C25 (40 kV/cm, 100 ns), spheroid volume increased after treatment (U-87 MG +24%, C6 +56%, GL261 +29%), likely reflecting disruption of cell-cell contacts and structural loosening of the tumor mass rather than growth [46]. In melanoma cells at C20 (8 kV/cm, 100 ns), sub-electroporation nsPEF induced membrane protrusions, intracellular vesicle shift to the cell periphery, and microvesicle release without causing gross cytotoxicity [47].

11.4.5 Metabolic reprogramming

TTFIELDS. Metabolic reprogramming was not assessed.

nsPEF. In MCF-7 cells at C24 (20 kV/cm, 50 ns), NADH autofluorescence lifetime increased by ~25% after nsPEF, indicative of changes in NADH-protein binding and altered mitochondrial metabolic activity associated with caspase-dependent apoptosis. In five melanoma cell lines at C20 (8 kV/cm, 100 ns), untargeted metabolomics and lipidomics revealed significant alterations in glutathione metabolism, amino acid biosynthesis pathways, and lipid composition [47]. These metabolic changes were consistent across cell lines and may reflect the cellular response to sub-lethal membrane perturbation and electro-exocytosis.

11.5 Molecular outcomes

11.5.1 TTFields

Molecular analysis was performed only in HCCC-9810 and RBE biliary tract cancer cells at C8 (150 kHz, 2.1 V/cm). Three markers were quantified, all associated with immunogenic cell death (ICD):

- **Calreticulin (CRT).** CRT is an endoplasmic reticulum chaperone that, when translocated to the cell surface, acts as an eat-me signal that facilitates the phagocytic uptake of dying tumor cells by dendritic cells. CRT-positive cells increased in HCCC-9810 and in RBE, indicating TTFields-induced ICD [17].
- **HMGB1 (High Mobility Group Box 1).** Nuclear protein released extracellularly during cell death, acting as a damage-associated molecular pattern (DAMP) that promotes immune activation. Extracellular HMGB1 levels increased significantly in the supernatant of both HCCC-9810 and RBE cells after treatment [17].
- **ATP.** Extracellular ATP released during apoptosis acts as a find-me signal for immune cell recruitment. ATP levels in the cell culture supernatant increased significantly in both cell lines after TTFields exposure [17].

The simultaneous up-regulation of all three ICD markers suggests that TTFields does not merely kill cancer cells but may trigger an immunogenic form of cell death capable of triggering an antitumor immune response.

11.5.2 nsPEF

Molecular analysis was performed in three separate studies that included breast cancer, glioblastoma, and melanoma.

Breast cancer (MCF-7; C24: 20 kV/cm, 50 ns)

- **Caspase-3/7.** Executor caspases that cleave cellular substrates during apoptosis, activated downstream of mitochondrial outer membrane permeabilization. Caspase-3/7 activity was significantly increased in MCF-7 cells, confirming caspase-dependent apoptosis. No activation was detected in MCF-10A normal cells under the same condition, indicating that the apoptotic cascade is cancer-selective [45].

Glioblastoma spheroids (U-87 MG, C6, GL261; C25: 40 kV/cm, 100 ns, 60 pulses)

- **N-cadherin.** Cell adhesion protein that facilitates cell-cell contacts essential for tumor cohesion. Down-regulated in U-87 MG, in C6 cells and in GL261, indicating disruption of tumor architecture [46].
- **ZO-1.** Tight junction protein that maintains paracellular barrier integrity and cell-cell cohesion. Down-regulated in U-87 MG, in C6 cells and in GL261 [46].
- **Ki67.** Nuclear protein expressed during active cell cycle phases, used as a proliferation marker. Down-regulated by in U-87 MG, in C6 cells and in GL261, consistent with the anti-proliferative effect observed in the cell fate outcomes [46].

Melanoma (A375, C32, MeWo, Colo-829, Me45; C20: 8 kV/cm, 100 ns, 100 pulses)

- **MAGE cancer-testis antigens (MAGE-A1, -A2, -A3, -A6, -A8, -A12, MAGE-C2).** Family of tumor-specific antigens, normally expressed only in germ cells and trophoblasts, aberrantly expressed in melanoma. They are recognized by cytolytic T-lymphocytes and therefore they are potential immunotherapy targets. Following nsPEF, MAGE antigen expression was up-regulated at the protein level in several cell lines (most consistently MAGE-A1, -A2, -A3 in A375 and MeWo), and mRNA up-regulation was confirmed by RT-PCR for MAGE-A3/A6

in MeWo and MAGE-A1/A2 in A375 and Me45. In 3D spheroids, MAGE-A1 shifted from the cell membrane to the cytoplasm and accumulated at the spheroid periphery, consistent with electro-exocytosis-driven antigen surface exposure [47].

- **PD-1 (glycosylated and deglycosylated).** Immune checkpoint receptor that suppresses T-cell activity when engaged by its ligand PD-L1. Both glycosylated and deglycosylated forms increased significantly after nsPEF in most cell lines, with the most significant increase in Me45. The authors interpret this as nsPEF-induced exocytosis of PD-1-containing intracellular vesicles to the cell surface, potentially modulating immune checkpoint signaling and making tumor cells more visible to the immune system [47].

11.6 Mechanism of action and connections

The two subgroups differ substantially and they are therefore described separately.

TTFields The mechanism of action of TTFields rests primarily on physical disruption of the mitotic process. The key point is the interaction between the alternating electric field and polar macromolecules that undergo large conformational changes during cell division. At intermediate frequencies (100–300 kHz) and low intensities (1–3 V/cm), the field exerts dielectrophoretic forces on the mitotic spindle apparatus: tubulin dimers, which carry a large electric dipole moment, experience oscillating torques that interfere with their polymerization into functional spindle microtubules. The morphological evidence for this mechanism is documented across multiple cell lines and conditions in the dataset: disorganized and multipolar spindle formation, asymmetric chromosome distribution, and the appearance of multinucleated cells are reported in HCCC-9810, RBE, and U251 glioblastoma cells following TTFields exposure. These findings are directly consistent with a mitotic disruption model.

The cell fate consequence of mitotic disruption is apoptosis rather than necrosis, the cell death program is caspase-dependent and programmed. The ICD markers measured in HCCC-9810 and RBE cells indicate that the form of apoptosis triggered by TTFields is immunogenic.

Because TTFields act preferentially on dividing cells, rapidly proliferating cancer cells are at risk during a far greater fraction of their time than slowly cycling or non-dividing normal epithelial cells. This is confirmed by the data, where healthy MCF-12 cells exposed to the same field parameters that eliminate MDA-MB-231 cells show only marginal and inconsistent viability changes up to 3 V/cm. Selectivity collapses at 6 V/cm, where healthy cells are more affected than cancer cells, establishing both the existence and the upper limit of the therapeutic window. Cells that survive the initial 24 h exposure under unmodulated fields begin to recover by 48–72 h, presumably because a subpopulation arrests in a non-dividing state and is therefore shielded. FM conditions prevent this recovery, consistent with the hypothesis that rotating the field direction eliminates the directional escape mechanism available to cells dividing outside the primary field axis. ROS production is documented in only one condition (C19, AEF/TENG), where it accompanies a strong reduction in mitochondrial membrane potential and an increase in intracellular calcium in B16F10 melanoma cells. Under conventional TTFields conditions, these mechanistic endpoints were not assessed, so the contribution of oxidative stress to the TTFields mechanism cannot be evaluated from the data.

nsPEF The mechanism of action of nsPEF is distinct from that of TTFields. Ultra-short pulses (5–300 ns) at high intensities (8–100 kV/cm) have high-frequency spectral content that allows the field to penetrate the outer plasma membrane, which acts as a low-pass filter, and deposit energy directly on intracellular membranes. The primary physical event is not spindle disruption, but electroporation of internal membrane compartments: the ER, lysosomes, and inner mitochondrial membrane are affected.

The mitochondrial consequence of intracellular electroporation is a dose-dependent depolarization of the inner mitochondrial membrane, documented as a function of both field intensity and pulse number. This mitochondrial damage provides a link to the downstream apoptotic response: in MCF-7 cells, the mitochondrial membrane potential collapses within 300 seconds of nsPEF application, and this is accompanied by a rise in cytosolic calcium and by a significant increase in ROS, while in MCF-10A normal cells all three events are markedly attenuated, confirming that cancer cells are preferentially susceptible to this form of intracellular perturbation.

The committed cell death program is caspase-dependent apoptosis, established by caspase-3/7 activation in MCF-7 cells and by Annexin V positivity under multiple conditions.

At sub-cytotoxic intensities (C20, 8 kV/cm), nsPEF acts through a different and non-lethal mechanism: electro-exocytosis, the field-driven translocation of intracellular vesicle contents to the cell surface, is documented in melanoma cells through the surface up-regulation and membrane redistribution of MAGE cancer-testis antigens and the surface exposure of PD-1. This condition does not reduce viability or proliferation but alters the immunophenotype of the cells in ways that could influence their visibility to cytotoxic T lymphocytes, opening a potential immunomodulatory route that is mechanistically distinct from direct cytotoxicity.

TTFIELDS act selectively on dividing cells, and cancer cells spend more time dividing; nsPEF acts selectively on cells whose intracellular membranes are more susceptible to electroporation, and the differential mitochondrial response between MCF-7 and MCF-10A confirms that this susceptibility is not equal between cell types. In both cases, the selectivity is conditional on the parameter window.

11.7 Limitations

The big limitation of the Electric Fields macro-category is the concentration of the dataset in a single cancer cell line: MDA-MB-231 accounts for 34 of the 89 experimental conditions, and a large fraction of these derive from a single study.

The two subgroups cannot be directly compared at the mechanistic level, and the mechanistic picture reconstructed in Section 11.6 is assembled from non-overlapping evidence.

The representation of healthy cell lines is markedly insufficient. Only two healthy lines are present in the entire macro-category, MCF-12 and MCF-10A, both breast epithelial cells. This means that selectivity can only be evaluated in the context of breast cancer: no cancer/healthy comparison is available for glioblastoma, biliary tract cancer, hepatocellular carcinoma, or melanoma. Whether the cancer-selective effects documented in breast cancer conditions generalize to other tumor types cannot be determined from the available data.

Several mechanistic outcomes are absent or almost absent from the dataset. Cell cycle analysis was not performed under any of the conditions in this macro-category, preventing any assessment of whether the anti-proliferative effects of both TTFIELDS and nsPEF are mediated by cell cycle arrest at specific checkpoints. Senescence and ferroptosis were not assessed under any condition. The antioxidant response and metabolic reprogramming were measured only under isolated conditions and in a single cell line. The absence of ROS quantification under the most effective TTFIELDS conditions represents a specific gap, as it prevents any assessment of whether oxidative stress contributes to the mechanism of action.

The TENG condition (C19) is not comparable to any other TTFIELDS condition.

The TTIF condition (C17) uses electromagnetic induction rather than direct electrode contact and represents a different delivery modality; the induced current density at the cell level was not reported, making comparison with other TTFIELDS conditions impossible.

In the nsPEF subgroup, pulse durations span two orders of magnitude (5 to 300 ns), intensities span more than one order of magnitude (8 to 100 kV/cm), and the number of pulses ranges from 60 to 10000. Comparing viability outcomes across these conditions would be misleading.

12 Mechanistic comparison between macro-categories

Despite the substantial differences in physical parameters and biological targets across the six macro-categories, several mechanistic themes recur independently in multiple subgroups, suggesting that electromagnetic fields of different types converge on a limited set of cellular vulnerabilities.

Mitochondria as a shared primary target.

Mitochondrial dysfunction is the most consistently documented upstream event across macro-categories. In ELF-EMF, LF-EMF, and RF-EMF (LTE subgroup), the mitochondrial electron transport chain is identified as the primary source of field-induced ROS, with converging pharmacological evidence: NAC rescues cell viability in all three subgroups, and KCN suppresses field-induced radical production in LF-EMF. In the nsPEF subgroup, inner mitochondrial membrane depolarization is directly quantified and precedes apoptotic commitment. The hypothesis that the field interacts with iron-sulfur clusters of Complex I is advanced independently in both the ELF-EMF and RF-MF+SMF subgroups, though it remains unproven in both.

The p53–p21 axis as a convergence point for cell cycle arrest.

G2/M or G0/G1 arrest mediated by p53 and p21 upregulation is documented in ELF-EMF (conditions C3, C32), LF-EMF (condition C6 at 227 kHz), and RF-EMF LTE (condition C25 at 1700 MHz, 2 W/kg). In all three cases the arrest is cancer-selective: healthy cells exposed to the same field show the same qualitative transcriptional signal (p53 and p21 upregulation) without undergoing cell death. This dissociation between stress signaling and death commitment is one of the clearest cross-cutting demonstrations of how selectivity emerges from the cellular context rather than from the field itself.

Metabolic reprogramming as a determinant of selective vulnerability.

An anti-Warburg shift (i.e., reduction of glycolytic activity and increase of oxidative phosphorylation) is reported independently in ELF-EMF (thermodynamically-calculated frequency conditions), LF-EMF (condition C6, 227 kHz), and proposed as a mechanistic framework in the MEMP condition of the ELF-EMF subgroup. In all cases, cancer cells are unable to compensate for the forced metabolic transition, while healthy cells accommodate it without functional consequences. This convergent finding across physically distinct fields supports the hypothesis that the Warburg phenotype is a genuine determinant of differential vulnerability, independent of the specific field type.

Calcium homeostasis as a parallel axis to oxidative stress.

Perturbation of calcium fluxes through voltage-gated channels is documented in ELF-EMF, and intracellular calcium elevation is reported under nsPEF and electric field (AEF/TENG) conditions as well. In the most characterized ELF-EMF study, calcium overload and ROS overproduction are shown to be independent and simultaneously required for cell death, suggesting a dual-trigger logic. A similar convergence of mitochondrial depolarization, calcium rise, and ROS increase is observed in nsPEF-exposed MCF-7 cells, though it operates through a physically distinct mechanism (intracellular electroporation rather than membrane channel modulation).

Cytoskeletal remodeling and differential resilience.

F-actin disruption in cancer cells with preservation of healthy cell cytoskeletal integrity is reported in ELF-EMF (LF-RMF, MEMP conditions) and inferred from morphological data in LF-EMF and nsPEF. The consistent finding that healthy cells are more resistant to cytoskeletal perturbation and that recover from it more rapidly, appears to be a property of normal cells that is independent of p53 status and may reflect intrinsic differences in cytoskeletal organization or mechanical stiffness.

Immune modulation as an emerging cross-cutting theme.

Two mechanistically distinct immune-mediated routes are documented at the extremes of the frequency spectrum. In ELF-EMF, a PEMF condition acts on CD8+ T cells rather than directly on

cancer cells, enhancing their cytotoxic activity through IL-37 suppression and NF- κ B activation. In nsPEF, sub-cytotoxic pulses trigger electro-exocytosis of MAGE antigens and surface exposure of PD-1 in melanoma cells, potentially modulating immune checkpoint interactions. In TTFields, immunogenic cell death markers (calreticulin, HMGB1, extracellular ATP) are upregulated following treatment. These three mechanisms are physically and biologically distinct, but their co-occurrence across macro-categories suggests that immune engagement may be a general, underexplored property of electromagnetic field exposure rather than a feature of any specific modality.

13 Global limitations

The limitations discussed in the dedicated sections for each macro-category (Sections 6.7, 7.9, 8.7, 9.7, 10.7, and 11.7) are specific to the parameter spaces, cell lines, and mechanistic coverage of each subgroup.

However, several further limitations are transversal to the entire dataset and must be made explicit before any conclusion is drawn.

Absence of a consistent nomenclature.

This field lacks a universally adopted and consistently applied naming convention, and this limitation is visible throughout the dataset. Authors frequently refer to the same type of field using different acronyms or descriptors: some studies refer to extremely low frequency fields simply as EMF followed by a frequency specification, whereas the more informative designation ELF-EMF would allow immediate classification. More problematic is the use of the acronym PEMF, which appears in the literature with two distinct and mutually incompatible meanings. In some studies, Pulsed EMF describes an exposure protocol in which the field is delivered discontinuously over time (e.g., one hour per day for ten consecutive days) where pulsed refers to the intermittent nature of the treatment schedule. In other studies, the same acronym refers to a field generated by a pulsed waveform, where pulsed describes the temporal shape of the electromagnetic signal itself rather than the treatment schedule.

This ambiguity is not merely terminological: it complicates the comparison between studies and can be misleading for researchers approaching these concepts for the first time.

Sparse and uneven coverage of outcomes across and within macro-categories.

A recurring theme of the gap identification analysis presented in Chapter 5 is that the coverage of mechanistic outcomes is markedly uneven, both between macro-categories and within them. The heat maps in Figures 6–9 make this visible: the gray cells that represent untested combinations are not marginal exceptions, but the dominant feature of the landscape. For the mechanistic picture reconstructed in each section to be genuinely informative, those gray cells would need to be replaced by measurements.

The fact that different studies measured different mechanisms under different conditions means that the connections between adjacent steps of a proposed mechanistic chain are often inferred by combining evidence from distinct experimental contexts rather than being traced within a single coherent experimental design. This limits the confidence that can be placed in any integrated mechanistic model, regardless of how internally consistent it appears.

Absence of a systematic biological rationale for parameter selection.

In the majority of the experimental conditions across the dataset, the frequencies and intensities tested were selected on the basis of prior literature, device capabilities, or regulatory reference values, rather than on a direct biological hypothesis. Only a small subset of studies approached parameter selection from a genuinely mechanistic starting point: for instance, deriving exposure frequencies from thermodynamic models of cellular characteristic times, from the hyperfine resonance frequencies of specific biochemical targets, or from the estimated resonance frequencies of calcium ions or cell membrane dynamics.

This is a fundamental limitation: identifying the optimal parameter window for a therapeutic effect cannot be achieved by systematic scanning of an arbitrary range or by replication of prior choices. Progress in the field requires parameter selection grounded in quantitative biophysical models of the interaction between the field and specific cellular targets.

Predominance of adherent monolayer cultures.

With few exceptions, the experimental conditions analyzed across all macro-categories were conducted on cells grown as two-dimensional adherent monolayers. Three-dimensional multicellular tumor spheroids provide a more physiologically relevant model; their limited representation in this dataset means that the results of most conditions analyzed here should be interpreted with caution.

Normalization of outcome data and heterogeneity of assays.

The percentages reported throughout this analysis for viability, proliferation, apoptosis, and other outcomes were extracted from the original studies and, in a number of cases, derived through mathematical normalization, since different studies expressed the same outcome in different units or relative to different baselines. A deliberate choice was made to normalize all data to the corresponding control group, so as to provide a common reference frame that enables at least a qualitative comparison across conditions.

However, this normalization does not eliminate the underlying heterogeneity. The control groups themselves were not treated identically across studies: some studies used sham-exposed controls, in which the cells were placed in the same position as the exposed group but with the device switched off, while others maintained control cells in a standard incubator without any sham procedure.

A parallel heterogeneity applies to the assays used to measure the same outcome.

Viability was quantified by MTT, XTT, Trypan Blue exclusion, and other methods across different studies. This assay-level heterogeneity is a source of apparent inconsistency between studies that cannot be resolved from the data available.

Heterogeneity of exposure devices and experimental geometry.

Even within the same macro-category, and sometimes within the same parameter window, the devices used to generate the field differ substantially. Some are commercially available clinical or research instruments; others are custom-built laboratory setups.

In addition, the direction from which the field reaches the cell layer, whether delivered vertically, horizontally, or at an angle relative to the culture surface, varies across studies and is rarely reported in a way that allows reconstruction. These variables were not incorporated into the comparative analysis, because doing so would have reduced the analysis to a study-by-study technical comparison rather than a synthesis of biological findings. They are real sources of variability that must be considered when interpreting differences between studies that nominally test the same parameters.

14 Conclusions

This report has analyzed 49 primary studies covering 434 non-thermal experimental conditions across six macro-categories of electromagnetic field, with the aim of assessing the evidence for electromagnetic fields as a mono-therapeutic modality against cancer cells in vitro and evaluating the selectivity of the observed effects with respect to healthy cell lines.

The dataset supports several conclusions, provided they are read in light of the structural limitations discussed.

An anti-tumoral effect of electromagnetic fields on cancer cells is documented across all macro-categories, but it is strictly conditional on the parameter window. The relationship between exposure parameters and biological response is non-linear and non-monotonic: within specific combinations of frequency, intensity, waveform, and duration, the effect is anti-tumoral; outside these windows, the same field can be neutral or actively pro-tumoral. There is no universal EMF effect on cancer, and any framing of electromagnetic fields as a uniformly beneficial modality is not supported by the evidence.

Selectivity for cancer cells over healthy cells is real and documented at multiple levels of the mechanistic chain, but it is not guaranteed. It depends on the parameter window and on the cellular context, with the metabolic phenotype and the functional status of P53 emerging as the two most consistent determinants across macro-categories. Cases of negative selectivity are documented and must be considered as informative boundaries rather than anomalies.

Common mechanistic axes like mitochondrial oxidative stress, calcium homeostasis, cytoskeletal remodeling, and metabolic reprogramming, recur across macro-categories, suggesting a conserved set of cellular targets. However, no complete causal chain is directly demonstrated in any single study, and the mechanistic frameworks reconstructed in this report should be treated as hypothesis-generating rather than conclusive. The available evidence is not sufficient to support therapeutic recommendations for any specific field type or parameter combination.

Independent replication under harmonized conditions, systematic use of three-dimensional culture models, parameter selection grounded in biophysical rationales, and a consistent measurement of a shared core set of outcomes represent the priorities that the field must address for this evidence base to mature.